Refugees' Right to Work: Efficiency and Equity in Host Country Labor Markets

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Abstract

Most refugees face legal barriers to work, pushing them into informal jobs or out of work altogether. Policymakers accept this underutilization of refugee labor to shield low-wage host workers from refugee competition. This paper studies whether this efficiency–equity trade-off holds in general equilibrium when workers can endogenously re-sort across occupations. Exploiting a large-scale work permit reform for Syrian refugees in Jordan, a novel harmonized dataset, and a shift–share measure of exposure to refugee competition, I document three facts. First, Jordanians exit the occupations most exposed to refugees, which are primarily lower-skill. Second, those who exit upgrade: college-educated Jordanians move into less-exposed, higher-paying occupations. Third, despite increased refugee labor supply, Jordanians' average wages rise in exposed occupations relative to less-exposed ones, consistent with positive selection among stayers. To rationalize these facts and separate the direct supply effect of refugee entry from hosts' re-sorting response, I build a general-equilibrium model of occupational choice in which workers sort on wages and homophilic preferences over co-worker composition, reflecting that refugee entry lowers both the wage and the non-pecuniary value of a job for hosts. As homophily amplifies wage-based sorting, re-sorting can dominate the direct supply effect, generating relative labor scarcity in low-skill occupations. Alongside sizable efficiency gains from improved refugee labor utilization, the estimated model implies work permits have a progressive incidence on host worker wages, as poorer Jordanians either (i) upgrade out of exposed low-paying, lower-skill occupations or (ii) remain and accrue scarcity-induced wage gains, with the poorest quintile gaining about 3%.

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1 Introduction

Today, one in every 200 people in the world is a refugee, a figure that has tripled since 2010 and is expected to grow with the onset of climate change (UNHCR, 2025, 2024a).[1] Displacement has also become increasingly protracted, with refugees displaced for 10 to 26 years on average (UNHCR, 2021; , IOM). Refugees are no longer a temporary humanitarian concern but a lasting presence in their host economies.

Yet most refugees live in low- and middle-income countries (LMICs), where the majority face substantial legal barriers to work (Ginn et al., 2022).[2] Restrictions on the right to work successfully deter some refugees from working, but push many others to work illegally, often in highly informal occupations (Fasani et al., 2021; Bahar et al., 2021; Ahrens et al., 2023; Bahar et al., 2024b). By confining refugees to a narrow set of occupations that rarely match their skills or excluding them from working altogether, these restrictions severely underutilize refugee labor, at a cost to both refugees and the aggregate productivity of the host economy. These efficiency losses are particularly acute for LMIC countries, where economic growth is first order.

Policymakers argue that such restrictions are necessary to shield local workers from the negative effects of competing with refugees (Clemens et al., 2018; Bahar et al., 2024b). The literature substantiates this claim: while average wage and unemployment effects of immigrant and refugee inflows are typically small, losses are concentrated among substitutable local workers, typically low-skill, low-wage workers (Borjas and Monras, 2017; Ottaviano and Peri, 2012). Restricting refugees' right to work may prevent worsening income inequality within the host community. However, this rationale assumes that local workers in exposed jobs do not respond to refugee entry.

This paper studies whether granting refugees the right to work presents an efficiency–equity trade-off when host workers can endogenously re-sort across occupations in response to refugee entry in general equilibrium. I leverage a natural experiment created by a large-scale work permit scheme for Syrian refugees in Jordan and build a novel harmonized labor-force dataset, covering both refugees and hosts before and after the policy change. Descriptively, Syrian refugees enter into employment and access a broader, less informal set of occupations. Using a shift-share measure of exposure to refugee competition, I find that re-sorting is the dominant margin of adjustment by Jordanians, as those in highly exposed, lower-wage occupations upgrade to higher-paying jobs. Exit is consistent with positive selection: the average wage for Jordanians in highly exposed occupations increases relative to less exposed occupations. To rationalize these facts and disentangle the direct supply effect of refugee entry from the effect of hosts' re-sorting, I develop a general equilibrium model of occupational choice. I augment the standard model to include homophilic preferences – a preference to work with members of one's own group – such that refugee entry lowers not only wages

[1] Statistic computed by author based on UNHCR's estimation that the global population of refugees was 42.7 million at the end of 2024 (UNHCR, 2025).

[2] Legal barriers to work include restricted access to work permits and limitations on freedom of movement (such as mandatory internment in refugee camps or assigned settlement locations within the hosting country). Barriers may be *de jure* or *de facto*.

for Jordanian hosts but also the non-pecuniary value of exposed jobs. The estimated model implies that work permits raise aggregate output, primarily through an improved utilization of refugee labor. Simultaneously, sorting on wages and homophily triggers sufficient re-sorting by host workers to offset the downward wage pressure created by refugees, generating a progressive incidence on host wages. After accounting for host workers re-sorting, the presumed efficiency–equity trade-off disappears: the right to work raises aggregate output while benefiting the poorest Jordanian workers most.

This paper exploits the introduction of a large-scale work permit scheme for Syrian refugees in Jordan, a natural experiment uniquely suited to causally identify the general-equilibrium effects of expanding refugees' right to work. First, the timing of the refugee arrival and policy change are distinct – Syrians primarily entered Jordan between 2011 and 2013 but did not gain the legal right to work until mid-2016.[3] Second, the policy exclusively reduced legal barriers to employment, unlike many other regularization programs, which also grant access to public services and benefits.[4] Third, the inflow of Syrian refugees – equivalent to 11.1% of Jordan's population – was likely large enough to produce detectable effects on local wages and employment (Jordanian Department of Statistics, 2015). Finally, because the policy was adopted under pressure from the European Union and international donors, its enactment was plausibly exogenous to domestic labor-market conditions.

I construct a novel dataset by harmonizing five distinct sources of government and non-governmental labor-force survey data, much of which remains publicly unavailable to date. The final dataset consists of representative samples for both groups before and after the work permit scheme, constituting roughly 950,000 individual Jordanian observations, spanning 2005-2023, and 22,000 individual Syrian refugee observations for the years of 2014 and 2017-2022. The data has two additional features that are key to identification and measurement. First, the 2014 survey of Syrian refugees includes recall data on their labor market outcomes, including their occupation prior to displacement, which provides a predictor of their occupation in Jordan once they have the right to work. Second, since the surveys are administered through home visits, all worker types, including formal, informal, and illegal workers, are included.

Jordan's labor market structure shapes how hosts are likely to respond to refugees' gaining the right to work. Unemployment is high and persistent throughout the period, and the public sector employs a large share of Jordanian men. Like workers in many LMICs, Jordanians change jobs and occupations frequently, and a substantial share of university graduates work in jobs that do not require a degree. Because workers expect to move, a job's value depends in part on how easily it leads to the next one, and job search runs largely through personal connections. These networks are segregated by nationality, in part because Jordanians prefer interacting with other Jordanians.[5]

[3]The Jordan–Syria border closed in 2016 following an ISIS attack, fixing the refugee stock before and after the policy change.

[4]For example, Colombia's regularization of Venezuelan forced migrants allowed access to legal work as well as a wide-range of public services and social assistance, which Ibáñez et al. (2024) evaluate.

[5]Using primary survey data I collected, I show the importance of networks in job search, the segmentation of networks, and the homophilic preferences Jordanians have for their networks.

Therefore, refugee entry can lower an occupation's value to Jordanians in two ways: (i) by pushing down wages, and (ii) by reducing its non-wage value, both because many Jordanians dislike working alongside refugees and because it thins Jordanian networks, making future search more costly.

Any such response by Jordanians requires that work permits actually changed whether and where Syrian refugees work. I therefore begin the empirical part of the paper by documenting a series of stylized facts showing how work permits reduced Syrian refugees' barriers to work, providing evidence of a first stage. Following the introduction of work permits, the employment rate among working-age Syrian men rose by 31 percentage points on a base of 20%.[6] Syrian refugees move away from highly informal occupations and spread across a broader range of occupations. Finally, the wage gap between observationally equivalent Syrians and Jordanians narrows from 31.2 percent to 10.5 percent, consistent with work permits reducing the wedges Syrian refugees face. Together, this descriptive evidence suggests that work permits alleviated a binding constraint on refugees' labor market decisions, consistent with a more efficient allocation of refugee labor post-permits.

To causally identify Jordanians' re-sorting response, I implement an event-study design that leverages variation in Jordanians' labor-market exposure to Syrian refugees. Exposure is defined using a shift–share measure that combines the spatial distribution of Syrian refugees across Jordan (shift) and their pre-displacement occupation distribution (share) (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2021). Using additional data on Jordanians prior to the arrival of Syrian refugees, I am able to benchmark the effect of work permits relative to the initial effect of refugee arrival, as well as rule out pre-trends. I find that a one–standard-deviation increase in exposure reduces the share of Jordanians working in exposed occupations by 7% on average for the post-period. Despite refugee entry putting downward pressure on wages, an additional standard deviation in exposure raises average wages among Jordanians in those occupations by 4.2%, suggesting positive selection in those who remain. Jordanians who are less productive in these highly exposed occupations leave, and the remaining workers are positively selected on ability (in that occupation).

Re-sorting also results in Jordanians upgrading from low-paid, highly exposed occupations into better-paid, less-exposed ones. Using a two-period panel (2010–2017) that tracks Jordanians before and after the policy change, I show that this upgrading is driven by college-educated Jordanians, who exit the highly exposed occupations that under-utilize their educational qualifications.[7] This pattern is consistent with labor-supply-driven growth. Work permits nearly tripled labor income per working-age Syrian refugee man, increasing demand for goods and services and allowing firms to expand production and absorb the additional refugee labor. As Syrian refugees fill low- and middle-skill jobs, firms' demand for the higher-skill work that complements them rises, creating opportunities for Jordanians to upgrade. In the model, this complementarity is formalized through the firm's CES production function.

[6] All analyses are restricted to working-age men due to low female participation among both groups. During the period, employment rates for Jordanian and Syrian refugee women stood at 7% and 9% respectively.

[7] The panel data observes individuals in 2010 (pre-refugee arrival) and 2017 (post-work permits). I assume effects estimated in the panel data are driven by the introduction of work permits, as there is no evidence of an effect from refugee arrival in the event study, which uses my main dataset of repeated cross-sections.

In the second part of the paper, I build a general equilibrium model of occupational choice consistent with my empirical findings. I extend the canonical Hsieh et al. (2019) model of labor misallocation by introducing *homophilic preferences* – a preference for working with members of one's own group – as an additional driver of sorting, capturing Jordanian's preference for social segmentation from Syrian refugees and the implications this has for network-based search.

Individuals choose the occupation (or unemployment) that maximizes their indirect utility, which depends on their effective wage – determined by their ability draw and the occupation-specific wage per efficiency unit – and their group-level amenity value for the occupation. This amenity value including an exogenous component and an endogenous component based on worker composition to capture homophilic preferences. Syrian refugees will face additional occupation-specific frictions, modeled as taxes on their effective wages, which will include all barriers faced by refugees in the labor market, including but not limited to not having the right to work. The work-permit scheme relaxes these wedges, prompting a reallocation of refugee labor toward a more efficient distribution across occupations.

The model illustrates how granting refugees the right to work impacts the host labor market. In partial equilibrium, refugee entry increases labor supply in exposed occupations, reducing the wage per efficiency unit, as is standard in models of migration shocks. In general equilibrium, however, Jordanians then respond to the wage effects by re-sorting out of these occupations. If the decline in relative labor supply from Jordanians' re-sorting outweighs the initial increase from refugee entry, the wage per efficiency unit in exposed occupations will increase.[8] Homophilic preferences magnify Jordanians' re-sorting response on wages alone, making the exit from exposed occupations more elastic and allowing for re-sorting to dominate the direct supply effect of refugee entry for some parameter values.

Subsequently, I estimate the model using a bootstrapped efficient GMM procedure, matching occupation shares and average wages from the model's log-linearized equilibrium equations to my data, before and after the introduction of work permits. This approach identifies the wedge, homophilic preference, and productivity parameters for each year that rationalize the observed employment and wages across occupations. To identify the effects of the work permit scheme and control for exogenous changes in occupation-level productivity over time, I compare the estimated equilibrium in the final year of my study period to a counterfactual in which the frictions faced by Syrian refugees are held at their pre-work permit levels.

Using this counterfactual, I quantify the aggregate and distributional effects of the work permit scheme, as well as decompose the relative importance of refugee entry (partial equilibrium) from local re-sorting (general equilibrium). The model implies that work permits increased total output by nearly 11%, driven primarily by an improved utilization of refugee labor. Syrian wages increase by 20% and their employment rate rises by 36.8 percentage points. Jordanians' average wages rise

[8] Additionally, re-sorting will have a compositional effect on the quality of the labor supplied, as lower ability Jordanians will be the ones to re-sort, leading to the remaining Jordanian workers in exposed occupations being positively selected.

modestly by 0.9%, accompanied by a 0.8-percentage-point increase in unemployment, on a base rate of 23.4%.[9] The work permits also reduces inequality within the host community, as the poorest quintile of Jordanians gain a 3% wage increase on average.

Accounting for re-sorting fundamentally alters the distributional effects of the work permit scheme. Without re-sorting, simulated by holding Jordanians' occupational choices fixed when refugees enter, Jordanians who compete with refugees experience wage losses. Given that Syrian refugees enter into lower-skilled low-wage occupations, the supply shock pushes down the wages for substitutable workers: already poor Jordanians. At the same time, Jordanians in high-paying occupations benefit as complementary segments of the labor force. Allowing Jordanians to re-sort reverses this pattern, generating a progressive incidence on host wages. These re-sorting gains arise through two channels. The first is occupational upgrading – some Jordanians exit the occupations most affected, switching to higher-paying occupations and earning more. Second, the re-sorting effect is large enough to reverse the refugees' relative labor supply shock in these low-skill occupations, creating a small scarcity-driven wage premium for those Jordanians who remain in these jobs. By accounting for both refugee and host reallocation, my estimates show allowing refugees to work improves both efficiency and equity, increasing total output while reducing wage inequality within the host community.

Contributions to the Literature: This paper contributes to the large literature on how refugee inflows affect host labor markets, beginning with Card (1990) and synthesized by recent meta-analyses (Ruiz and Vargas-Silva, 2018; Becker and Ferrara, 2019; Verme and Schuettler, 2021; Rozo and Grossman, 2025). The most common finding across contexts is that average employment and wage effects are typically null.[10] When negative effects are found, they fall on substitutable local workers, typically those in low-skill or informal jobs. In cases of positive effects, gains accrue to complementary workers, namely those in high-skill or formal jobs.[11] I extend this literature by showing that accounting for occupational re-sorting of local workers overturns this canonical pattern.[12] Without re-sorting, high-skill locals gain while low-skill ones lose; once accounting for re-sorting, the opposite occurs. Low-skill workers benefit through two channels of re-sorting: (i) upgrading into higher-paying occupations, consistent with evidence of occupation change as a response margin to migration shocks (Foged and Peri, 2016; Lebow, 2024; Caiumi and Peri, 2024); and (ii) relative-scarcity effects that raise wages in low-skill occupations as others exit. In this way, my paper highlights how general equilibrium responses by local workers – often overlooked, and certainly not separately identified, in reduced-form analyses – fundamentally reshapes the

[9] The total wage bill accrued by Jordanians also increases, implying the gains in average wages is not offset by the rise in unemployment.

[10] Fallah et al. (2019) finds a null average effect of Syrian refugees' arrival on Jordanian workers.

[11] E.g., Ottaviano and Peri (2012); Maystadt and Verwimp (2014); Calderón-Mejía and Ibáñez (2016); Tumen (2016); Morales (2018); Borjas and Monras (2017); Altindag et al. (2020); Caruso et al. (2021); Fasani et al. (2021) (negative efects) and Foged and Peri (2016); Dustmann et al. (2016); Peri and Yasenov (2019) (positive effects).

[12] While I study the introduction of the right to work for refugees already residing in a country, a substantial share of Syrian refugees in Jordan were deterred from working by legal barriers. This allows my results to speak directly to the literature on the local impacts of refugee arrivals.

distributional consequences of refugee integration.

Second, I contribute to the emerging literature studying the impacts of reducing refugees' legal barriers to work, instead of the impacts of their initial arrival. Existing work, which primarily focuses on developed countries, has shown that right to work policies improve refugees' employment and wage outcomes (Fasani et al., 2021; Foged et al., 2022; Ahrens et al., 2023; Schuettler and Caron, 2020; Ibáñez et al., 2024).[13] While I also find positive impacts on employment and wages, my primary contribution is embedding refugees' labor market decisions into a structural model, which allows me to quantify how the right to work improves the allocation of refugees across occupations, and the aggregate implications this has for productivity. Most similar to my paper are Bahar et al. (2021) and Bahar et al. (2024a), which study the spillover effects of a regularization program for undocumented forcibly displaced Venezuelan migrants onto local Colombians.[14] These papers focus on how the formalization of refugees affects local employment across the formal and informal segments of the economy. I, instead, focus on how refugees' right to work impacts the allocation of refugee and local labor across occupations through sorting.

Third, I contribute to the literature on the misallocation of labor in three key ways (Hsieh et al., 2019; Bryan and Morten, 2019; Birinci et al., 2024). First, I extend the Hsieh et al. (2019) canonical model for labor misallocation by introducing homophilic preferences, which amplify wage-based sorting patterns. Homophily can account for the persistence of occupation-based segregation of groups, documented in the literature on social networks (Calvó-Armengol and Jackson, 2004; Charles and Guryan, 2008; Currarini et al., 2009; Patacchini and Zenou, 2012; Bolte et al., 2020; Jackson, 2021), even when frictions are reduced for the marginalized community. Second, I extend the analysis of the model, going beyond the aggregate productivity effects of reducing frictions, to understand how barriers faced by one group (refugees) indirectly distort the allocation of another (locals). Third, rather than relying solely on counterfactuals to quantify effects, I am able to leverage a natural experiment that exogenously relaxed refugees' legal barriers to work, which allows me to estimate changes in misallocation directly and to decompose the resulting welfare gains into two channels: the direct supply effect from refugee entry and the local re-sorting effect.

Finally, I contribute to a growing body of work, demonstrating how large refugee and migration waves can catalyze aggregate gains through productivity growth. Using historical migration episodes in developed countries, Peters (2022), Tabellini (2020) and Sequeira et al. (2020) link economic growth, through mechanisms of innovation and industrialization, to the influx of migrants.

[13] There is a corresponding literature on regularizing undocumented migrants, which holds some parallels. However, these programs also risk encouraging new migrants to come irregularly in the hopes of future rounds of amnesty. Devillanova et al. (2018) and Amuedo-Dorantes and Bansak (2011) demonstrate that amnesty programs for undocumented migrants positively impact beneficiaries. Additionally, Elias et al. (2025) and Borjas and Edo (2023) consider how these policies impact local workers and aggregate outcomes, such as tax revenues and GDP.

[14] Colombia's regularization program for forcibly displaced Venezuelans combined work rights with access to social protection, education, and healthcare, whereas I study a work permit scheme alone to isolate the effect of easing legal barriers to employment. Additionally, Venezuelans are classified by UNHCR as "others in need of international protection" rather than refugees, they remain eligible for UNHCR assistance, but lack some protections and rights that refugees are given (UNHCR, 2025).

In contemporary settings, Galaasen et al. (2025) and Alix-Garcia et al. (2018) provide evidence that migrant or refugee arrival specifically can generate growth and employment opportunities through increased demand. I add an additional mechanism that fuels migration-driven growth – improvements in the utilization of refugee labor and the resulting productivity gains.

Outline: The remainder of the paper is organized as follows. In Section 2, I describe the study setting and my data. Section 3 characterizes the Jordanian labor market. Section 4 presents stylized facts documenting how Syrian refugees responded to work permits. In Section 5, I present causal evidence of Jordanians re-sorting in response to Syrian refugee entry. I present my model in Section 6 and its subsequent identification and estimation in Section 7. Section 8 quantifies the relative magnitude of the *direct supply effect* from the *re-sorting effect* and evaluates the aggregate and distributional effects of the policy. Finally, Section 9 concludes.

2 Setting and Data

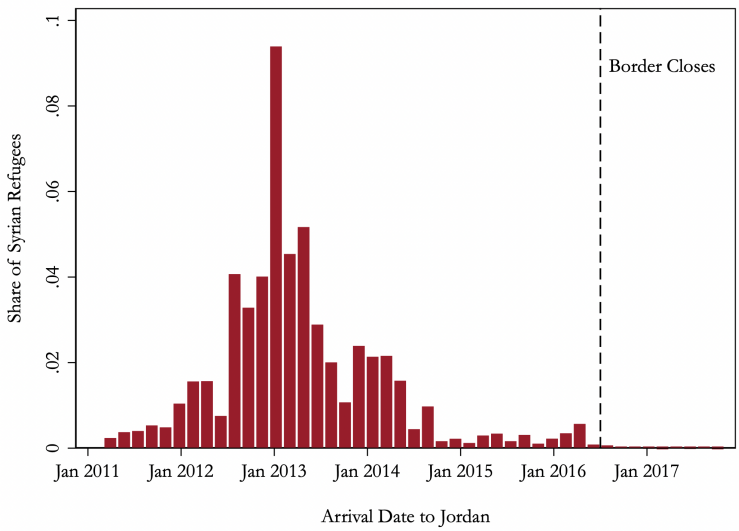
2.1 The Arrival of Syrian Refugees

Since the outbreak of the Syrian Civil War in March 2011, more than 62 percent of Syria's pre-war population has been internally or externally displaced. Of the 5.5 million Syrian refugees who fled the country, UNHCR estimates 670,000 sought refuge in Jordan (UNHCR, 2018). Jordan's 2015 census found the Syrian refugee population to be closer to 950,000 (Jordanian Department of Statistics, 2015). According to Jordanian census data, the Syrian refugee crisis represented an 11.1% increase to Jordan's total population of 8.55 million. Restricting the local population to only Jordanian citizens, the population increase is closer to 14.3%.[15]

Beyond its magnitude relative to Jordan's existing population, the crisis was exacerbated by the speed of refugees' arrival. The majority of Syrian refugees arrived to Jordan in the two-year period between 2012 and the end of 2013. By the end of 2014, 95% of refugees had already arrived (as illustrated in Figure 1). Following an ISIS car-bomb attack in June 2016, Jordan closed its border with Syria to new refugee entries, effectively ending the flow (Sweis, 2016).

[15] Using more conservative estimate of UNHCR-registered Syrian refugees, the population increase was 7.7% and 10.2% respectively. For comparison of scale, Venezuelan refugees in Colombia increased the host population by 5.7%, Syrian refugees in Turkey by 3.9%, and all refugees in Germany by 3.0% (UNHCR, 2024b). (These figures are calculated using the UNHCR-estimated refugee population hosted by that country as the percent change in the total population.)

Figure 1: Distribution of Syrian Refugee Arrivals to Jordan



Note: Distribution constructed from refugees' self-reported arrival dates in Fafo/DoS 2018 survey data. (Observations: 32,881)

Syrian refugees settled primarily among the host community in Jordan's urban and peri-urban areas, resulting in roughly 86% of Syrian refugees residing outside refugee camps. Out-of-camp refugees were expected to secure housing and support themselves economically, despite limited humanitarian assistance and no legal right to work.[16] In 2015, UNHCR estimated that 86% of Syrian refugees were below the Jordanian poverty line of 68 JOD per capita per month and 67% below the abject poverty line of 28 JOD (UNHCR Jordan, 2015). Refugee household expenditures were largely financed by accumulating debt and engaging in negative coping strategies. Over 60% of Syrian refugee families had debts of at least 40 JOD per capita and more than a third had debts exceeding 100 JOD per capita; nearly all refugees had exhausted their savings and sold household assets.[17]

Access to legal employment was extremely restricted. Only 1% of Syrian refugee households had a member with a work permit (UNHCR, 2014).[18] By 2015, it was estimated that 160,000 Syrian refugees had resorted to working illegally in Jordan's informal sector, despite the risk of detention,

[16] Since the onset of the crisis, only 20% of out-of-camp Syrian refugees received cash assistance from UNHCR. The majority received minimal support in the form of food vouchers from the World Food Program (World Food Programme, 2015). Based on their level of food insecurity, qualifying refugee households received vouchers valued between 10 and 20 JOD per capita per month, both in and out of camps.

[17] 80% of Syrian refugee households engaged in at least one of the following negative coping strategies: selling productive assets, decreasing food intake, resorting to high-risk or socially degrading jobs, or sending family members to beg (UNHCR Jordan, 2015).

[18] In rare cases, some Syrian refugees were able to obtain work permits intended for migrant workers, which required documentation that most refugees lacked and for employers to complete an extensive application process to hire the worker at the cost of 500 to 800 JOD per application. The cost of work permits often far exceeded what a Syrian refugee or employer would be willing to pay (Jordanian Ministry of Labor, 2012; Tobin and Alahmed, 2019).

fines, or deportation if caught (Stave and Hillesund, 2015; Tobin and Alahmed, 2019).[19]

2.2 The Policy Change: the Jordan Compact

Jordan has a long history of hosting refugees, including successive waves of Palestinians (1948, 1967, 1991, 2000), Iraqis (1991, 2003), and, most recently, Syrians (Chatelard, 2010). Apart from granting citizenship to Palestinians who arrived between 1948 and 1967, Jordan has not provided refugees with the legal right to work. However, under mounting pressure from the European Union and the broader international humanitarian community, the government of Jordan signed the Jordan Compact in February 2016, which committed Jordan to issuing 200,000 work permits for Syrian refugees.[20] The Compact represented a major shift in both Jordan's policy toward hosting refugees and represented one of the world's first efforts to reform refugees' right-to-work policy.[21]

In return, the European Union committed to extensive foreign grants and loans as well as trade concessions that facilitated Jordanian manufacturing exports to Europe (Hagen-Zanker et al., 2018; Kattaa et al., 2021). The Compact aimed for preferential trade deals to help Jordan expand its existing economic zones and generate sufficient labor demand to absorb the labor supply created by the work permits. As of 2018, the zones had failed to grow with firms in those zones exporting negligible amounts to the EU and other markets abroad (Hagen-Zanker et al., 2018). Moreover, Syrian refugees did not take up the few jobs that existed in these zones. The zones are located far from the major cities and the jobs offer worse pay and longer hours than working in the informal economy (Hagen-Zanker et al., 2018). Despite the failure of the Compact to kick-start Jordanian manufacturing exports, it did lead to the issuance of over 430,000 work permits (including renewals) between 2016 and the end of 2023 (UNHCR, 2024c). However, work permits are valid for only one year and must be renewed annually. Internal UNHCR estimates correspond to roughly 60,000 Syrian refugees holding valid work permits each year.

The Jordan Compact did not contain an implementation plan for providing work permits to Syrian refugees. As a result, the Jordanian government enacted a series of changes to improve take-up of work permits. Initially, the Jordanian Ministry of Labor (MoL) extended the work permit scheme for migrant workers to include Syrian refugees. As with other non-Jordanian workers, Syrian refugees had to find an employer willing to apply and pay for a employer-specific work permit on their behalf (ILO, 2017). They were also restricted to the set of occupations open to foreign workers, which pre-dated the arrival of Syrian refugees (ILO, 2012; Jordan et al., 2023).[22]

[19] Syrian refugees apprehended for working illegally were often detained in a designated section of the Azraq refugee camp, which served as a de facto detention facility.

[20] The 2015-2016 refugee crisis in Europe fueled the political impetus behind the Compact. It was largely thought that improving conditions for Syrian refugees in their existing host countries would make them less likely to migrate to Europe. Notably, this policy did not extend to other refugee communities in Jordan, such as Iraqis (Huang and Gough, 2019).

[21] In many ways, the Compact heralded in a new approach to refugee integration in cases of protracted displacement, with the goal of transitioning refugees from dependence on humanitarian aid into economic self-sufficiency (Huang and Gough, 2019).

[22] For a complete list of occupation closed to non-Jordanian workers, see Appendix G.

Given the prohibitive cost of work permits, which often exceeded the wages refugees would earn, few permits were issued in the initial following the Compact's enactment (Tobin and Alahmed, 2019). In response, MoL waived all work permit fees for Syrian refugees. To improve mobility across employers, particularly in sectors characterized by temporary or seasonal work, sector-wide work permits, called "flexible work permits," were introduced for agriculture in October 2016 and construction in June 2017. In late 2018, MoL introduced its first reform toward refugee self-employment: permits for Syrian refugees to open home-based businesses in the sectors of food processing, handicrafts, and tailoring (Kattaa et al., 2021). In January 2019, Syrian refugees and employers became exempt from paying the fines associated with late renewals of work permits (Tobin and Alahmed, 2019). In July 2021, flexible work permits were extended to cover all open occupations. For a complete list of reforms enacted during this time period, see Appendix F.

These policy reforms to the work permit scheme improved access for refugees. Before the Compact's introduction, only 10% of working Syrian refugees did so with obtained work permits, originally designated for migrant workers, similar to a work visa in other contexts, despite not being technically eligible. In the year following the enactment of the Compact in mid-2016, work permit coverage expanded 25 percentage points. The subsequent years continued to see increased take-up with 53% of Syrian refugee workers holding a work permit in 2020. Appendix Figure A1 plots, over time, the share of employed working-aged Syrian refugee men holding an active work permit.

2.3 The Data

A key contribution of this paper is the construction of a novel dataset, comprising representative repeated cross-sections for the labor market outcomes of the Jordanian host community and Syrian refugees respectively. Most of these data sources are not publicly available and required substantial relationship-building with their respective owners to obtain access for use in this project. For a more detailed description of the various data sources, construction of key variables, and descriptive tables of the result samples, see Appendix D.

For Jordanians, the dataset spans from 2005 to 2023, encompassing several years pre-refugee arrival (2005-2011), post-arrival but pre-permit years (2012-2015), and post-permit periods (2016-2023). To build it, I harmonize labor market survey data from several different sources, including the Economic Research Forum (ERF), the International Labor Organization (ILO), Fafo Institute, and the Government of Jordan's Department of Statistics (DoS) (OAMDI, 2017, 2018; Department of Statistics, 2024; Stave and Hillesund, 2015). Notably, all labor market surveys in Jordan employ random sampling of households and interview all household members aged 15 or older about their labor market outcomes, ensuring coverage of both formal and informal workers. The resulting dataset contains 950,111 observations of working-aged Jordanian men.

For Syrian refugees, the dataset includes one pre-permit year (2014) and several post-permit years (2017-2022), but lacks continuous coverage for the pre-period, unlike the dataset for Jordanians, due to limitations in data availability.[23] I harmonize labor market survey data from several

[23]Until 2017, the Jordanian Department of Statistics did not collect representative data for Syrian refugees in

different sources, including the International Labor Organization (ILO), Fafo Institute, and the Government of Jordan's Department of Statistics (DoS) (OAMDI, 2017; Department of Statistics, 2024; Stave and Hillesund, 2015; Tiltnes et al., 2019). Restricting the sample to working-aged men, the dataset has over 27,447 individual observations of working-aged Syrian refugee men. Importantly, the survey data from ILO and Fafo Institute in 2014 contains recall data on one's labor market outcomes in Syria in the beginning of 2011, prior to refugees' displacement.

The harmonized dataset is restricted to working-aged men, given that female labor force participation rates being among the lowest in the world at 18% for Jordanian women and 14% for Syrian refugee women in Jordan.[24] Due to high structural unemployment, the actual share of working-aged women who are working is 9% and 7% for Jordanians and Syrian refugees respectively.

In addition to this new labor market dataset for Syrian refugees and Jordanians, I used two additional sources: (i) census data to construct the population distribution of both groups across governorates (Jordanian Department of Statistics, 2015) and (ii) primary survey data on 1,192 Syrian refugees and Jordanians that I collected in Mafraq governorate in spring 2025 to measure network-based job search, attitudes towards refugees, and zero-sum thinking.

3 Defining Characteristics of the Jordanian Labor Market

This section describes Jordan's labor market prior to the introduction of work permits. Key institutional features include structural unemployment, prevalent occupational mismatch and mobility, and a heavy reliance network-based search where networks are highly segmented by nationality.

3.1 Jordanians and Syrian Refugees Before Work Permits

Table 1 provides summary statistics for working-aged male Jordanians and Syrian refugees prior to work permits in 2014, for which data exists for both populations. The third column provides statistics based on available recall data for Syrian refugees' outcomes in Syria prior to displacement (in 2011). Both Jordanians and Syrian refugees predominantly live in urban areas in Jordan, while Syrian refugees originate from much more rural locations in Syria (as evidenced by the recall data). Working-aged male Jordanians have 2.9 years more of education (11.3 on average) than Syrian refugees and a much higher rate of tertiary education (15%), nearly four times that of Syrian refugees. Economically, Syrian refugees are worse off than Jordanians across every margin. On average, they are roughly fives times more likely to be unemployed, earn 44% less than Jordanians (in monthly wages), and work 7.2 hours more a week.

The Jordanian labor market is characterized by high structural unemployment (13%), minimal self-employment (9%), and a high reliance on public sector employment (44%). Jordan's labor market also maintains high levels of informality with only 58% of Jordanian workers covered by social security, which drops to 35% after excluding those employed in the public sector. Due to not

their Employment and Unemployment Survey, as refugees were not legally allowed to work.

[24] Working age is defined as between 18-60 years old. In Jordan, 18 is the minimum age for full-time work and 60 is the official retirement age.

having the right to work, Syrian refugees workers are clustered in the informal sector under insecure employment conditions, including high rates of temporary work, a lack of written contracts, and exclusion from social security.

Table 1: Jordanian Labor Market Characteristics Prior to Introduction of Work Permits

	Jordanians	Syrian refugees	
	2014	2014	2011 (recall)
Age	34.0 (0.05)	32.7 (0.21)	32.5 (0.22)
Urban	0.71 (0.002)	0.80 (0.009)	0.34 (0.010)
Education (years)	11.3 (0.01)	8.4 (0.07)	8.3 (0.07)
Tertiary degree	0.15 (0.002)	0.04 (0.004)	0.04 (0.004)
Full-time student	0.11 (0.001)	0.01 (0.002)	0.04 (0.004)
Employment rate	0.65 (0.002)	0.20 (0.008)	0.56 (0.010)
Labor force participation rate	0.75 (0.002)	0.67 (0.009)	0.75 (0.009)
Unemployment rate (of LF)	0.13 (0.002)	0.69 (0.011)	0.26 (0.011)
Own-account (self-employed)	0.09 (0.002)	0.08 (0.012)	0.13 (0.009)
Outcomes not collected in recall data			
Monthly wage (JOD)	342 (4.9)	193 (5.2)	–
Weekly hours	43.3 (0.05)	50.5 (0.98)	–
Public sector	0.44 (0.003)	0.02 (0.005)	–
Social security coverage	0.58 (0.014)	0.02 (0.005)	–
Written contract	0.59 (0.014)	0.08 (0.012)	–
Temporary work	0.15 (0.010)	0.69 (0.020)	–
Observations	55,122	2,560	2,260

Note: Sample restricted to males of working age (18-60 years old). Each statistic reported is the mean with standard errors in parenthesis. Labor force participation is the share of either employed or unemployed people relative to total population. Unemployment rate is the share of labor force participants that are unemployed. 2011 data is for Syrian refugees prior to displacement, constructed through recall.

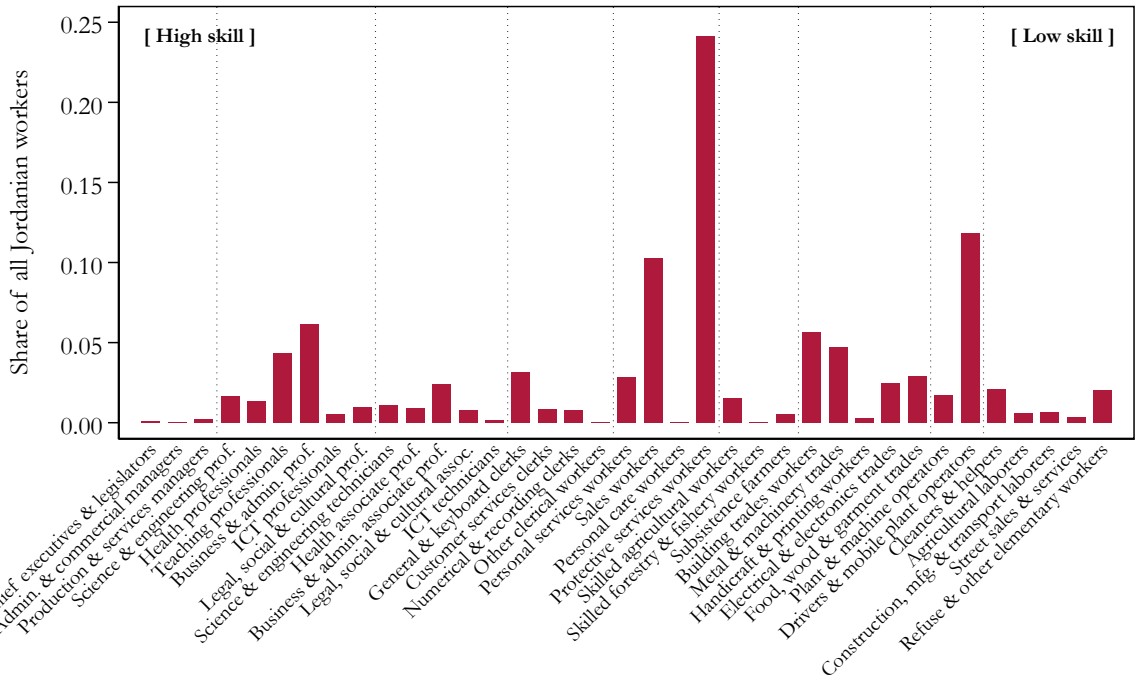
3.2 Occupations, Skill Mismatch, and Mobility

Jordanians primarily work in medium-skill jobs, with high shares in sales, protective services, various technical trades, and driving and machine operation.[25] Figure 2 plots the distribution of Jordanian workers across the occupation space, where each bar corresponds to the share of all Jordanian workers working in a given 2-digit ISCO code occupation. Comparatively, Syrian refugees also worked in many of these occupations prior to displacement, as can be seen in Appendix Figure A2. Consistent with Jordanians' higher rates of educational attainment, 15% of Jordanians are employed in occupations that require tertiary degrees while few Syrian refugees held such jobs

[25] 74% of Jordanian men work in ISCO 1-digit codes 4-8, which are classifed by ILO (2008) as "Skill Level 2" or medium-skill occupations.

in Syria.

Figure 2: Occupation Distribution of Jordanian Workers - Prior to Work Permits



Note: The y-axis plots the share of all Jordanian workers working in a given occupation group in 2014, prior to the introduction of work permits. The x-axis plots occupation groups, which correspond to ISCO 2-digit occupation codes. For a complete mapping to 2-digit ISCO codes, see Appendix H. Dotted lines on the graph correspond to cutoffs between 1-digit ISCO codes, which are ordered by skill-level.

Highly educated Jordanians experience a substantial rate of occupational mismatch; in 2014, one in six employed Jordanian graduates worked outside of occupations that required a college-degree.[26] Half were in protective services, which is largely public and closed to refugees, while about a quarter were in sales and personal services, among the occupations where Syrian refugees are most concentrated prior to displacement (20% of Syrian workers).

In line with many LMICs, Jordan's labor force experiences high rates of job switching and occupational mobility, even in the absence of labor supply shocks (Donovan et al., 2023). From employment spells recorded in the JLMPS 2010 survey wave, 17% of employed Jordanians transitioned between jobs in the last two years (2008-2010), of which over half also switched occupations. Among the workforce as a whole, occupational downgrading was as common as upgrading. However, 9% of mismatched college graduates – defined as those working in occupations that do not require tertiry degrees – upgraded within the past two years at nearly twice the rate of non-graduates in the same occupations (see Appendix Table A1). This suggests that mismatched Jordanians with

[26] Educational requirements of occupations are classified by the ILO. This rate of mismatch is stable throughout the period prior to work permits (2012-2015), with an average of 18.2% of tertiary-educated Jordanians working in jobs that did not utilize tertiary degrees.

tertiary degrees in occupations exposed to Syrian refugee competition may be more likely to adjust by changing occupations, as explored in Section 5.

3.3 Job Networks and Search

Personal connections, often family-based – known as *wasta* or *ma'aref* in Jordanian Arabic – are deeply rooted in Arab societies and govern many social and professional transactions (Baranik et al., 2023). In Jordan, these personal connections play a substantial role in both job search and matching (see Appendix Table A2). From primary survey data I collected in Mafraq, the governorate most affected by Syrian refugee arrival, half of Jordanian respondents found their current or most recent job through a friend (31%) or family member (20%). Syrian refugees rely even more on personal networks to find work (47% through friends and 13% through family). For both groups, contacts primarily help by connecting respondents directly to their eventual employer.

Jordanians' networks are highly segregated. Restricting to friends, since family ties are almost always within nationality, only 1% of friends who helped Jordanian respondents find their jobs were Syrian, even though Mafraq has roughly one Syrian refugee for every 3.5 Jordanians (outside Za'atari refugee camp).[27] If Jordanians formed network ties without regard to nationality, roughly 22% of their contacts would be Syrian. By contrast, Syrian refugees rely substantially on Jordanian connections in job search: 37% of the friends who helped Syrian respondents find work were Jordanian. This asymmetric network segmentation is consistent with network formation models, which predict that minority groups (Syrian refugees) form more ties with the out-group than the majority (Jordanians) does (Currarini et al., 2009).

3.4 Jordanian Preferences for Homophily

Why are Jordanians' networks so homophilic when Syrian refugees' are not? I show that a preference among Jordanians for working with and relying on other Jordanians in job search plays a substantial role. I first document Jordanians' self-reported disutility from interacting with Syrian refugees, then use an incentivized willingness-to-pay elicitation to separate this preference from differences in how useful Jordanian and Syrian contacts are.

Many Jordanians report a discomfort from close contact with Syrian refugees, including at work (Figure A3). In 2014 ILO/Fafo survey data, only 48% of Jordanians stated they were comfortable working with a Syrian, which was lower than the proportion who were comfortable living in the same village (53%) or having children at the same school (64%), but only slightly above being close neighbors (46%). These attitudes suggest that Jordanians have homophilic preferences over co-worker composition.

While preferences may drive network segmentation among Jordanians, formed networks may also reflect long-standing ties that pre-date refugee arrival or that Jordanian contacts are simply more

[27] This ratio is computed from the 2015 census count for Jordanians and Syrian refugees in Mafraq governorate, excluding the population of Za'atari camp (**?**), which my survey did not sample. Including refugees in Za'atari, there are roughly 1.8 Jordanians for every Syrian refugee in Mafraq.

useful in job search. To separate a *homophilic preference* for Jordanian contacts from variation in the usefulness of different types of contacts, I elicited survey respondents' incentivized willingness to pay (WTP) for a job-search connection.[28] In partnership with a local Mafraq-based NGO, respondents chose between a one-time cash transfer for job-search costs and having the NGO connect them to a contact in a sector they want to work in, for a varying range of cash amounts. Each respondent valued four types of contacts: a Jordanian worker, a Syrian worker, a Jordanian employer, and a Syrian employer.

Because Jordanians and Syrian refugees earn very different wages, I express WTP as the share of the average monthly wage for the respondent's group.[29] Both groups value Jordanian contacts more than Syrian contacts, but by very different margins, as illustrated in Figure 3. Jordanian respondents are willing to pay 13.2% of a monthly wage for a connection to a Jordanian worker but only 8.1% for a Syrian refugee worker, or 38% less. For employers, the gap is 41% (15.7% versus 9.1%). Comparatively, Syrian refugee respondents value a Syrian contact only 8% less than a Jordanian one for workers (15.0% versus 16.4%) and 10% less for employers (17.7% versus 19.7%).

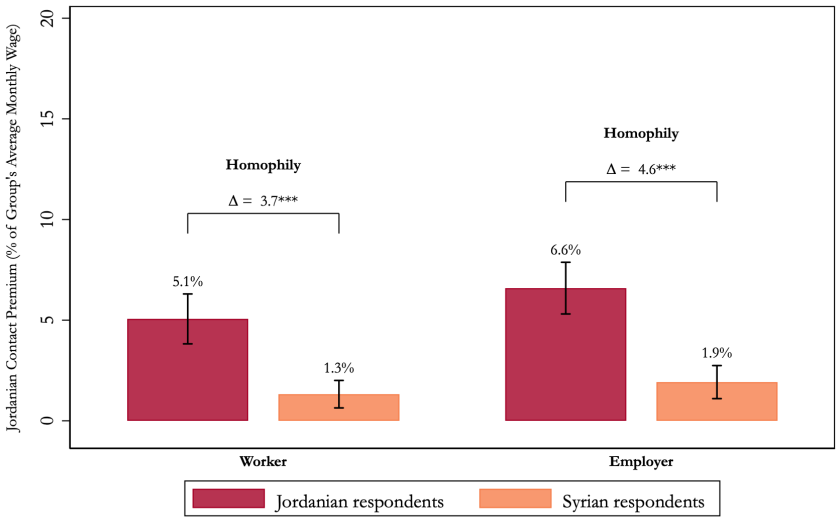
Jordanian contacts may be worth more to everyone simply because they are more useful – Jordanians' networks cover more sectors, better placing them to refer job seekers to employers.[30] Assuming this usefulness premium of a Jordanian over a Syrian contact is the same for both groups, the gap for Syrian respondents identifies this difference in contact value, while the additional gap among Jordanian respondents (3.7 percentage points of a monthly wage for workers and 4.6 points for employers) reflects a preference for Jordanian contacts that contact value alone cannot explain.

[28] Choices were incentivized by informing respondents that 10% of those surveyed would be chosen to have one of their choices enacted.

[29] Average monthly wages of employed men in Mafraq from the DoS Employment and Unemployment Survey in 2023: 396 JOD for Jordanians and 258 JOD for Syrian refugees.

[30] In my survey, respondents were asked about five randomly selected sectors. Randomizing the sectors means the comparison reflects how far networks reach, not which sectors each group happens to work in. Jordanians know someone in 31% of these sectors, compared with 25% for Syrian refugee respondents, and know 40% more people per sector (1.24 versus 0.88). Syrian refugee respondents who had not heard of any recent openings in a sector were more than twice as likely to attribute this to not knowing anyone there (12% versus 5%). Syrian refugees were, however, slightly more likely to have heard of an opening in the past 30 days (5.3% versus 3.4%), likely reflecting more active job search consistent with unemployment rates being higher.

Figure 3: Difference in Willingness to Pay for a Jordanian vs. Syrian Connection



Note: The y-axis plots the value premium of Jordanian contacts, which is defined as the additional percent of one's group-average monthly wage that one is willing to pay to be connected to a Jordanian rather than a Syrian refugee. This is plotted for workers on the left and employers on the right. Dark red bars correspond to this premium among Jordanian respondents and orange bars for Syrian refugee respondents. The difference in the computed premium between Jordanian and Syrian refugee respondents then corresponds to the added preference Jordanians have for being connected within their own group.

There are two key threats to this assumption that would bias the estimated homophilic preference among Jordanians. First, relative to their incomes, Syrian refugee respondents value a Jordanian contact more than Jordanian respondents do (16.4% versus 13.2% of a monthly wage for a worker), consistent with refugees having weaker networks overall. This would suggest that the value difference between Jordanian and Syrian contacts is larger for Syrian respondents, such that subtracting it to estimate Jordanian homophilic preferences constitutes a lower bound. Second, if Syrian refugees themselves prefer Syrian contacts, this would attenuate the true premium of a Jordanian contact such that the decomposition would overstate the importance of Jordanian homophily. However, since Syrian refugees rely substantially on Jordanians for job search in practice, any such in-group preference are likely weak and the estimates are more plausibly a lower bound.

Together, these facts shape how Jordanians are likely to respond when Syrian refugees enter their occupations. Jordanians change jobs and occupations frequently, and mismatched graduates are the most likely to move up (Section 3.2). Jordanians find work through networks that are almost entirely Jordanian, and they value Jordanian contacts beyond what those contacts' usefulness can explain. As Syrian refugees enter an occupation, the occupation becomes less attractive to Jordanians for reasons beyond falling wages: through direct reluctance to work alongside Syrians, and because job search within the occupation becomes more costly with fewer in-group contacts. To account for this in the model in Section 6, I include an occupation-specific amenity that depends on the composition of workers in an occupation and allow the strength of this preference to differ between

Jordanians and Syrian refugees. Thus, Jordanian workers will re-sort in response to changes in the pecuniary and non-pecuniary value of an occupation.

4 Stylized Facts – Syrian Refugees' Response to Work Permits

I next document how Syrian refugees' labor market outcomes in Jordan changed in response to the work permit scheme. Because the work-permit scheme was implemented nationally, I cannot causally identify its direct effects in reduced form. I instead recover them through the estimated structural model.

The arrival of Syrian refugees expanded Jordan's labor force and reshaped the equilibrium allocation of workers across occupations. In principle, this new equilibrium could be efficient if observed employment or occupational differences reflected preferences and endowments rather than constraints. In practice, refugees face numerous potential frictions, including psychosocial and physical trauma (Ashraf et al., 2025), restrictions on spatial mobility (Fasani et al., 2021; Foged et al., 2022), social exclusion (Loiacono and Silva-Vargas, 2025), and legal barriers to employment (Ginn et al., 2022). Ex ante, it is unclear which constraint binds most tightly. The introduction of work permits isolates one such constraint: the legal barrier to work. If this constraint limited refugees' labor-market participation and occupational choice set, relaxing it should generate observable changes in their employment and occupational allocation.

Three stylized facts support this interpretation of how work permits affected Syrian refugees' labor market outcomes. First, Syrian refugee employment increased by roughly 30 percentage points following the policy change. Second, refugees systematically exited the most informal occupations, and those employed in more formal sectors were far more likely to hold a work permit, indicating that permits served as the main channel to accessing employment in more formal occupations. Finally, the wage gap between observationally equivalent Syrian and Jordanian workers narrowed after the introduction of work permits.

Fact 1: Employment rates of Syrian refugees increased after work permits.

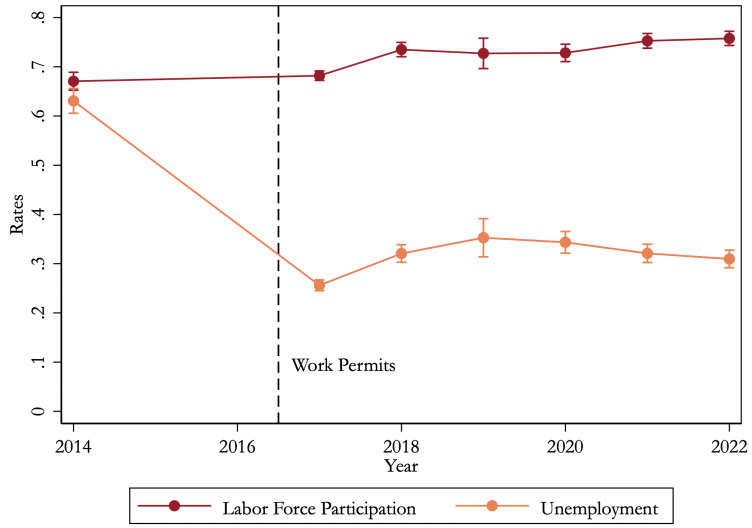
The majority of Syrian refugees who has been employed before displacement are pushed into unemployment or out of the labor force entirely, once they arrive in Jordan.[31] While this could be an efficient employment rate for Syrian refugees in their new labor market, there is strong evidence that refugees' employment decisions were severely constrained by the absence of legal work rights.

This is most evident by the high costs Syrian refugees faced when working. If caught working illegally, refugees were at risk of being fined, forcibly relocated to Azraq refugee camp, or deported back to Syria. From the employer side, hiring refugees illegally exposed firms to high fines if caught (ILO, 2022; Tobin and Alahmed, 2019). In 2014, 52.6% of unemployed refugees cited the "lack of work permits" as the reason they were not working. After work permits were introduced, that share

[31] To show this, I exploit a unique feature of the 2014 survey data – recall data on Syrian refugees' labor market outcomes in Syria in 2011 (prior to displacement). I then construct an artificial panel between 2011 and 2014. Only 25.4% of those employed in Syria in 2011 were working in Jordan in 2014.

fell to 8.8%. (see Appendix Figure A5). I refer to these individuals who abstained from working due to potential legal repercussions as *labor law compliers*.

Figure 4: Labor Force Participation and Unemployment Rates for Syrian Refugees



Note: Sample restricted to male Syrian refugees of working age (18-60 years old). Labor force participation is the share of either employed or unemployed people relative to total population. Unemployment rate is the share of labor force participants that are unemployed. (Observations: 27,341)

Figure 4 underscores the decline in unemployment among Syrian refugees following the introduction of the work permit scheme. Unemployment fell 37.5 percentage points from 63.1% in 2014 to 25.6% in 2017 because *labor law compliers* obtained work permits and entered employment. In terms of the effective employment rate among all working-aged Syrian refugee men, this is a 30 p.p. increase. In subsequent years (2018-2022), the unemployment rate among Syrian refugees increases to an average of 32.8%, which is largely offset by a corresponding increase in the labor force participation rate.[32] This pattern aligns with broader macroeconomic trends in Jordan, particularly rising unemployment among young people. Appendix Figure A4 compares extensive margin labor market outcomes between Syrian refugees and Jordanians.

Fact 2: Syrian refugees move out of highly informal occupations after work permits.

Before the introduction of work permits, Syrian refugees were disproportionately concentrated in a few highly informal occupations -— construction, sales, and personal services – where enforcement of labor regulations is limited. In 2014, 57 percent of employed refugees worked in these three occupations alone.[33] In the post-period (2017-2022), that share declined to 47 percent, a nearly 20-percent reduction. Consistent with legal barriers confining refugees to a narrow subset of the

[32]Using the common denominator of all working-aged men, the adjusted unemployment rate rose from 17.7% to 23.7% between 2017 and 2018 and remained relatively constant.

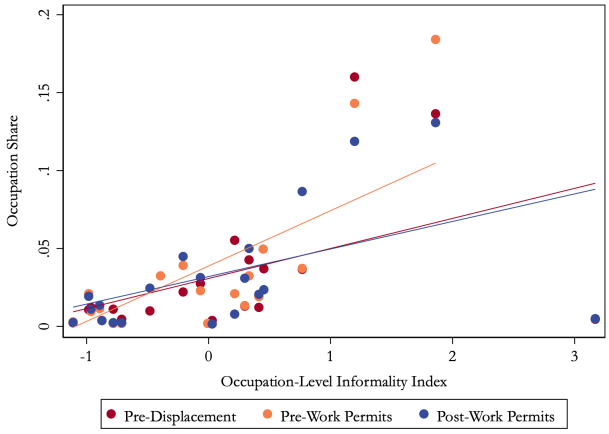
[33]Occupations are classified using the 2-digit ISCO 08 codes from the ILO. For a full characterization of these codes and specific examples, see Appendix H.

economy, namely highly informal occupations, I also find that work permits correspond to an increased dispersion of Syrian refugees across occupations and entry into a wider set of occupations (see Appendix Figure A6). This diversification decreases the average share of Syrian refugees working in a given occupation by 5 percentage points.[34]

Figure 5 is the resulting binned-scatter plot of the share of all employed Syrian refugees working in a given two-digit ISCO occupation code relative to that occupation's informality index. Notably, the slope shifts sharply upward when Syrian refugees do not have the right to work. This is consistent with larger shares of Syrian refugees working in occupations with high levels of informality. Following the introduction of work permits, the allocation shifts away from these highly informal occupations, mirroring a distribution across occupations that is more in line with that observed before displacement.

Work permits appear to drive this shift. Refugees in more formal occupations were substantially more likely to hold a work permit, and permit coverage rose systematically with an occupation's formality. A one-standard-deviation increase in occupation informality is associated with a 4.8 percentage-point lower share of refugees holding a permit (Appendix Figure A7). Together, these patterns indicate that work permits operated as a gateway toward accessing a wider set of occupations, enabling refugees to move beyond the narrow, highly informal segment of the economy accessible to them prior to the reform.

Figure 5: Occupation Shares vs. Occupation-Level Informality



Note: Binned scatter plot of the correlation between the share of Syrian refugee workers in a 2-digit ISCO occupation code and the z-scored index of informality for the occupation, based on informality of Jordanian workers in that occupation. (Observations: 13,052)

Fact 3: Wage gap narrows between observationally equivalent Syrians and Jordanians.

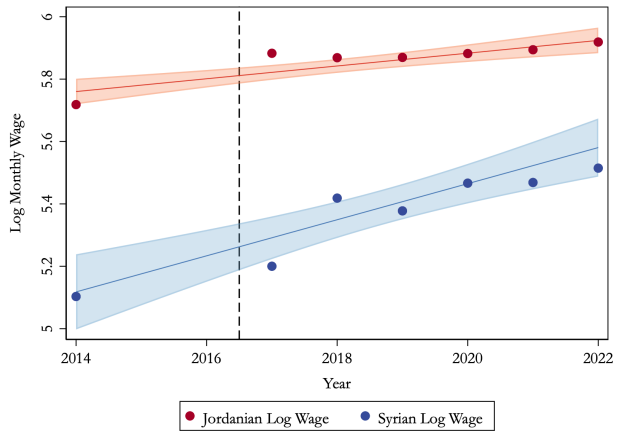
Figure 6 shows that the occupational reallocation, demonstrated in Fact 2, is assortative. For

[34]The skewness of the distribution nearly doubles, suggesting selection into occupations that were previously difficult to access. See Appendix Table A4 for the statistics governing the distribution across occupations pre- and post-work permits.

estimates of the wage gap over time, see Appendix Table A5, which contains the coefficient estimates for the wage penalty faced by Syrian refugees relative to Jordanians that are observationally equivalent: working in the same occupation and industry, living in the same location (governorate), and having the same years of education and experience (age). Prior to the introduction of work permits, a Syrian refugee earned 31.2 percentage points less than a comparable Jordanian.[35] After work permits are introduced, the wage gap narrowed steadily, reaching a low of a 10.5 percentage points in 2022.

These wage gain estimates cannot disentangle the effect of Syrians' occupational reallocation from other benefits to legalization, such as reducing the monopsony power of employers over refugees (Amior and Stuhler, 2023; Amior and Manning, 2024).[36] Both effects should positively impact wages, reducing the gap between Syrian refugees and their Jordanian counterparts. In the structural model, I will be able to isolate the role of reallocation for Syrian refugees in capturing a larger share of their marginal product of labor and thus the evolution of the wage gap between Syrian refugees and Jordanians.

Figure 6: Wage Differences between Syrian Refugees and Jordanians



Note: Log monthly wage regressed on year x refugee status, includes human capital controls as well as governorate, occupation, and industry FEs. Standard errors clustered at the unit of 2-digit ISCO occupation codes. Points represent average log wage by group. Line corresponds to best fit from regression with 95% confidence bands. (Observations: 147,193)

Together, these stylized facts reveal substantial changes in Syrian refugees' labor-market outcomes following the introduction of work permits. First, these findings suggests that legal barriers to work acted as a binding constraint on Syrian refugees' labor supply optimization problem, resulting in an inefficient allocation of refugee labor. Second, we should view the introduction of work permits as a relaxation of this constraint, potentially resulting in a better allocation, both along the extensive margin (employment vs. unemployment) and the intensive margin (across occupations).

[35] Column 1 in Table A5 reports β, the difference in the log wage, which can be converted to percentages using the following transformation: $\left(e^{\beta} - 1\right) \cdot 100$.

[36] Intuitively, a greater degree of monopsony power allows employers to exploit and underpay refugee employees by exacting greater markdowns.

I will formally test this through the quantification of my model.

5 Causal Evidence of Jordanians Re-sorting

Syrian refugees' entry into and reallocation across occupations under the work permit scheme created differential supply shocks by occupation. This section provides causal evidence on how Jordanians respond to these supply shocks, primarily by re-sorting into less-exposed occupations. I first present my shift-share measure for exposure and corresponding event-study design. Next, I report my main findings for how Jordanian occupation shares and average wages are affected by exposure to Syrian refugees. Given the upgrading pattern in Jordanians' occupation reallocation, I use my two-period panel data to examine the mechanisms driving these responses. I find that Jordanians with college degrees are the most likely to exit highly exposed occupations and upgrade to occupations that better utilized their educational qualifications.

5.1 Empirical Strategy

I estimate how the introduction of work permits for refugees affected local workers by exploiting variation in exposure to the resulting refugee labor-supply shock. I leverage two sources of exogenous variation to construct a shift–share measure of exposure at the occupation–governorate level that reflects both (i) the relative concentration of Syrian refugees across Jordan's governorates (the *shift*) and (ii) the pre-displacement occupational composition of Syrian workers (the *share*). Following Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2021), identification requires that at least one component – either the shift or the share – be exogenous. In my setting, both components are plausibly exogenous, and orthogonal to each other (Borusyak and Hull, 2023), as refugees do not sort spatially based on their pre-displacement occupation.

Shift Component: The shift captures the relative size of the refugee inflow to each local labor market, defined at the governorate level.[37] I measure geographical exposure using the ratio of Syrian refugees to the total population in each governorate, using the 2015 population census. As seen in Figure 7, there is substantial variation in the geographic distribution of Syrian refugees across Jordan. The share of Syrians in a governorate's total population ranges from 1.3% in the southern governorate of Al-Tafileh to 35.7% in the northern governorate of Al-Mafraq. Consistent with most refugee populations, Syrian refugees predominantly settled in northern Jordan near the border of their country of origin.[38]

The key identifying assumption for the shift to be valid is that Syrian refugees do not strategically choose a location to settle based on local labor market conditions. Nearly 95% of Syrian refugees had arrived in Jordan by the end of 2014, before the work permit scheme was annoucned and at a time when no right to work was anticipated, making it unlikely that settlement decisions

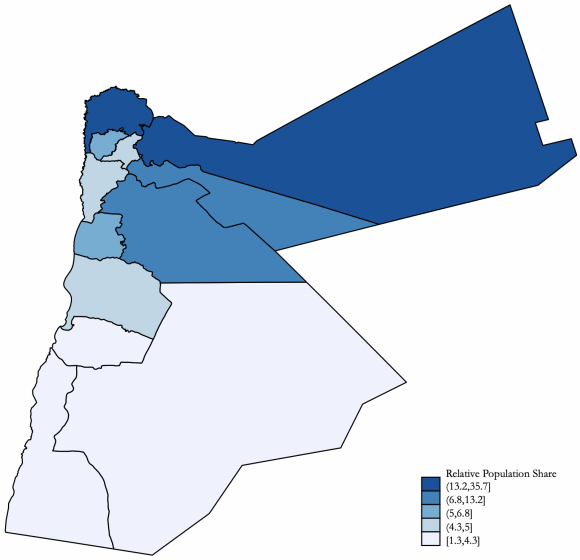
[37] Due to data limitations, I can only disaggregate to the administrative unit of the governorate, of which there are twelve in Jordan.

[38] Distance to the Syran broder is a standard proxy for geographic exposure to refugee inflows in the forced migration literature; see Angrist and Kugler (2003) and Bahar et al. (2024a).

were influenced by local employment opportunities. Furthermore, Jordan closed its border with Syria in June 2016, following an ISIS attack at Rukban (Sweis, 2016), fixing the stock of refugees shortly after the policy change. Thus, there is little concern that additional refugees arrived and subsequently selected locations based on local employment opportunities.

In the case of any strategic relocation in response to the introduction of work permits, I use the 2015 census distribution of refugees, which predates the permit scheme, to measure geographic exposure. However, there is minimal evidence of refugee relocation within Jordan after initial arrival (Fallah et al., 2019).

Figure 7: Distribution of Syrian Refugees to Jordanian Population



Note: Relative population shares based on the Jordanian Department of Statistics population census conducted in 2015. For each population's distribution separately, see Appendix Figure A8.

Share Component: To capture exposure by occupation, I exploit a unique feature of my data: the occupations that Syrian refugees held in Syria in 2011, shortly before displacement, measured through recall. Appendix Figure A2 plots this pre-displacement occupation distribution for Syrian refugees.[39] I define the occupation-level share, Share_o, as the fraction of Syrian refugees who worked in occupation o in Syria before displacement. This share proxies for the relative likelihood that refugees will work in occupation o once they gain the legal right to work. To verify the relevance of this instrument, I show that pre-displacement occupation shares strongly predict refugees' observed occupational distribution after the introduction of work permits, as reported in Appendix Figure A9. The exclusion restriction is that Jordanians' occupational outcomes are affected by these pre-displacement shares only through the occupations that refugees ultimately

[39] Given that this recall data comes from a representative sample of Syrian refugees in Jordan in 2014, I do not face the usual concerns that refugees are a selected sample from the general occupations distribution in the country of origin. The plot includes the distribution in 2011 (pre-displacement) alongside the distribution in 2014.

enter into once in Jordan.

Jordanians face substantial exposure to Syrian refugees because of substantial occupational overlap between the two populations. Appendix Figure A13 plots the Jordanian occupation distribution pre-work permits with quartiles of exposure to Syrian refugees. Jordanians in the middle-to-lower end of the skill distribution (based on the ISCO occupation codes ordering) are most exposed to Syrian refugee labor market competition.

My shift-share measure of exposure can then be expressed as:

$$\text{Exposure}_{o,g} = \left(\frac{\text{Syrian Refugees}_g}{\text{Population}_g} \right) \cdot \text{Share}_o \tag{1}$$

where Share_o measures the share of Syrian refugees working in occupation o in Syria prior to displacement. Thus, a Jordanian is more exposed to the Syrian refugee labor supply shock if they (i) live in a governorate with a higher concentration of Syrian refugees and (ii) work in an occupation where more Syrian refugees have worked in prior to displacement. I then use a z-scored version of the previously defined exposure measure in my main specification.[40]

Event Study Design: To estimate the effects of the work permit scheme on Jordanians, I use the following event study specification with two-way fixed effects and an unstaggered continuous treatment (the shift-share):

$$Y_{o,g,t} = \sum_{t=-6}^{12} \beta_t \left(\text{Exposure}_{o,g} \times \mathbb{1}\left(\text{Year} = t\right) \right) + \gamma_{o,g} + \delta_t + \epsilon_{o,g,t} \tag{2}$$

where $Y_{o,g,t}$ denotes the outcome of interest (Jordanians' occupation share or occupation-average wage), $\gamma_{o,g}$ are occupation-by-governorate fixed effects, and δ_t are year fixed effects. The unit of observation is an occupation–governorate–year cell, constructed by aggregating individual-level labor-force survey data. Standard errors are clustered at the unit of treatment: an occupation-governorate. Because the underlying surveys are repeated cross-sections rather than panels, changes in these aggregated outcomes reflect reallocation across occupations (and into unemployment) at the population level.

This specification identifies how local outcomes evolve differentially across occupations and locations with greater exposure to refugees. The key identifying assumption is that, conditional on fixed effects, no unobserved factors correlated with both exposure and local labor-market dynamics evolved differentially around the introduction of work permits.

5.2 Main Results

My event study specification covers three periods: before Syrian refugees arrive (2005-2011), after Syrian refugees arrive but before work permits (2012-2016), and after work permits are introduced (2017-2023). This allows me to test for pre-trends in my outcome variables, the effect of

[40] For a plot of z-scored exposure across the occupation distribution, see Appendix Figure A10.

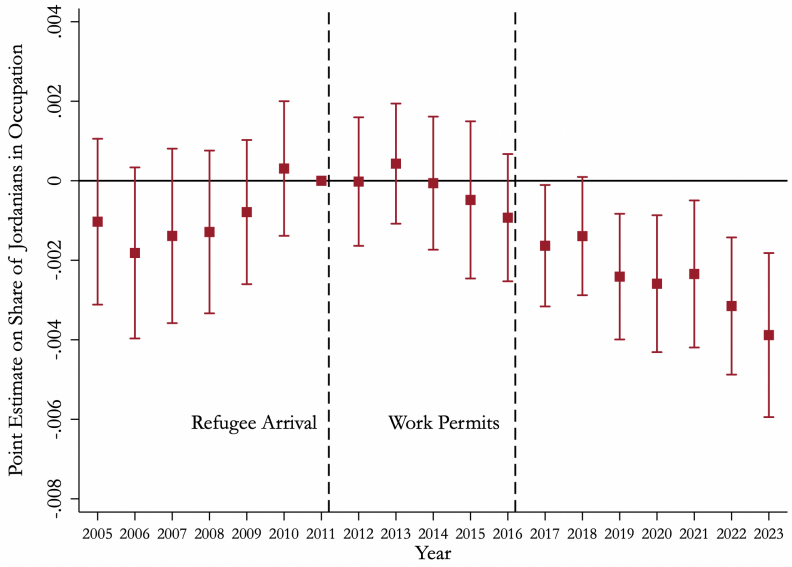
Syrian refugees' arrival, and the subsequent effect of introducing work permits.

Occupational Shares: I estimate how exposure to Syrian refugees affected Jordanians' reallocation across occupations and exit into unemployment. To do so, I measure reallocation across occupations by looking at changes in Jordanians' occupation shares over time. I construct occupation shares by location and measure them as the number of Jordanians working in a given occupation relative to the number of Jordanians participating in the labor market.

$$\text{Occ Share}_{o,t} = \frac{\text{Jordanian Workers}_{o,t}}{(\sum_o \text{Jordanian Workers}_{o,t}) \; + \; \text{Unemployed Jordanians}_t} \tag{3}$$

I include unemployed Jordanians in the denominator of my preferred specification to account possible exit into unemployment in addition to reallocating across occupations. The exclusion of unemployment results in a downward bias in the estimated coefficient on occupation shares in cases where there is exit into unemployment.[41]

Figure 8: Effect of Exposure on Jordanian Occupational Shares Including Unemployment



Note: Point estimates from a regression of occupation shares, computed including unemployment from the denominator, for Jordanians on a z-scored shift-share measure of exposure to Syrian refugees, measured as share of refugees in that occupation x share of refugees in a given governorate. Regression includes year and occupation-by-governorate fixed effects. Standard errors are clustered at the unit of treatment: the occupation-governorate.(Observations: 6,696 occupation x governorates)

Figure 8 plots the resulting coefficients, β_t, estimated from running my main event study specification on Jordanian occupational shares (including unemployment). The yearly estimates for the period of 2005 to 2011 shows no evidence of pre-trends. After Syrian refugees arrive to Jordan, I

[41]This is caused by exit causing both the number in the occupation (the numerator) and the number of workers overall (the denominator) to fall simultaneously, resulting in a seemingly larger effect than when unemployment is included in the denominator, as only the numerator will change.

find that higher levels of exposure do not affect the occupation shares of Jordanians. Given my definition of exposure, this is not a surprising result. As previously detailed, Syrian refugees are disproportionately clustered in informal occupations prior to the introduction of work permits relative to where they worked pre-displacement. If Syrian refugees are mismatched relative to their occupations prior to displacement, then my exposure measure should not be a strong predictor of the competition Jordanians face in the labor market as a result of Syrian refugee entry. Additionally, only 20% of working-aged Syrian refugee men were working during this period, deterred by the lack of work permits, creating smaller labor supply shock.

However, once work permits are introduced, I find a strong, increasingly negative effect on share of Jordanians in occupations that are highly exposed to refugees' entry. On average over the post-period, an additional standard deviation of exposure decreases the share of the Jordanian labor force working in an occupation by of 0.29 percentage points. Given the average occupation share size is 3.54%, this corresponds to an average exit rate of 7% out of highly exposed occupations.[42]

I report the estimated coefficients for my alternative measure of occupation shares (excluding unemployment from the denominator) in Appendix Figure A11. Absolute effect sizes are slightly smaller in this alternative specification, consistent with the main estimate of exit including transitions into unemployment as well as reallocation to other occupations. To decompose the relative importance of reallocation vs. unemployment in exit from highly exposed occupations, I compare the relative difference in the event-study coefficients with and without unemployment in the denominator. As my data is repeated cross-sections and my exposure variable is defined using one's occupation, I cannot directly estimate unemployment using my main specification. Appendix Figure A12 plots the share of occupational exit accounted for by reallocation vs. unemployment in the post-permit period, showing that roughly 85–90% of adjustment occurs through reallocation while 10-15% can be attributed to exit into unemployment.[43] I will provide additional evidence that occupational exit is driven by reallocation, not unemployment, using my 2-period panel data (see Table 2).

Given exit is driven primarily by reallocation across occupations, I find a pattern of occupational upgrading – Jordanians leave highly exposed, lower-paying occupations for less exposed, higher-paying occupations. As Appendix Figure A13 shows, Jordanians in the middle to lower end of the skill distribution are most exposed to labor market competition with Syrian refugees. By exiting these occupations, Jordanians re-sort to higher-skill jobs. Appendix Figure A15 documents average log wages by quartiles of exposure to show that these highly exposed, lower-skill occupations are in fact lower paying on average.

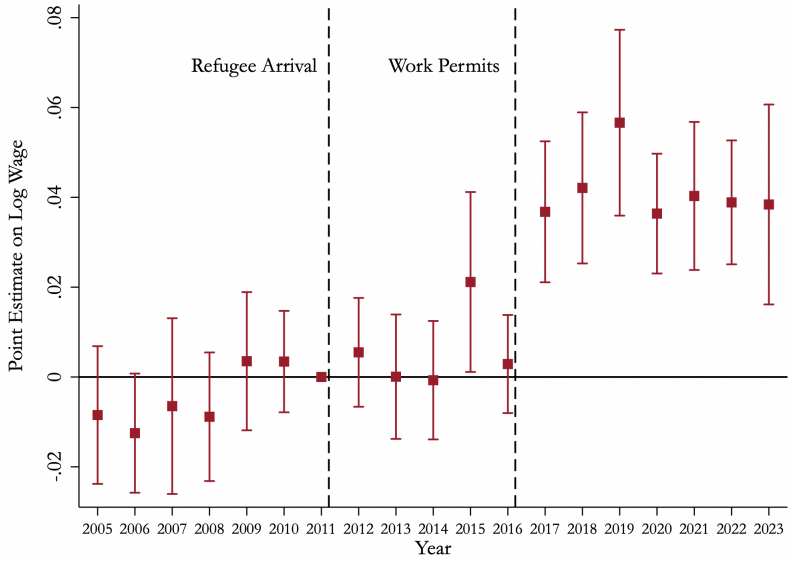
Occupation Average Wages: My regression results present strong evidence of Jordanians exiting

[42] Using the mean occupation share size as a scaling factor, the yearly coefficient implies a decrease in the occupation share of 3.9% in 2018, growing to 11% in 2023.

[43] Appendix Table A6 reports the full set of coefficients and corresponding decompositions for the entire study period. Decomposition values before 2016 are noisy because both specifications yield coefficients close to zero, mechanically inflating ratios. After 2016, the decomposition stabilizes, with less than 15 percent of exit reflecting unemployment.

high exposure occupations in response to Syrian refugees obtaining work permits. To affirm that this reallocation is assortative, thus indicative of selection, I estimate the the effect of exposure on the average wage of Jordanians by occupation. Figure 9 presents the effect of exposure on the logged average wage of Jordanians at the occupation-governorate level. Similarly to the event study for occupational shares, the yearly estimates for the period of 2005 to 2011 shows no evidence of pre-trends. After Syrian refugees arrive to Jordan, I find that exposure does not affect the occupation-level wages of Jordanians (consistent with the null for shares). Once work permits are introduced toward the end of 2016, a strong positive and persistent treatment effect emerges for Jordanians in occupations (in governorates) that are more exposed to Syrian refugees. A standard deviation increase in the exposure level increases occupation-level wages by an average of 4.2% for the period of 2017-2023.

Figure 9: Effect of Exposure on Occupation Average Wages for Jordanians



Note: Point estimates from a regression of log occupation-average wages for Jordanians on a z-scored shift-share measure of exposure to Syrian refugees, measured as share of refugees in that occupation x share of refugees in a given governorate. Regression includes year and occupation-by-governorate fixed effects. Standard errors are clustered at the unit of treatment: the occupation-governorate. (Observations: 6,696 occupation x governorates)

The event study coefficients estimate changes in the relative average wages between high- and low-exposure occupations. Combined with the previous finding that Jordanian occupation shares fall in high-exposure occupations, these estimates provide causal evidence that Jordanians re-sorted across occupations following Syrians' gaining access to work permits.

In a standard sorting model, individuals sort to the occupation where they have a comparative advantage, resulting in an inverse relationship between the share of workers in an occupation and the average ability (positive selection). The entry of Syrian refugees will drive the wage per efficiency unit (marginal product of labor) down in high exposure occupations and Jordanians with

the lowest ability in these occupations will exit in response. As a result, the average ability of the remaining Jordanians in the high exposure occupation will increase, offsetting to some degree the wage reduction caused by Syrians' entry. Comparatively, the low exposure occupations will now receive a larger share of Jordanians, which will be lower in ability than the current stock of Jordanians in the occupation, thus lowering the average ability and, in turn, the occupation's average wage among Jordanians. In addition to this compositional effect induced by re-sorting, I will also use the model to test for whether the re-sorting of Jordanians can be large enough to offset the direct supply effect of refugee entry, amplifying these wage gains.

As additional evidence of positive selection in Jordanians' re-sorting, I compare the distribution of residuals from an individual-level wage regression across occupations in the first and fourth exposure quartiles. Following the introduction of work permits, the mean and upper quantiles of these occupation-average residuals rise systematically in high-exposure occupations relative to low-exposure ones. This pattern indicates that higher-ability Jordanians increasingly remain in high-exposure occupations, while lower-ability workers exit, consistent with assortative reallocation. See Appendix Subsection C for specification and results.

My estimates can only detect changes in the relative wage gap between occupations differentially exposed to Syrian refugees, which creates two limitations in my analysis. First, as with all shift-share estimates, I have a "missing intercept" problem.[44] The estimated coefficient on exposure estimates relative changes, but not whether aggregate wages have shifted up or down as a result of the policy. Second, I cannot assign the directionality of these relative wage effects by exposure, as the reduction in the wage gap could be driven by (i) increases in the average wages of highly exposed occupations, (ii) decreases in the average wages of less exposed occupations, or most likely, (iii) a combination of the two. I will use the estimated model to overcome both of these challenges, allowing me to speak to the general equilibrium level effects on aggregate wages as well as within exposure level wages that the reduced form cannot identify.

5.3 Mechanism: Occupational Upgrading from Re-Sorting

Jordanians are switching out of highly exposed occupations, which tend to be lower-paying and typically require less formal education, into less exposed occupations, which tend to be the opposite. To understand how this re-sorting to more education-intensive occupations occurs, I leverage my one source of panel data, the Jordan Labor Market Panel Survey (JLMPS), which contains observations on Jordanian individuals in 2010, prior to refugees' arrival, and again in 2017, after work permits are introduced. For comparability, I compute the same exposure measure as the main event study specification but based on an individual's occupation and location in 2010, prior to refugees' arrival. Moreover, since my event study does not detect an effect from exposure prior to the introduction of work permits, I interpret the effects found using this panel data to be

[44] The "missing intercept" problem arises for regression with a continuous treatment – such as exposure to refugees – to infer aggregate or level effects from relative coefficients. The regression estimates how outcomes vary with exposure but not the overall level shift in the economy, as there is no instrument for the intercept.

attributed to the effect of work permits, not refugees' arrival. The panel data analysis has two main benefits: (i) I can explicitly test for the effect of exposure on exit into unemployment and (ii) I can look at *who* upgrades occupations or exits in response to being exposed to Syrian refugees after work permits are introduced.

As I am primarily interested in transitions – namely in reallocation across occupations and from employment to unemployment – I define my outcome variables as indicator variables that take the value of 1 if the transition occurred and zero if else. Importantly, since my exposure measure is defined in terms of one's occupation, the sample is restricted to those working-aged Jordanian men employed in 2010, prior to the arrival of refugees. For the extensive margin, I define two measures: employment among the working-aged and unemployment among those still active in the labor force. For the intensive margin of occupational reallocation, I look at both occupational upgrading and downgrading. To do this, I rely on the occupation's corresponding ILO ISCO 1-digit code, which are ordered based on the occupation's educational requirement, such that higher numbers requires less formal education.[45] An individual has upgraded (downgraded) their occupation if they work in an occupation with a lower (higher) 1-digit ISCO code in 2017 than in 2010.[46]

With these outcome variables in mind, I implement the following regression for working-aged Jordanian men:

$$Y_{o,g,i} = \beta \cdot \text{Exposure}_{o,g,t=2010} + \gamma \cdot \text{Age}_{i,t=2010} + \epsilon_{o,g,i}$$

The regression includes baseline age as a control and clusters standard errors at the unit of exposure (occupation x location).

To understand *which* Jordanians change occupations or exit employment, I run the above regression specification and include a categorical variable for educational attainment, which is defined by the highest level of education individuals have completed at baseline (2010). Education levels are defined into the following mutually exclusive categories: primary education or less (up to 5th grade), lower secondary (up to 10th grade), secondary education, post-secondary diploma (associate's degree), and tertiary education (bachelor's or more).[47] In 2010, the education levels of the Jordanian workforce were distributed as follows: 15.7% had completed primary education or less, 41.1% had completed lower secondary, 21.1% had completed secondary education, 7.3% had completed a diploma, and 14.7% had completed some form of tertiary education. For a yearly distribution of educational attainment, please see Appendix Figure A17.

Table 2 presents the results from running these pair of regression specifications over different

[45] For example, ISCO Occupation Group 1 corresponds to "Managers," which typically require workers to have at least a bachelor's degree. Comparatively, ISCO Occupation Group 9 corresponds to "Elementary Occupations", which require no formal education.

[46] This definition further restricts the sample to those employed in both periods.

[47] In Jordan, education up to grade 10 is considered basic education, making it a clear level for demarcation. The final two years of secondary education are defined by specializing in a specific academic track and culminate in a national exam called *Tawjihi*, which one must complete to attend university. This is similar structure to A-Levels in the United Kingdom.

outcome variables. As evidenced by the coefficient in Columns (1), (3), (5), and (7), I find that the average treatment effect of an additional standard deviation of exposure does not increase the likelihood of being employed, unemployed, upgrading one's occupation, or downgrading one's occupation respectively.

Table 2: Effect of Exposure on Occupational Outcomes by Education

	Employed (All)	Employed (All)	Unemployed (Only LFPs)	Unemployed (Only LFPs)	Stayed in Occupation	Stayed in Occupation	Upgraded Occupation	Upgraded Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exposure	0.003	-0.003	0.009	0.014	0.008	0.008	0.001	-0.027
	[0.008]	[0.013]	[0.007]	[0.014]	[0.010]	[0.015]	[0.006]	[0.014]*
Exposure x Tertiary Educ		-0.056		0.017		-0.099		0.266
		[0.021]***		[0.037]		[0.038]**		[0.076]***
Tertiary Education		0.119		-0.048		0.054		0.074
		[0.033]***		[0.020]**		[0.050]		[0.059]
Exposure x Diploma		0.066		-0.021		0.046		0.096
		[0.042]		[0.020]		[0.046]		[0.044]**
Diploma		0.116		-0.051		-0.042		0.114
		[0.039]***		[0.023]**		[0.048]		[0.050]**
Exposure x Secondary Educ		0.005		-0.006		0.002		0.033
		[0.017]		[0.018]		[0.017]		[0.030]
Secondary Education		0.055		-0.041		-0.041		0.047
		[0.035]		[0.021]**		[0.042]		[0.040]
Exposure x Lower Secondary		0.026		-0.012		0.005		0.042
		[0.014]*		[0.013]		[0.016]		[0.015]***
Lower Secondary		0.033		-0.029		0.040		-0.062
		[0.030]		[0.019]		[0.035]		[0.032]*
Var Mean	.71	.71	.11	.11	.39	.39	.22	.22
Obs	2417	2417	1922	1922	2313	2313	1704	1704

Notes: Reported regressions regress different dummy variables for labor force outcomes in 2016 on a measure of exposure, constructed at the level of the ISCO 2-digit occupation code x governorate, that accounts for both the share of refugees in a given location and the share of refugees that have experience working in a given occupation. Columns (1) and (2) look at the probability of being employed in 2017, conditional on being employed in 2010. The outcome variable takes a 1 if the individual is still employed in 2017 and 0 if they have either become unemployed or exited the labor force entirely. Columns (3) and (4) next consider transition to unemployment where the variable takes the value 1 if the individual is unemployment in 2017 and restricts the sample to exclude those that exit the labor force. Column (5) and (6) look at whether an individual has upgraded occupations, which is defined as switching into an occupation that has a lower 1-digit ISCO code. Column (7) and (8) look at whether an individual has downgraded occupations, which is defined as switching into an occupation that has a higher 1-digit ISCO code. Regressions control for age and are restricted to working-aged men. "Primary education or less" is the excluded category in the set of education dummies included in the regression. Standard errors are clustered at the unit of exposure - the ISCO 2-digit occupation x governorate.

However, there are important differential effects by level of education in terms of occupational change as seen in Column (6) and Column (8) of Table 2. While additional education (relative to the omitted category of "primary education or less") increases the likelihood of upgrading in response to higher levels of exposure, the effects are much larger among those with a tertiary education. For those with bachelor's degrees or post-graduate degrees, an additional standard deviation of exposure increases the likelihood of upgrading one's occupation by 26.6 p.p. relative to an average rate of occupational upgrading of 22%. Those with higher levels of education are also less likely to downgrade occupations in response to higher levels of exposure. These effects are generally smaller in magnitude relative to the coefficient estimates for the effect of exposure on occupational upgrading by education level.

For the extensive margin outcomes, I am also able to run a standard Difference-in-Differences (DiD) specification with individual fixed effects where the outcome variable is a dummy for the extensive margin labor market outcome in each period. This alternative specification does not change the results, as seen in Appendix Table A7.

6 The Model

In the first half of the paper, I have provided empirical evidence of occupational reallocation by Syrian refugees once work permits were introduced, and subsequent re-sorting by Jordanians in response. However, without some additional structure, I cannot assess either aggregate effects of the policy or the distributional implications. By then estimating the corresponding model, I can quantify the reduction in misallocation caused by the work permit scheme and, in turn, the distribution of these gains across Syrians and refugees. I can further decompose the effects on Jordanians, to understand the relative importance of the *direct supply effect* of Syrians' entry vs. the *re-sorting effect* of Jordanians, in determining in what cases Jordanians can aggregately benefit from refugees' obtaining the right to work.

To do this, I utilize an adaptation of the model developed in Hsieh et al. (2019), which nests an occupational choice model in a general equilibrium framework. Such a model is ideal for my setting. The presence of frictions faced in the occupational choice of one group has the potential to not only distort the allocation of that group's labor but, through general equilibrium wages, the allocation of other groups. These group-specific occupation-level frictions in the model map clearly to the introduction of the work permit scheme, which reduced barriers overall and differentially across occupations.

This section outlines the defining features of my model and provides intuition, through the lens of the model, for how the introduction of work permits can reduce misallocation, improving aggregate labor productivity.

6.1 Individual's Occupational Choice

Human Capital

I begin with how the model assigns individuals occupation-specific productivities. Using the standard specification first outlined in McFadden (1974), individuals draw an iid occupation-specific ability draw $\epsilon_{i,o}$ from each occupation's Fréchet distribution with shape parameter θ:

$$F_o(\epsilon) = \exp\left(-\epsilon_{i,o}^{-\theta}\right) \tag{4}$$

Note that the ability draw does not depend on the individual's group – whether they are Syrian or Jordanian. This implies that, in the absence of any frictions or differences in preferences over occupations, Syrians and Jordanians are perfect substitutes. The Fréchet distribution is defined by a single shape parameter θ, which governs the dispersion of talent across occupations. A lower value of θ generates a fatter-tailed distribution, which results in an individual's productivity in one occupation being less predictive of their productivity in another.[48] In practical terms, this means that someone who is highly productive in one type of job (e.g., accounting) is not necessarily productive in another (e.g., carpentry), resulting in individuals choosing occupations based on their comparative advantage. For very high values of θ, being productive in one occupation corresponds to being productive in all occupations, leading to sorting based on absolute advantage.

An individual i in occupation o from group g accumulates human capital stock $h_{i,o,g}$ through the following production function:

$$h_{i,o,g} = s_i^{\phi_o} \tag{5}$$

Human capital is thus a function of an individual's educational attainment s_i (measured in years of schooling). The term ϕ_o denotes the elasticity of human capital with respect to time invested and is occupation-specific.[49] Finally, the human capital stock $h_{i,o,g}$ is scaled by one's ability draw $\epsilon_{i,o}$ to get their total skill for a given occupation.

Wages and Wedges

In the model, any frictions that Syrian refugees face relative to a Jordanian worker in that same occupation are captured in an occupation-specific wedge, denoted as $\tau_{o,\text{Syrian}}$. The wedge operates similar to a tax on the wage a Syrian refugee would have received given their individual productivity draw and human capital in absence of any frictions, which is equivalent to the wage a Jordanian with the same profile receives. In addition to refuges not having the legal right to work, this wedge also includes other labor market barriers faced by Syrian refugees such as discrimination,

[48] When productivity draws are independent across occupations and follow an extreme value distribution (such as Fréchet), a lower θ implies greater dispersion in draws and weaker correlation in productivities across occupations.

[49] ϕ_o can be interpreted as how useful a year of education is for generating skill in a given occupation. For example, attending higher-level education does not improve a worker's productivity in occupations where skills are learned primarily through on-the-job experience.

limited networks, higher search costs, etc. Since Syrians' wedges are measured in relative terms to Jordanians, I set $\tau_{o,\text{Jordanian}} = 0 \ \ \forall o$ without loss of generality. Thus $\tau_{o,\text{Syrian}}$ can be interpreted as the composite labor market frictions faced by Syrian refugees in each occupation relative to that of a local Jordanian worker.

An individual i in occupation o from group g will then be paid the following wage:

$$\text{wage}_{i,o,g} = (1 - \tau_{o,g}) w_o \cdot h_{i,o,g} \epsilon_{i,o}$$

Where w_o corresponds to the wage per efficiency unit of labor in that occupation, equivalent to the marginal product of an efficiency unit of labor. The individual's wage is then w_o scaled by the individual's stock of human capital units and, in the case of Syrian refugees, discounted by the occupation-specific wedge.

Homophilic Preferences

Individuals also hold preferences over occupations, which affect their choices. I define preferences $z_{o,g}$ at the occupation-group level, which is then comprised of an exogenous component, that is group-occupation specific but does not vary within group, and an endogenous component reflecting preferences for homophily. Let $N_g * p_{o,g}$ be the number of workers from one's own group in the occupation, where N_g is the total number of workers in each group g and $p_{o,g}$ is the share working in occupation o. Then the expression for preferences among a group for a given occupation is:

$$z_{o,g} = \alpha_{o,g} \cdot \left(\frac{N_g p_{o,g}}{\sum_g N_g p_{o,g}} \right)^{\lambda_g} \tag{6}$$

where $\alpha_{o,g}$ can be viewed as an exogenous amenity value of working in a given occupation (which I allow to vary by group to accommodate differential social norms around what jobs are acceptable).[50]

The second term captures group-specific preferences for homophily, which are based on the share of in-group workers relative to the total workers in an occupation (summed across both groups). For simplicity of notation, I define this compositional share $\delta_{o,g} = \left(\frac{N_g p_{o,g}}{\sum_g N_g p_{o,g}} \right)$. The relative importance of homophily compared to the amenity value of an occupation depends on the preference parameter λ_g. As λ_g approaches 0, individuals no longer care about the group-based composition of the workers in their occupation. Higher values of λ_g correspond to a greater taste for homophily, leading to more segregation of groups across occupations.

Homophily can arise from several mechanisms, and the model is agnostic about which one drives it: λ_g captures any force that makes an occupation more valuable to group g when more of its workers are from group g. The evidence in Sections 3.3 and 3.4 points to two channels.

The first is a preference over co-workers. In Becker (1957), prejudice is a disutility from in-

[50] The amenity term $\alpha_{o,g}$ captures unobservable variation in group's distributions across occupations that cannot be explained by the wedge or the distribution of individual's comparative advantages (as determined by θ). I allow to vary by group to accommodate differential social norms, which shape the valuation of amenities.

teracting with the out-group, which pushes workers away from occupations the out-group enters. Equivalently, affinity bias gives a utility bonus from working with one's own group; the two are observationally equivalent here, since only relative valuations matter. As previously shown, Jordanians report substantial discomfort with close contact with Syrians, including at work (Figure A3), and are willing to pay 3.7 to 4.6 percentage points of a monthly wage more than Syrians for a Jordanian rather than a Syrian contact, a gap that differences in contact value do not explain (Figure 3).

The second is referral-based hiring (Calvó-Armengol and Jackson, 2004; Patacchini and Zenou, 2012; Bolte et al., 2020; Jackson, 2021). Because workers find jobs through contacts and networks are largely homophilic, workers sort into occupations where members of their own group already work. Jordan's labor market relies heavily on such referrals: half of Jordanians found their current or most recent job through friends or family, almost always Jordanian ones (Section 3.3).

Importantly, λ_g can differ across groups. Models of network formation predict that minority groups form more out-group ties than the majority (Currarini et al., 2009), implying weaker homophily among refugees. My survey results substantiate this – 37% of the friends who helped Syrians find work were Jordanian, compared with fewer than 1% of those who helped Jordanians being Syrian. Additionally, Syrians are willing to pay more for a Jordanian over a Syrian contact, which suggests that self-segregation by refugees to avoid discrimination, while plausible, is not a dominant force at play in this setting.

Indirect Utility of a Choice

I define the indirect utility for choosing an occupation as a function depending on an individual's wage and their preferences. The indirect utility for individual i from group g in occupation o is:

$$U_{i,o,g} = z_{o,g}\left(1-\tau_{o,g}\right) w_o h_{i,o,g} \epsilon_{i,o} \tag{7}$$

Utility is determined by preferences over occupations, which vary by group, and the individual's realized wage, which is a function of the effective wage rate w_o, the wedge they face $\tau_{o,g}$, their human capital stock $h_{i,o,g}$, and their ability draw $\epsilon_{i,o}$. The term $z_{o,g}$ represents group-level preferences or amenity values over occupations. Individuals then choose occupation o^* that yields the highest indirect utility value.

I also include unemployment in the choice set of "occupations". Unemployment will capture the effect of work permits on entry of Syrian refugees as well as potential exit of Jordanians. To measure the role of reservation wages, the unemployment sector will have a common wage rate w_u, which is then scaled by human capital $h_{i,u}$ to account for more educated individuals having higher reservation wages.

Aggregation of Choices

The probability that individual i in group g chooses occupation o can be aggregated to compute the share of individuals i in group g in each occupation o, denoted as:

$$p_{o,g} = \frac{\tilde{w}_{o,g}^{\theta}}{\sum_j \tilde{w}_{j,g}^{\theta}} \tag{8}$$

where $\tilde{w}_{o,g} = z_{o,g}\,(1-\tau_{o,g})\,w_o \overline{h}_{o,g}$.[51] Equation 8 serves as the main sorting equation for workers over occupations. As is standard in occupational choice models, an individual's choice of occupation is determined by the value of the chosen occupation relative to the sum of the value of all occupations. Thus, the distribution of individuals across occupations is determined by the occupation's relative preferences and realized wages, as opposed to its absolute value.

Using the occupational shares equation where workers sort based on their ability draw $\epsilon_{i,o}$, the average quality of workers in an occupation is then:

$$\mathbb{E}[\epsilon_{i,o} h_{i,o,g} |\text{ choose } o] = \overline{s}_o^{\phi_o} \left[\left(\frac{1}{p_{o,g}} \right)^{1/\theta} \tilde{\Gamma} \right] \tag{9}$$

where $\tilde{\Gamma} = \Gamma(1-(1/\theta))$ and $\Gamma(\cdot)$ refers to the Gamma function and $\overline{s}_o$ can is the occupation-specific average schooling level. Equation 9 relates average quality of workers in group g in occupation o inversely to the number of workers from group g in said occupation, which can be understood as the selection effect. Intuitively, for either group, the most productive individuals select into an occupation first. Thus, an increase in $p_{o,g}$, the share of group g in occupation o, can only be achieved by drawing down the ability distribution, resulting in a decline in the average ability of the workers in that occupation.

Let $\overline{\text{wage}}_{\text{o,g}}$ represent the average earnings in occupation o for group g, which is a function of frictions $\tau_{o,g}$, the efficiency wage w_o, and the average quality of workers in that occupation.

$$\begin{aligned} \overline{\text{wage}}_{o,g} &= (1-\tau_{o,g}) w_o \cdot \mathbb{E}[\epsilon_{i,o} h_{i,o,g} |\text{ choose } o] \\ &= (1-\tau_{o,g})\, w_o \overline{s}_o^{\phi_o} \left(\frac{1}{p_{o,g}} \right)^{1/\theta} \tilde{\Gamma} \end{aligned} \tag{10}$$

6.2 Production and General Equilibrium

The Firm's Problem

To close the model, I follow Hsieh et al. (2019). A representative perfectly competitive firm uses labor from each occupation to produce an aggregate good Y, under a CES production function:

$$Y = \left[\sum_o (A_o H_o)^{\rho} \right]^{\frac{1}{\rho}} \quad \forall o \neq u \tag{11}$$

where H_o is the total efficiency units from labor in occupation o and A_o is an exogenous occupation-

[51] This result stems from the properties of the Fréchet distribution. See Appendix E.1 for a complete derivation. Note, $\overline{h}_{o,g} = \overline{s}_o^{\phi_o}$ where $\overline{s}_o$ is the average schooling of workers in occupation o.

specific productivity parameter.

Note, the production function excludes unemployment, as unemployed individuals do not generate any output. The inclusion of unemployment is a permutation of including home production as an occupation (often to account for female selection into the labor force), which is a typical extension of the Roy Model for occupational choice. While the "wages" of both home production and unemployment are unobserved in the data, I deviate from the standard assumption that this sector contributes to the overall production function in the economy. Given my setting, where unemployment is characterized by idle labor, excluding unemployment from the firm's labor demand and thereby the production function is a more natural assumption.[52]

Importantly, the substitution parameter, ρ, governs how easily the firm can substitute labor across occupations.[53] Intuitively, when substitution across occupations is easier, the negative effects of labor being misallocated across occupations on aggregate production will be lessened.

Normalizing the price of good Y to 1, the firm's maximization problem can be written as:

$$\max_{\{H_o\}} \left(\sum_{o=1} (A_o H_o)^\rho \right)^{\frac{1}{\rho}} - \sum_{o=1} w_o H_o \quad \forall o \neq u$$

The first-order condition with respect to H_o implies the wage:

$$w_o = A_o^\rho \cdot \left(\sum_{k=1}^{n} (A_k H_k)^\rho \right)^{\frac{1-\rho}{\rho}} \cdot H_o^{-(1-\rho)} \tag{12}$$

Definition of Equilibrium

The set of components in the economy needed to characterize equilibrium are: an individual's occupational choice o^*, the total efficiency units of labor in each occupation H_o, final output Y, and efficiency wage w_o. A competitive equilibrium in this economy is therefore defined as:

I. Given education s_i and idiosyncratic ability draw $\epsilon_{i,o}$, each individual chooses occupation o^* that maximizes their utility, taking $\{\tau_{o,g}, w_o, \epsilon_{i,o}, z_o\}$ as given

$$o^* = \text{argmax} \max_o U\left(\tau_{o,g}, w_o, \epsilon_{i,o}, z_o\right)$$

II. A representative firm hires H_o^* in each occupation to maximize profits

$$\max_{\{H_o\}} \left(\sum_{o=1}^{n} (A_o H_o)^\rho \right)^{\frac{1}{\rho}} - \sum_{o=1}^{m} w_o H_o \quad \forall o \neq c$$

[52] This assumption does not affect the process of estimating the model but does impact the counterfactuals, which require an additional assumption that the reservation wage in unemployment does not change relative to the equilibrium used as the starting point. These points will discussed more in Section 7 and Section 8.

[53] The elasticity of substitution is then $\sigma = \frac{1}{1-\rho}$. For $\rho > 0$ (i.e. $\sigma > 1$), occupations are substitutes in the production function. In the limit, as $\rho \leftarrow 1$, occupations become perfect substitutes.

III. The occupational wage w_o^* clears the labor market for each occupation

$$H_o = \sum_{g=1}^{G} q_g p_{o,g} \cdot \mathbb{E}[h_{o,g}\epsilon_{i,o} \mid \text{choose o}] \quad \forall o \neq u$$

IV. Total output Y is given by the firm's production function

$$Y = \left[\sum_{o=1}^{m} (A_o H_o)^{\rho}\right]^{\frac{1}{\rho}}$$

The general equilibrium solution of the model is then the set of vectors $\left\{p_{o,g}, H_o^S, H_o^D, w_o, Y\right\}$ such that the following hold:

I. The probability of an individual in group g working in occupation o, $p_{o,g}$, satisfies

$$p_{o,g} = \frac{\widetilde{w}_{o,g}^{\theta}}{\sum_{j=1}^{m} \tilde{w}_{j,g}^{\theta}} \quad \forall o \tag{13}$$

II. The supply of efficiency units of labor for occupation o, H_o^S, aggregates individual occupational choices such that

$$\begin{aligned} H_o^s &= \sum_{g=1}^{G} p_{o,g} \cdot \mathbb{E}[h_{o,g} \mid \text{choose o}] = \sum_{g=1}^{G} p_{o,g} \left[s^{\phi_o} \left(\frac{1}{p_{o,g}}\right)^{\frac{1}{\theta}} \tilde{\Gamma} \right] \\ &= s^{\phi_o} \tilde{\Gamma} \left(\sum_{g=1}^{G} \left[\frac{\tilde{w}_{o,g}^{\theta}}{\sum_j \tilde{w}_{j,g}^{\theta}} \right]^{\frac{\theta-1}{\theta}} \right) \end{aligned}$$

III. Demand for efficiency units of labor for occupation o, H_o^D, satisfies the firm's profit maximization problem such that

$$H_o^D = \left[\frac{A_o^{\rho}}{w_o}\right]^{\frac{1}{1-\rho}} \cdot Y \quad \forall o \neq u$$

IV. Pre-distortion efficiency wage, w_o, clears each occupational labor market such that

$$H_o^S = H_o^D \quad \forall o \neq u$$

V. Total output, Y, is equal to total wages plus "revenue" from wedges $\tau_{o,g}$

$$\left[\sum_{o=1}^{m} (A_o H_o)^{\rho}\right]^{\frac{1}{\rho}} = Y = \sum_{o,g} \left(\text{Pop.}_g \cdot p_{o,g}\right) \cdot \left(\overline{wage}_{o,g} + \tau_{o,g}\right)$$

By Walras' Law, if all occupational labor markets clear through matching the firm's labor demanded and the workers' labor supplied except for in unemployment then the market for them will also clear. The general equilibrium will be solved numerically.

6.3 Model Intuition for the Effect of Work Permits

To illustrate how the introduction of work permits impacts the allocation of labor, consider a simplified version of the model with only two occupations. To start, assume groups possess the same preferences over occupations and do not exhibit homophilic preferences. Individuals will then choose the occupation that yields the highest utility, $U_{i,o} = w_o \epsilon_{i,o}$ based on their own draws for $\epsilon_{i,o}$. Finally, I normalize the exogenous productivity parameters A_o in each occupation such that, in the absence of frictions, workers are evenly divided between occupations and $w_1^* = w_2^*$.[54]

Equilibrium Without Work Permits

Figure 10 shows the distribution of $\epsilon_{i,o}$, the ability draws of Syrian refugees and Jordanians, across Occupations 1 and 2.[55] Line A, a 45-degree line, represents the ratio of wages in the allocatively efficient equilibrium, where there are no wedges. The efficient allocation of labor is the one where all workers, Syrian and Jordanian, above Line A, work in Occupation 2, where they have a comparative advantage and thus will be paid more.[56] By symmetry, those below A sort into Occupation 1.

Without work permits, assume Syrian refugees face substantial frictions only in Occupation 2, denoted as the wedge τ.[57] Line B then represents the wage ratio experienced by Syrian refugees pre-work permits. The most talented Syrian refugees in Occupation 2, those in region I, still work in Occupation 2, as they did in the case without frictions. But Syrian refugees who are still comparatively better suited for Occupation 2, those in region II, now sort to Occupation 1, where they receive higher realized wages due to the friction-induced penalty on wages in Occupation 2. In terms of maximizing productivity, Syrian refugees in region II are thus misallocated. Syrian refugees in regions III and IV remain in Occupation 1.

In general equilibrium, the concentration of Syrian refugees in Occupation 1 causes w_1 to fall relative to w_2, as labor is now overly supplied to Occupation 1 relative to Occupation 2. [58] Line C represents the resulting wage ratio faced by Jordanians in the pre-work permit labor market. The most talented Jordanians in Occupation 1, those in region IV, stay in Occupation 1. However, the relative increase in w_2 causes Jordanians in region III to switch to Occupation 2. These Jordanians

[54] It follows a 45-degree line as the sectors are assumed to be equally productive and w_o is equal to the marginal productivity of labor.

[55] For the purposes of building intuition, assume all individuals have the same amount of human capital up to the idiosyncratic ability draw. Moreover, Syrian refugees and Jordanians draw from the same ability distribution.

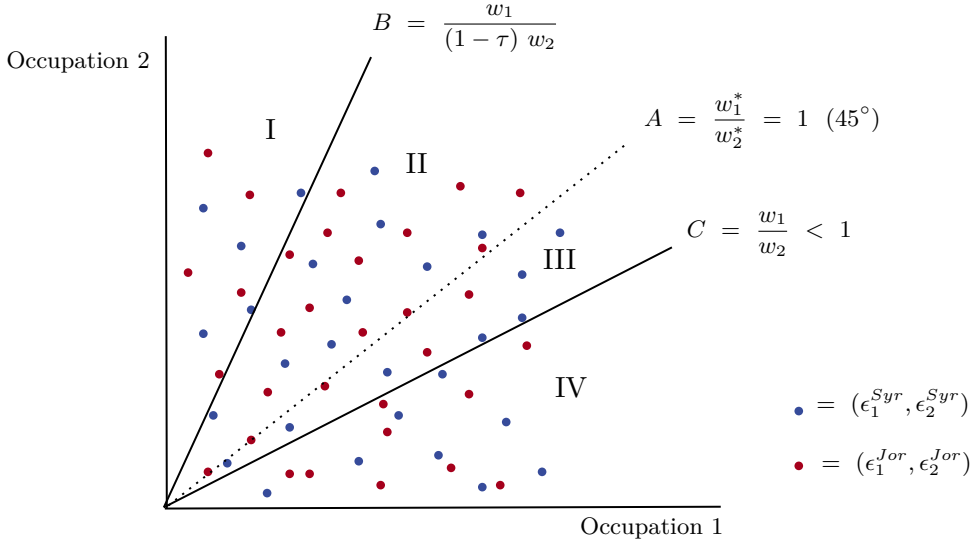
[56] This is because $\epsilon_{i,1} w_1 > \epsilon_{i,2} w_2 \iff w_1/w_2 > \epsilon_{i,2}/\epsilon_{i,1}$.

[57] Having a wedge in only one occupation is isomorphic to taking the relative wedge between both sectors in the case that both have frictions.

[58] Technically, under CES with constant-returns to scale, a supply shock to Occupation 1 will result in an absolute fall in Occupation 1's wage per efficiency unit (w_1) and Occupation 2 will experience a slight increase. For a proof of each result, see Appendix E.4.

are those that have a comparative advantage in Occupation 1 in the absence of frictions and thus are now misallocated. Jordanians in regions I and II remain in Occupation 2.

Figure 10: Allocation of Labor Pre-Work Permits



Thus, relative to the case of allocative efficiency, a labor market without work permits creates misallocation of Syrian refugee workers and, importantly, also creates misallocation among Jordanian workers. In the aggregate, this lowers the average skill of workers in both occupations, as Syrians in region II, formerly in Occupation 2, have been replaced with Jordanians in region III and vice versa.

$$\mathbb{E}[\epsilon_{\text{Syrian},2} \mid II] > \mathbb{E}[\epsilon_{\text{Jordanian},2} \mid III]$$
$$\mathbb{E}[\epsilon_{\text{Syrian},1} \mid II] < \mathbb{E}[\epsilon_{\text{Jordanian},1} \mid III]$$

Equilibrium With Work Permits

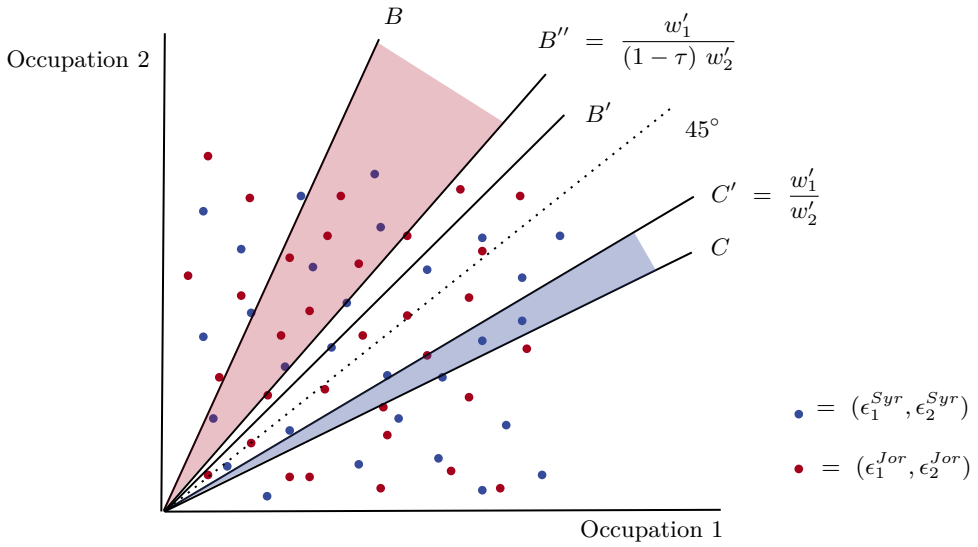
Now consider the introduction of work permits, which improves Syrian refugees' access to Occupation 2, modeled as a decrease in τ. In Figure 11, I visually decompose the impact of work permits on the misallocation of workers into the *direct supply effects* of Syrian entry into Occupation 2 and the *re-sorting effects* of Jordanians exiting.

Beginning with the Syrian refugees, when τ falls to τ', the effective wage Syrians receive in Occupation 2 increases, represented by the shift in the line of indifference between occupations from Line B down to Line B' (with slope $w_1/(1-\tau')w_2$). In turn, Syrian refugees between Lines B and B' will move into Occupation 2, reducing the misallocation of refugee skills. In general equilibrium, wages will adjust in response to the supply shock to Occupation 2, causing w_2 to fall relative to w_1, which I refer to as the *direct supply effect* of Syrian entry. For completeness, this is denoted as the Line B'', which reflect the true line of indifference that Syrian refugees will face in

general equilibrium.[59]

Comparatively, Jordanians will re-sort in response to the relative increase of w_1'/w_2', as reflected by their new line of indifference C'. Specifically, Jordanians between Lines C and C' will exit Occupation 2. This re-sorting will also have wage effects (i) compositionally from selection (those left in Occupation 2 will be relatively higher quality and increasing the occupation's average wage) and (ii) from an increase in w_2' from the reduction of labor supplied to Occupation 2. I refer to this as the *re-sorting effect*.

Figure 11: Allocation of Labor Post-Work Permits



From this simple example, the overall effect on allocative efficiency from the work permit scheme becomes clear. The two shaded regions represent the reduction in each group's misallocation. Prior to the work permit scheme, refugees between Line B and the 45-degree line were misallocated. After work permits, Syrians in the red shaded region between B and B'' have switched from Occupation 1 to 2 and constitute a reduction in misallocation. For Jordanians, those between the 45° line and Line C were misallocated. After work permits, those in the blue shaded region between C and C'' have switched from Occupation 2 to 1 and similarly constitute a reduction in misallocation.

Homophily as an Amplifier of Sorting

To understand the role homophily plays in the allocation of labor, let κ be the premium on wages accrued on the occupation that has the higher share of workers from the in-group.[60] For the purposes of this example, I will assume that only Jordanians have strong preferences for homophily.[61] As before, prior to the introduction of work permits, Syrians are overly concentrated in Occupation 1, due to high frictions in Occupation 2. With homophilic preferences, Jordanians will

[59] Misallocation persists as long as $\tau' > 0$, such that lines B'' and A do not have the same slope.

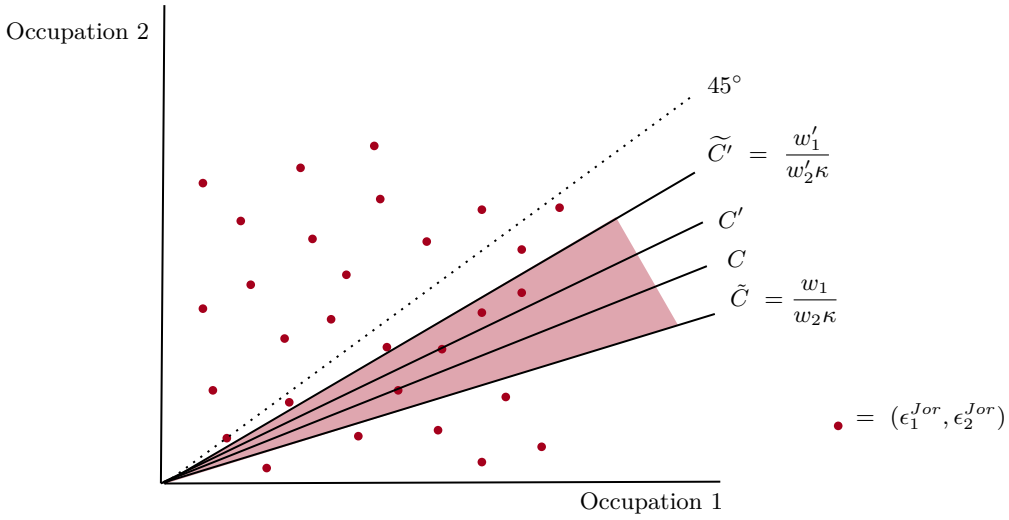
[60] The line of allocative efficiency does not change under homophilic preferences as workers are evenly divided across occupations so neither occupation is preferrable on the basis of differences in the composition of the workers.

[61] If refugees have negligible preferences for homophily ($\lambda \approx 0$), which is consistent with the literature on minority groups, their sorting will be driven by changes in frictions and wages.

face an additional premium on the wages earned in Occupation 2, from working mostly with other Jordanians. This is represented in Figure 12, where the indifference line for Jordanians decreases from Line C (no homophily) to $\tilde{C}$ (homophily).

After work permits are introduced, Syrians will again respond to the reduction in frictions in Occupation 2 by entering. Jordanians will now re-sort in response to both the change in relative wages (as represented by Line C') as well as their preference for homophily. Since work permits increase the share of Syrians working in Occupation 2, Jordanians will derive less utility from working in Occupation 2, represented as a decrease in the value of κ to $kappa'$. As before Jordanians will exit Occupation 2, but the magnitude of the response will be much larger. Those Jordanians in the shaded red region between $\tilde{C}$ and $\tilde{C}'$ will re-sort to Occupation 1, as opposed to the base case where only those between C and C' switched. For a formal proof that homophily amplifies re-sorting relative to a case of wage-based sorting alone, see Appendix E.3.

Figure 12: Allocation of Jordanian Labor Under Homophilic Preferences



6.4 Mapping Model to Event Study - Comparative Statics for Wage Gaps

The model delivers a constant wage gap between groups (across all occupations) up to a relative difference in occupation-level preferences. From Equation 10, the expression for the occupation-specific wage gap between Jordanians and Syrian refugees becomes:

$$\frac{\overline{wage}_{o,\text{Jordanian}}}{\overline{wage}_{o,\text{Syrian}}} = \frac{z_{o,\text{Syrian}}}{z_{o,\text{Jordanian}}} \cdot \left(\frac{\sum_j \tilde{w}^{\theta}_{j,\text{Jordanian}}}{\sum_j \tilde{w}^{\theta}_{j,\text{Syrian}}} \right)^{1/\theta}$$

If the relative value of all alternatives ($\sum_j \tilde{w}^{\theta}_{j,g}$) changes, such as reducing the wedges faced by Syrians as a result of the work permit scheme, this will also affect the relative wage gap across occupations.

The model pins down a relationship between occupation-average wages within a group. Without

homophilic preferences, a change in w_o in one occupation (or more) is resolved via re-sorting, resulting in the equilibrium ratio of average wages between any two occupations invariant to changes in w_o. Substituting the definition for shares (Eq. 8) into the average wage equation (Eq. 10):

$$\overline{wage}_{o,g} = z_{o,g}^{-1}\tilde{\Gamma}\left(\sum_k \tilde{w}_{k,g}^{\theta}\right)^{1/\theta}$$

The average wage only depends on preferences, a scalar, and the total value of all occupations, which is a result of the Roy Model's multinominal discrete choice structure. Finally, taking the ratio between two occupations yields:

$$\frac{\overline{wage}_{o,g}}{\overline{wage}_{j,g}} = \frac{z_{j,g}}{z_{o,g}} = \frac{\alpha_{o,g}\cdot\delta_{o,g}^{\lambda_g}}{\alpha_{j,g}\cdot\delta_{j,g}^{\lambda_g}}$$

Thus, without homophilic preferences ($\lambda_g \neq 0$), the average wage gap would be constant between occupations, which contradicts the empirical fact outlined in Figure 9 – average wages in high exposure occupations increase relative to low exposure occupations, following the introduction of work permits. In practical terms, homophilic preferences induce more sorting in response to refugee entry than would occur if sorting was only based on wage changes alone.

7 Identification and Estimation

In this section, I discuss the identification and estimation procedures used to estimate the exogenous parameters of the model.

7.1 Identification

The model has the following parameters: $\{\theta, \rho, w_o, \alpha_{o,g}, \lambda_g, \tau_{o,g}, A_o\}$. For identification, I add the normalization of $\phi_o = 1\ \forall o$, so that education has constant returns to scale across occupations. Finally, it is not necessary to pin down the substitution parameter ρ nor the exogeneous occupation-level productivity term A_o. Although, the estimates of these parameters will matter when implementing counterfactual analyses and will be discussed then.

Identifying Frechet shape parameter: $\{\theta\}$

As previously discussed in the model section, I follow the McFadden (1974) assumption that idiosyncratic ability ϵ is drawn from a single-parameter Frechet distribution, governed by shape parameter θ. This assumption is used to make the model tractable. To empirically estimate this parameter, I exploit the following relationship between an individual's observed wage and their ability draw.

$$wage_{i,o,g} = \tilde{w}_{i,o,g}\epsilon_i \quad \text{where} \quad \epsilon_i \sim F_i(\epsilon) = \exp\left(-\epsilon_i^{-\theta}\right)$$

Taking the log of both sides and plugging in for $\tilde{w}_{i,o,g}$ delivers:

$$ln(wage_{i,o,g}) = ln\left(z_{o,g}\right) + ln\left(1-\tau_{o,g}\right) + ln\left(w_o\right) + \phi_o ln\left(s_i\right) + ln(\epsilon_i) \tag{14}$$

This equation can then by estimated by the regression of log wages on a set of occupation, occupation-group, and group dummy variables (which approximate the respective terms of $ln(\tilde{w}_{i,o,g})$). The resulting residuals can then be exponentiated to back out the distribution of ϵ_i, which should theoretically follow a Frechet distribution.

To then back out the Frechet shape parameter θ, I numerically estimate the following equation using the variance and mean of the observed distribution of the exponentiated ϵ_i in the data:[62]

$$\frac{\text{Variance}}{\text{Mean}^2} = \frac{\Gamma\left(1-\frac{2}{\theta}\right)}{\left(\Gamma\left(1-\frac{1}{\theta}\right)\right)^2} - 1 \tag{15}$$

Identifying parameters governing wages: $\{w_o, \tau_{o,g}\}$

Taking the log of Equation 10, I get the resulting log-linear equation for occupation-average wages:

$$ln\left(\overline{\text{wage}}_{o,g}\right) = \underbrace{ln\left(\tilde{\Gamma}\right) + ln\left(w_o\right)}_{\text{occupation fixed effect}} + \underbrace{ln\left(1-\tau_{o,g}\right)}_{\text{occ x group fixed effect}} + \phi_o\, ln\left(\bar{s}_o\right) - \frac{1}{\theta}\, ln\left(p_{o,g}\right) \tag{16}$$

Having separately identified θ, observing occupation average wages, shares by group, and average education, Equation 16 identifies the relative values of $\tau_{o,g}$ between groups. To use equation 16 to identify $\tau_{o,g}$ in levels, I require the additional assumption that $\tau_{o,\text{Jordanian}} = 0 \;\; \forall\; o$. As previously described in the introduction of the model, $\tau_{o,\text{Syrian}}$ captures the pecuniary wedge Syrians face in a given occupation relative to Jordanians. Intuitively, this is the difference in wages received by Syrians and Jordanians in the same occupation, after controlling for differences in human capital, $\bar{s}_o$, and its usefulness in a given occupation, ϕ_o.

Finally, I can use equation 16 to identify the wage per efficiency unit w_o in levels by parsing it from the occupation fixed effect using the previously estimated θ to compute $ln(\tilde{\Gamma})$.[63] Intuitively, after controlling for the role of selection through the inclusion of $p_{o,g}$, wedges faced by Syrian refugees through $\tau_{o,g}$, and human capital, the only difference in wages must be due to exogeneous productivity differences across occupations.

For interpretability, I assume the corresponding wedge for unemployment $\tau_{u,g} = 0 \;\; \forall g$. Barriers to entering employment for refugees can be measured by the average occupation-specific friction across the other occupations. Thus, when work permits change other occupation-specific wedges, the relative utility of unemployment will shift, inducing churn in and out of unemployment.

[62] This follows a similar procedure implemented in Hsieh et al. (2019) and Bryan and Morten (2019) to recover the value of the wage dispersion parameter.

[63] Recall that $\tilde{\Gamma} = \Gamma\left(1-\frac{1}{\theta}\right)$.

Identifying occupation-specific group-level preference terms: $\{\alpha_{o,g}, \lambda_g\}$

To estimate the occupation-specific preferences of each group, I rely on the log-linearized sorting equation (Equation 8):

$$ln\left(p_{o,g}\right) = \theta\ ln(w_o) + \theta\Big(\phi_o\ ln(\overline{s}_o)\Big) + \theta\ ln(1-\tau_{o,g}) + \theta\ \underbrace{ln\left(\alpha_{o,g}\cdot\delta_{o,g}^{\lambda_g}\right)}_{z_{o,g}} - \underbrace{ln\left(\sum_j \tilde{w}_{j,g}^{\theta}\right)}_{\text{group fixed effect}} \tag{17}$$

Having already estimated $\{\theta, w_o, \phi_o, \tau_{o,g}\}$, the above equation identifies the collective preference term $z_{o,g}$. To then separately identify the amenity and homophily components of preferences, I use the relative wage gap between Syrians and Jordanians for each occupation:

$$\begin{aligned} ln\left(\overline{wage}_{o,\text{Jor}}\right) - ln\left(\overline{wage}_{o,\text{Syr}}\right) = &\left[ln\left(\alpha_{o,\text{Jor}}\right) + \lambda_{\text{Jor}}\cdot ln\left(\delta_{o,\text{Jor}}\right) - \left(\sum_j \tilde{w}_{j,\text{Jor}}^{\theta}\right)\right] \\ &- \left[ln\left(\alpha_{o,\text{Syr}}\right) + \lambda_{\text{Syr}}\cdot ln\left(\delta_{o,\text{Syr}}\right) - \left(\sum_j \tilde{w}_{j,\text{Syr}}^{\theta}\right)\right] \end{aligned} \tag{18}$$

I directly observe the share of Jordanians (Syrians) relative to the total workers in an occupation, which will determine $\delta_{o,g}$. Together with Equation 17, I can then separately identify $\alpha_{o,g}$ from λ_g.

Summary of identifying assumptions

I have to make several identifying assumption to estimate the model. Table 3 reviews each.

Table 3: Identifying Assumptions

Parameter	Definition	Value
$\tau_{o,\text{Jor.}}$	Labor market frictions for Jordanians	0
$\tau_{u,\text{Syr.}}$	Unemployment frictions for Syrians	0
ϕ_o	Returns to schooling by occupation	1

7.2 Estimation

I follow a static general equilibrium approach similar to Hsieh et al. (2019), estimating a separate equilibrium for each year of data. This allows me to recover the evolution of labor market distortions faced by Syrian refugees and occupational misallocation following the introduction of work permits.[64] While an alternative would be to use the yearly data to estimate a transition path model with exact hat algebra, such an approach would require additional structural assumptions about expectations and adjustment speeds (Dekle et al., 2008).[65] In my setting, work permits

[64]The resulting assumption is that each year corresponds to a unique static equilibrium in the model and that differences across years are driven by changes in fundamentals (such as the wedges, productivities, or wages) rather than transitional dynamics.

[65]These structural assumptions would be necessary to parse what changes are driven by changes in the model's fundamentals from the transition dynamics, adding an additional margin for error without clear returns.

represent a discrete policy change with marginal adjustments over time, my estimation approach allows for a more flexible identification of time-varying frictions and more transparent evaluation of the policy's effects.

To estimate each year's static equilibrium, I first externally estimate θ separately for each year, using the procedure described in its identification (see Section 7.1). As is consistent in this type of model estimation, I then take the average value for each year to get a more robust estimate that has standard errors and is less sensitive to idiosyncrasies across years. The use of a common θ parameter in the main estimation of the model, assumes that there is a constant distribution of ability draws over my time period, which implies that individuals only draw their occupation-specific abilities once and keep them over time. This assumption is reasonable from a modelling standpoint. Individuals are unlikely to experience frequent, random shifts in their comparative advantages across occupations over time. Additionally, there is minimal concern about compositional change in the labor force over the eight-year period covered by my data, which further supports the stability of the underlying distribution. Estimating a single θ improves statistical precision and facilitates counterfactual analysis by allowing a fixed population of simulated individuals to make occupational choices in each year, responding only to changing wedges and wages.[66]

I then implement an over-identified efficient Generalized Method of Moments (GMM) estimator that recovers the remaining structural parameters of the model by minimizing the distance between model-implied and observed moments. These moments are constructed from the model's equations that determine each group's occupation shares and occupation-average wages (Equations 16, 17, and 18) I use bootstrapping twice in the estimation procedure: first, to construct the weighting matrix for efficient GMM and, second, to estimate standard errors and confidence intervals for the final parameter estimates.[67]

For the purposes of estimation, I define an occupation o to be a one-digit ISCO code plus unemployment. The aggregation from the two-digit to one-digit serves two purposes: to avoid occupation cells with zero observations and to constrain the dimensionality of the parameter space. Given the small share of Syrian refugee workers in ISCO groups 1 (managers) and 2 (professionals) throughout the period, I combine the two into a single occupation group for the purposes of estimation, resulting in a final set of nine occupations (including unemployment). This addresses the concern that small sample sizes in a given cell might produce imprecise estimated wages.[68]

[66] However, I will also estimate a single θ through the alternative strategy of a pooled regression with year fixed effects, as a measure of robustness.

[67] The complete estimation procedure is as follows. First, I estimate an initial stage of GMM on the original data of a given year to recover an initial estimate of the parameters. Looping over each bootstrap sample, I then recompute the distance between each model moment (as estimated in the first stage) and the observed moment in the bootstrapped sample. I then use the inverse diagonalized covariance matrix of these deviations to create the efficient weighting matrix. Next, I implement efficient GMM using this weighting matrix on the observed moments in the original data to get my final parameter estimates. Finally, I re-estimate efficient GMM on each bootstrapped sample to construct standard errors and confidence intervals for each parameter estimate.

[68] Under this definition, I am still left to estimate 36 parameters with GMM each period.

7.3 Estimation Results

I now present the parameter estimates from the implementation of the estimation procedure described above. The goal of this section is to understand how work permits affected the model's fundamental parameters – namely the wedges faced by Syrian refugees and the occupation-level wages per efficiency unit. As I can only estimate the model for periods that I have data for Syrian refugees and Jordanians, I present results for the following years: 2014 (pre-work permits) and 2017-2022 (post-work permits).

Fréchet Parameter (θ)**:** I estimate θ separately for each year of my data and arrive at an average value of $\theta = 3.46$ with a standard error of 0.798. This serves as my baseline specification for the shape parameter, governing the dispersion of ability draws. Alternatively, estimating a common value across periods, I arrive at a value of $\theta = 3.19$, which is not statistically different from my preferred specification. Moreover, my estimate is largely consistent with the broader literature. Hsieh et al. (2019) report a baseline value of $\theta = 3.44$, which they derive from re-estimating θ for each decade of U.S. census wage data from 1960 onward.

Occupation Wedges (τ_o)**:** I estimate substantial wedges for Syrian refugees prior to work permits with an average $\tau_o = 0.53$ (with a standard deviation of 0.024). This is equivalent to Syrian refugees being paid an average 47% of their marginal product of labor prior to gaining the right to work through permits. By the end of the study period, wedges had fallen 29 p.p on average, corresponding to a 61% increase in refugees capture of their marginal product.

Figure 13 presents the resulting estimates for the wedges or frictions faced by Syrian refugees in each occupation (excluding unemployment) over time, split by occupation-level informality. Panel A plots the frictions for the four occupations that are less informal (below average in the z-scored informality index) while Panel B plots the frictions for the remaining four more informal occupations. After work permits are introduced, Syrians experience non-linearly decreasing frictions across all occupations.

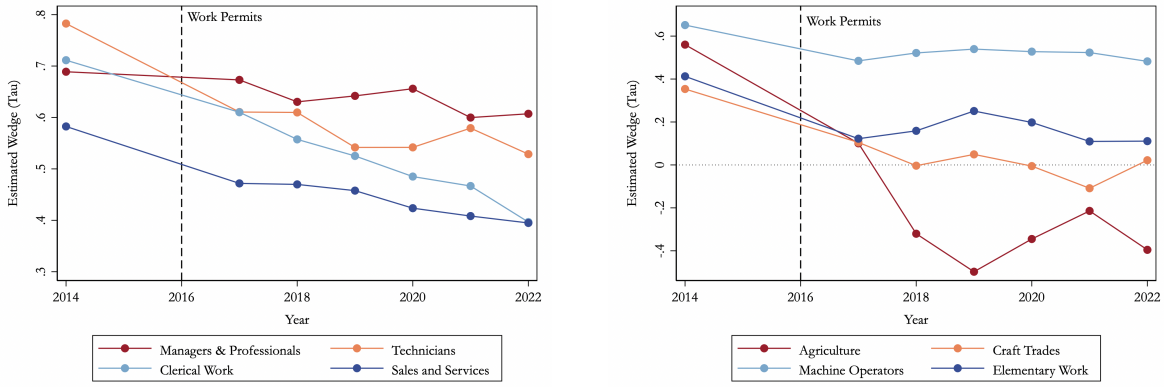
I also find a pronounced effect of the Jordanians government's introduction of "flexible" work permits in the occupations of agriculture and construction. This type of permit was designed to promote Syrian refugees' entry into targeted occupations. Agriculture appears directly in the set of estimated occupations, but construction is grouped within the broader ISCO 1-digit category of "Craft Trades." Consequently, the estimated effect of flexible permits on construction is attenuated by the inclusion of other occupations within the same category, which are not directly affected by the policy. Despite this, the policy was sufficiently effective that the estimated wedges become negative for agriculture and approach zero in craft trades – effectively creating subsidies for these occupations.[69] This result highlights the potential efficacy of refugee integration schemes that target specific occupations.

As expected, the initial occupation wedges faced by Syrian refugees are negatively correlated

[69] While this has a positive effect on Syrian refugee wages, it creates a new margin of misallocation by over-incentivizing Syrians' entry into the occupations of agriculture and craft trades.

with an occupation's informality, such that refugees could more easily enter informal jobs before work permits. Appendix Table A8 lists each occupation's z-scored informality index (measured among Jordanians in 2014, Section 4) alongside the estimated initial wedge, τ. Contrary to expectations, the decline in frictions after the reform is not systematically larger for formal occupations. Reductions in τ are broadly similar across the formality spectrum, except for agriculture where the government deliberately promoted refugee entry. This likely reflects persistent barriers that limit the permits' effectiveness, such as occupations closed to non-nationals, lack of credential recognition, and employer discrimination in formal sectors.[70] Differences in the education of refugees also contributes to the persistence of high wedges: relative to Jordanians, fewer Syrians can access high-skill occupations that require tertiary degrees, such as "Managers" and "Professionals."

Figure 13: Evolution of Wedges (τ_o) After Work Permits



(a) Below Average Informality

(b) Above Average Informality

Note: Each panel plots the model estimated occupation-specific wedges in each year. Panel A plots wedges for occupations with below average informality, as defined by the informality index for the occupation. Panel B plots wedges for occupations above average.

Homophilic Preference Parameters (λ_g)**:** I find that Jordanians hold much stronger preferences for homophily than Syrians, nearly twelve times stronger on average. Syrians have an estimated mean value of $\lambda = 0.13$, with a mean standard error of 0.0074. Comparatively, Jordanians have a mean value of $\lambda = 1.64$, with a corresponding mean standard error of 0.0074. Beyond level differences, there are important changes in group-level homophily preferences over time. Syrians' preference for homophilly increases over time, nearly doubling from their pre-work permit levels (2014) to post work-permits (2017-2022). Jordanians see a more continuous weakening of preferences over time, decreasing by 45% from 2014 until 2022.

[70] See Appendix G for a list of closed occupations and corresponding ISCO 1-digit codes.

Table 4: Homophilic Preference Parameter Estimates (λ)

Parameter	2014	2017	2018	2019	2020	2021	2022
λ - Syrians	0.079	0.153	0.128	0.130	0.130	0.143	0.149
	(0.006)	(0.007)	(0.005)	(0.013)	(0.005)	(0.008)	(0.008)
λ - Jordanians	2.253	2.117	1.709	1.094	1.239	1.803	1.235
	(0.006)	(0.007)	(0.005)	(0.013)	(0.005)	(0.008)	(0.008)

Notes: The top row corresponds to estimated value of the preference parameter for homophily among Syrian refugees and the second row for Jordanians. Columns correspond to the year of data being estimated. Each cell reports the main GMM estimate and the bootstrapped standard error is below in parentheses.

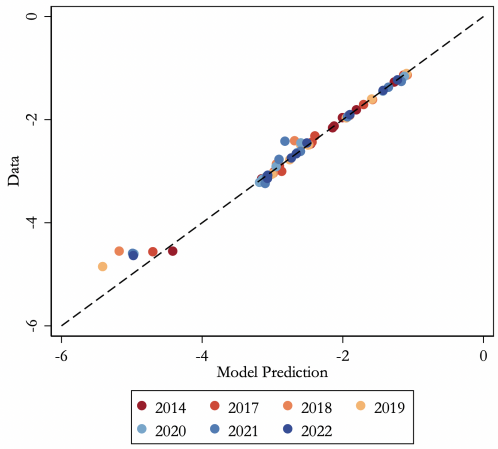
Intuitively, once Syrians have the right to work, frictions (τ_o) are less binding and Syrians are better able to sort along preferences for homophily. The trend for Jordanians is more consistent with a story of refugee integration – as Syrians integrate into Jordanians society over time, Jordanians' discriminatory sentiments weaken, reducing their preference for homophily. It should be noted that, even accounting for this trend, I find that homophily still plays a large role in Jordanians' sorting decisions.

Remaining Parameter Estimates: Appendix I.1 reports the full set of estimates for parameters $\{\tau_{o,g}, \alpha_{o,g}, \lambda_g\}$ and equilibrium object $\{w_o\}$ with boot-strapped standard errors for each year the model is estimated. From the demand-side equations, I am able to back out $\{A_o\}$ for each year as well but cannot report standard errors.

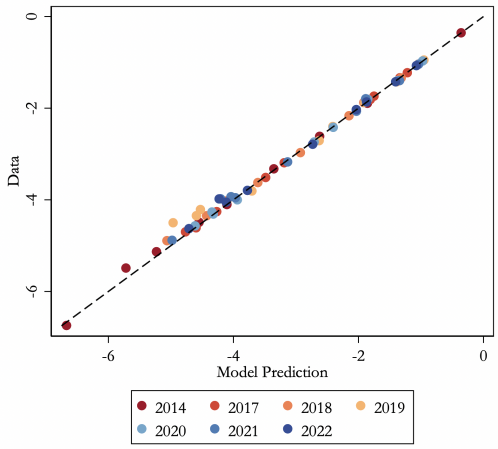
7.4 Model Fit

I present an overview of the model's fit to the moments targeted in the data. Figure 14 plots the model's predicted value for the log occupation share and log occupation average wage respectively against their observed value in the data for each year the model is estimated. For a complete assessment of model fit relative to targeted moments, see Appendix I.1 for a full set of results.

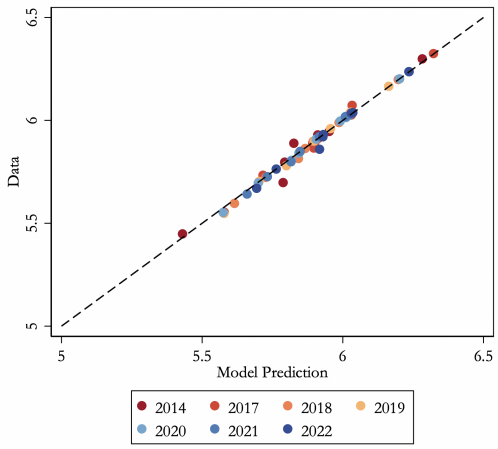
Figure 14: Key Moments - Data vs Model Prediction



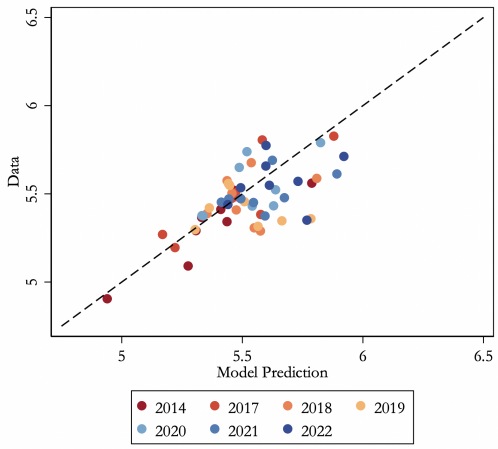
(a) Occupation Shares - Jordanians



(b) Occupation Shares - Syrians



(c) Occupation Avg. Wages - Jordanians



(d) Occupation Avg. Wage - Syrians

Note: The first row of panels plots the observed value in the data versus model predicted log share of each group in each occupation. Panels A and B correspond to Jordanian and Syrian occupation shares respectively. The second row plots the observed value in the data versus model predicted log average wage in each occupation. Panels C and D correspond to Jordanian and Syrian occupation average wages respectively. All moments are logged as the GMM estimation uses log-linearized model equations.

8 Evaluating the Effects of Work Permits

This section turns to answering the central question of the paper – what are the aggregate and distributional effects of reducing refugees' barriers to work? In my stylized facts, I show evidence of refugees' labor market outcomes improving after the introduction of permits. Using my event study design, I estimate relative effects between Jordanians in occupations that are highly exposed to Syrian refugees and those that are less exposed. Only through the estimated model, consistent with my empirical results, can I quantify the aggregate and distributional consequences of the work permit scheme and ascertain the importance of re-sorting in driving these effects.

8.1 Identifying Effects of the Work Permit Scheme

To determine the aggregate effects of the work permit scheme, I have to isolate changes in the wage per efficiency unit that are caused by the supply of labor (driven by the permits) from changes caused by the non-fixed exogenous productivity parameter.[71] To do this, I will first have to take a stand on the general equilibrium parameter, ρ, which governs the substitution of labor in the firm's production function. I set this parameter based on the literature, and then utilize other values to evaluate the robustness of my benchmark results.

The few papers that estimate the elasticity of substitution across types of labor, typically do so with a narrower set of labor types – namely skilled and unskilled labor or education levels – rather than occupations. For my baseline specification, I thus select an elasticity of substitution $\sigma = 3$, which corresponds to $\rho = 2/3$, in line with the preferred specification of Hsieh et al. (2019). Importantly, a higher value of ρ will increase the substitutability of labor across occupations. Intuitively, when substitution across occupations is easier, the negative effects of labor being misallocated across occupations on aggregate production will be lessened. From the side of wages, when occupations are closer substitutes, relative wages will be less elastic to labor supply shocks.

In my main counterfactual, which isolates the effect of work permits, I will re-estimate the model with the following parameters – exogenous amenity parameters $\{\alpha_{o,g}\}$, the homophily parameter $\{\lambda_g\}$, and productivity parameters $\{A_o\}$ – fixed at their final estimated values in 2022, as denoted in Appendix Table A9, and the initial estimated wedges from 2014, prior to work permits. Using endline productivities allows the economy to evolve as it would have, regardless of the work permit scheme, as changes in the occupation-level productivity parameter are orthogonal to the introduction of the work permit scheme. This provides the alternative where work permits were never introduced, which I can then compare against the observed equilibrium with work permits.

Additionally, I simulate a set of individuals based on the estimated distribution of ability, governed by shape parameter θ. I estimate their occupational choice based on their utility maximization problem under the two equilibria, as pinned down by the following model parameters

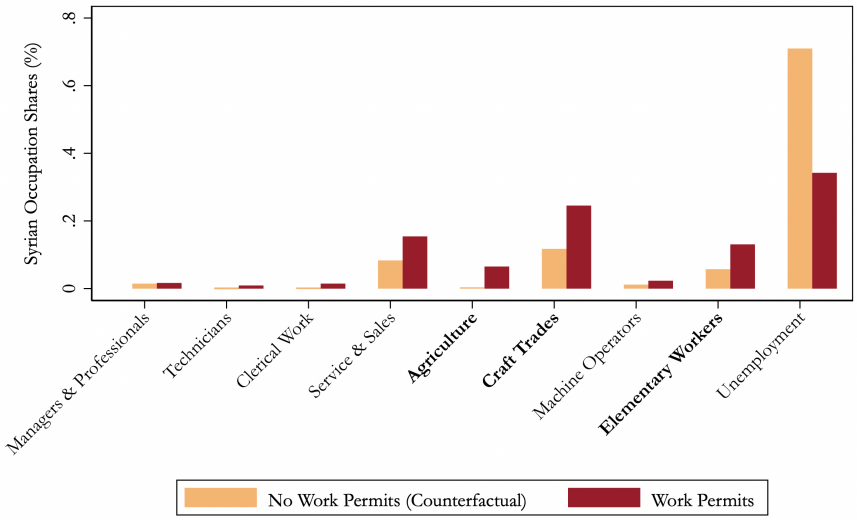
[71] Recall the perfectly competitive firm's optimization problem results in the wage per efficiency unit being equal to the marginal product of labor, which is a function of the occupation's productivity parameter (A_o), the supply of labor to that occupation (H_o), and the elasticity of substitution across occupations (ρ).

$\{w_o, \tau_{o,g}, \alpha_{o,g}, \lambda_g\}$. This allows me to compare the occupational choices and wages of the same set of individuals with and without work permits to determine the policy's winners and losers.

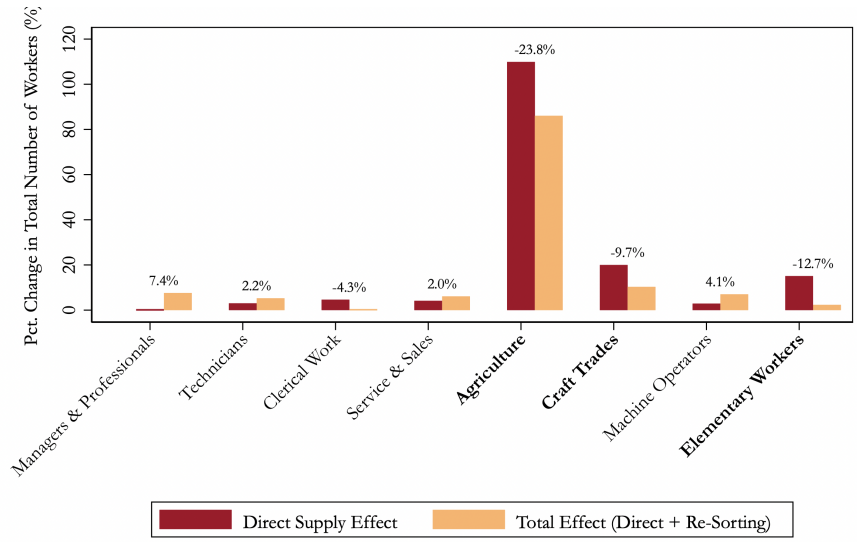
8.2 Effects on Labor Supply and Occupational Re-Sorting

Figure 15: Estimated Effect of Work Permits on Labor Supplied

(a) Effect on Syrian Refugees' Labor Supply



(b) Total Effect on Occupation-Specific Labor Supply



Note: Panel A plots the share of Syrians in each occupation under the counterfactual permits relative to the calibrated model with permits. Bolded occupations correspond to those that see the largest percentage point increase in Syrian refugees from introducing permits. Panel B plots the percentage change in an occupation's size from (i) the direct supply effect from Syrian entry under the work permit scheme and (ii) the total effect of the work permit scheme, including Syrian entry and the subsequent re-sorting of Jordanians. Agriculture is a small occupation in Jordan, such that Syrian entry has a very large effect in percentage terms, while a reasonable effect in levels.

First, I look at how the work permits affected the relative supply of Syrians across occupations. Figure 15a plots the share of Syrians in each occupation with and without work permits, where the latter is the previously described counterfactual estimated by the model. Syrian refugees leave unemployment and enter into all occupations; however, they do not do so uniformly. For the full set of occupation transitions for Syrian refugees, see Appendix Figure A18. Agriculture, Craft Trades, and Elementary Workers see the largest relative increase in Syrian workers.

Figure 15b then shows the percentage change in each occupation's size (total number of workers), as a result of this *direct supply effect* from Syrian entry, denoted by red bars. The second set of bars plots the total effect on the labor supply to each occupation, after accounting for the *re-sorting effect* of Jordanians. While entry by Syrian refugees increases the number of workers in all occupations relative to a labor market without work permits, re-sorting has a substantial effect on the magnitude. Consistent with the theory, Jordanians exit from occupations that see the biggest increases in Syrian refugees (those bolded) and re-sort into those with relatively fewer.

In addition to re-sorting across occupations, the model generates some exit among Jordanians as a result of the work permit scheme. The unemployment rate increase by 0.6 percentage points on a base rate of 23.4%, which is equivalent to a 2.6% increase. For the estimated effect on all Jordanian occupation shares, see Appendix Figure A19.

8.3 Effects on Wages

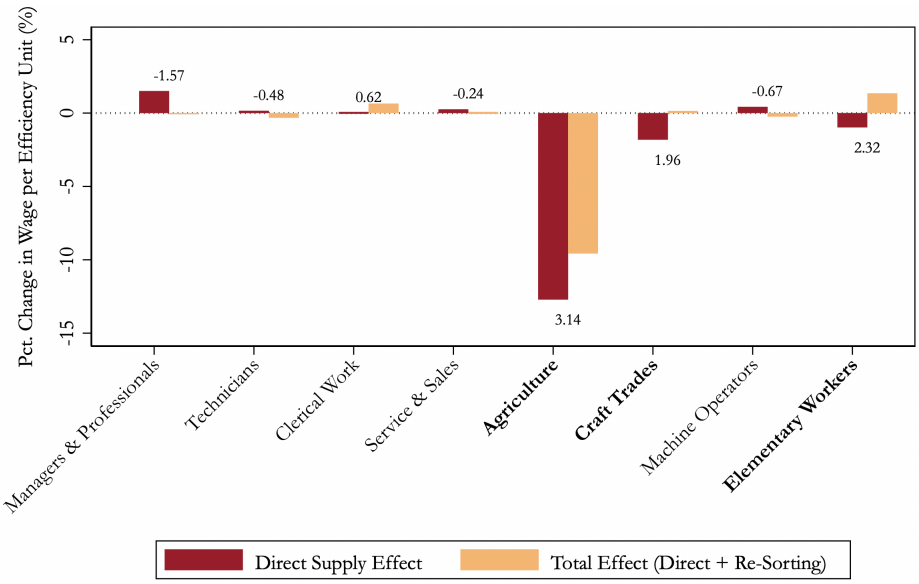
Figure 16 plots the effect on the wage per efficiency unit in each occupation decomposed into the partial equilibrium effect caused by refugee entry (*direct supply effect*) and the total effect after accounting for the general equilibrium re-sorting of Jordanians (*re-sorting effect*). Under the direct supply effect alone, Syrians enter more into Agriculture, Craft Trades, and Elementary Workers relative to other occupations, pushing the wage per efficiency unit down in those occupations. This produces the standard effect often cited in the migration literature where a labor supply shock has a negative effect on the wages of those that are directly substitutable with the migrants that arrive.

However, after accounting for the general-equilibrium re-sorting of Jordanians, in response to Syrian entry, these effects largely flip – the wage per efficiency unit actually decreases in occupations Syrians do not enter.[72] This is because the re-sorting of Jordanians is substantial enough to offset the relative supply shock of Syrian entry, reversing the overall directionality of the labor supply effect. Thus, labor in occupations that initially received Syrian refugees becomes relatively scarce due to Jordanians exiting.

These effects are then magnified when taking the occupation-average wage for Jordanians. Under the *direct supply effect* alone, occupation-average wages fall for Jordanians in the highly exposed occupations (where Syrians enter most), as the wage per efficiency unit has fallen. Once accounting for Jordanians re-sorting, occupation-average wages among Jordanians increase in the highly exposed occupations as a result of the increase in the wage per efficiency unit and the additional se-

[72] With the exception of agriculture, where the de facto subsidization of Syrians who work there creates too large of an effect to be off-set.

Figure 16: Estimated Effect on Wage Per Efficiency Unit



Note: This figure plots the percent change in the wage per efficiency unit from the direct supply effect of Syrian entry once work permits are introduced and the total effect after accounting for Jordanians re-sorting. The wage per efficiency unit is equivalent to the marginal product of labor in the model.

lection effect – where the lowest quality Jordanians exit, raising the average productivity of workers remaining in the occupation. The corresponding plot is shown in Appendix Figure A20.

8.4 Winners and Losers

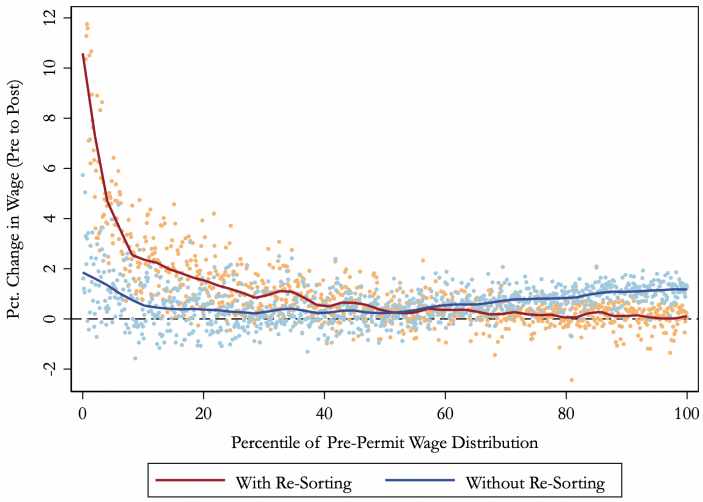
To understand the distributional effects of the work permit scheme and the importance of re-sorting as an adaptive response, I look at the changes in the wages across the model simulated Jordanian population in an economy with and without work permits. Figure 17 plots the percent change in individuals' wages along the income distribution in the counterfactual without work permits. For wage changes in levels, see Appendix Figure A23.

First, I consider what happens to wages when the work permits are introduced but Jordanians are not allowed to re-sort. The light blue dots correspond to percent changes in the wage among individuals binned at the 0.1 percentile of the income distribution and the dark blue line as a local polynomial fit of these values. Without re-sorting, the Jordanians who lose (those that see wage decreases) are clustered at the bottom of the income distribution. Comparatively, those that benefit the most from the introduction of work permits are at the top of the income distribution, working in the occupations that are relatively less exposed to Syrian entry. Without re-sorting, I find a story consistent with the literature: (i) there is broadly a null effect on average wages, (ii) the losers among the local workers are clustered at the bottom of the income distribution, facing the brunt of the labor supply shock, and, (iii) in cases where there are winners among locals, they are in the high-paying occupations that are largely complements to those occupations that received

refugees.

However, the outcomes flip once I account for the role of locals' re-sorting. The orange dots plot the binned percent changes in individuals' wages from introducing work permits and allowing Jordanians to re-sort in response, with the dark red line fitting these values. Allowing for re-sorting, the winners are those at the bottom of the pre-work permit income distribution. The poorest 20% of Jordanians experience an average increase of 3% from the work permit scheme. Relative to the case without sorting, the losers are now concentrated at the top of the income distribution. Ultimately, re-sorting changes who wins and loses from granting refugees the right to work, reducing income inequality among the host community.

Figure 17: Simulated Effect of Re-Sorting on Wage Inequality

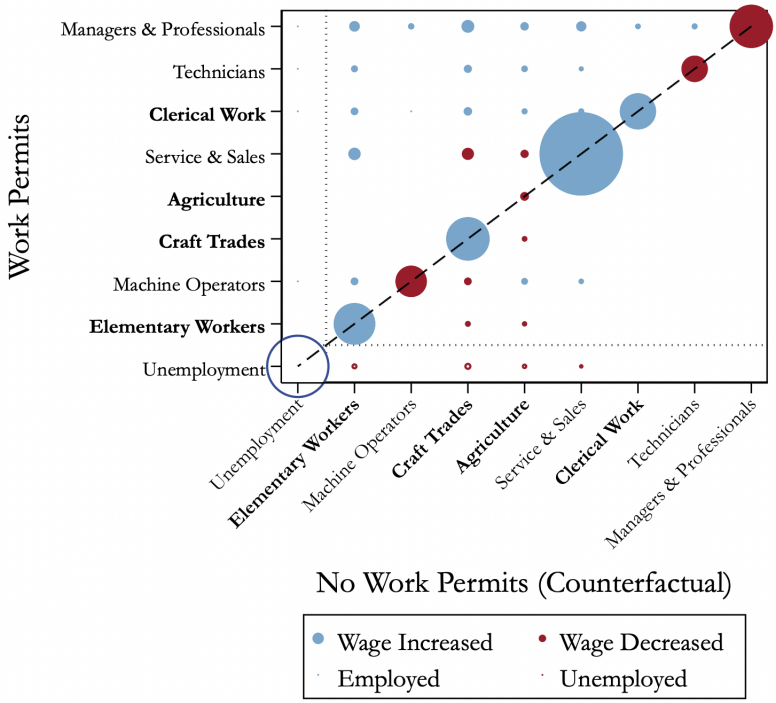


Note: The y-axis measures the percent change in the wage caused by introducing work permits. The x-axis represents a simulated individuals' percentile in the wage distribution before work permits are introduced. The dots then correspond to the percent change in an individual Jordanian's wage, binned at the 0.1 percentile for visibility. The light blue corresponds to the percent change in wages under the counterfactual where Jordanians cannot re-sort after work permits are introduced while the orange dots correspond to when they can. The maroon and dark blue lines plot the best fit line resulting from a kernel-weighted local polynomial regression of the percent change in the wage relative on the percentile in the pre-work permit wage distribution.

To understand how re-sorting drives these results, Figure 18 plots the occupation transitions of Jordanians from the counterfactual without work permits relative to the calibrated model with work permits. The size of circles represents the number of Jordanians in each transition cell and the colors correspond to the average wage effect for that group. Wage gains among the poorest workers come from two sources. First, some exit low-paying occupations, where Syrians enter once there are work permits, and upgrade to higher paying occupations, namely Technicians and Managers and Professionals. Those that upgrade to these occupations see wage increases. The second source of wage gains for the poorest workers comes from the increase in the wage per efficiency unit in low-paying occupations, as these types of labor become relatively scarce after Jordanians re-sort

away.

Figure 18: Simulated Occupation Transitions by Winners and Losers



Note: The y-axis plots the occupational choice of simulated individuals after work permits and the x-axis plots the occupational choice in the counterfactual without work permits. The size of the circles correspond to the number of individuals in each transition cell. Those circles off the diagonal represent individuals who switched occupations in response to the work permit scheme. Maroon colored circles correspond to those in that given cell having a wage loss on average while light blue corresponds to an average wage gain. The open circles correspond to unemployment and are maroon when individuals exit an occupation and dark blue when they remain unemployed in both cases. Bolded occupations correspond to those that receive the most Syrian refugees after work permits are introduced.

8.5 Aggregate Effects and Efficiency Gains

In the aggregate, the model predicts that the introduction of work permits resulted in a 0.92% increase in the average wage of Jordanians and a 20% increase in the average wage of Syrians. Without re-sorting, Jordanians gain 0.90% on average, reflecting that sorting is driven largely by homophily, as opposed to wage gains. For density plots of the full wage distributions under each scenario, see Appendix Figure A21 for Jordanians and Appendix Figure A22 for Syrians. The wage gains for Jordanians are partially offset by a modest rise in Jordanian unemployment of 0.8 percentage points on a base rate of 23.4%. Even accounting for the increase in unemployment, the total wage bill among Jordanians increases 0.11% as a result of the work permit scheme.

In terms of aggregate output, the model implies a 10.9% increase in total output following the introduction of work permits, driven almost entirely by improved allocation of Syrian refugee labor (10.8%) rather than improvements in the allocation of Jordanian labor, again consistent with

sorting driven largely by homophilic preferences rather than wages.

Overall, the work permit scheme improves allocative efficiency, raising total output by nearly 10.9%, and simultaneously compresses the Jordanian wage distribution by shifting gains toward the poorest workers. Both channels contribute positively to aggregate welfare under standard concave utility.

9 Conclusion

This paper examines how granting refugees the right to work reshapes the allocation of refugee and local labor across occupations and the subsequent aggregate and distributional impacts. Exploiting a unique natural experiment created by Jordan's introduction of a large-scale work permit scheme for Syrian refugees, I show that relaxing legal barriers to work triggered substantial occupational reallocation by both refugee and local workers. Syrian refugees entered into employment and gained access to a broader, less informal set of occupations. In response, Jordanians re-sorted out of occupations that were most affected by Syrian entry, generating occupational upgrading and positive selection among those Jordanians who remained. This upgrading was driven by the over-educated Jordanians in highly exposed occupations, where their college-level skills were underutilized.

To formalize these dynamics, I develop a general equilibrium model of occupational choice that embeds homophilic preferences – a preference for working alongside one's own social or ethnic group – alongside standard wage-based sorting. Homophily transforms the right to work into a social as well as economic shock: as refugees enter new occupations, the change in worker composition amplifies Jordanians' re-sorting. The model shows that preferences for homophily amplify wage-based sorting responses among Jordanians, supporting equilibria where re-sorting by locals outweighs the direct supply effects of refugee entry.

Estimating my model against data, I find that the work permit scheme reduced the misallocation of refugee labor and generated large aggregate gains. Total output increased by roughly 11%, driven primarily by improved utilization of refugee labor. Jordanians' average wages rose modestly as did unemployment but income inequality declined. Accounting for re-sorting reverses the canonical distributional effects of refugee arrival – instead of lower-skill locals losing, they benefit through occupational upgrading and relative-scarcity effects from others' re-sorting. The model implies that the poorest quantile of Jordanians gain the most from work permits, with average wage rising by three percent.

A natural extension of this research is to study how social forces, such as homophily, continue to constrain refugees' access to job opportunities, once legal barriers are removed. In corollary work, I design an intervention to experimentally vary Syrian refugees' exposure to local Jordanian networks – through matching treated refugees to Jordanian volunteer partners – to identify the causal effect of cross-group connections on job search and occupational mobility for refugees.[73] The experiment

[73]Personal connections, often familial-based – known as *wasta* or *ma'aref* in Jordanian Arabic – are deeply

aims to quantify how reducing social distance between refugees and hosts can mitigate the misallocation of refugees that is induced by segmented networks and the corresponding segmentation of information. Simultaneously, the experiment will speak to the relative importance of mechanisms, such as discrimination and network-based job search, in driving homophilic preferences among Jordanians. Together, these results aim to micro-found homophily in the model and clarify its welfare implications.

The implications of this research extend beyond the labor markets of Jordan. As climate change, conflict, and demographic pressures drive ever-larger and -longer waves of displacement, the policy question of how to best integrate refugees into labor markets will become increasingly central. This paper demonstrates that the right to work is not simply a humanitarian entitlement for refugees but a structural lever that can catalyze reallocation across occupations, unlocking productivity gains for both refugee and host workers. Yet effective integration requires understanding not only how refugees respond to policy, but how host communities adjusting in turn – through their re-sorting, wage responses, and social adaptations – can fundamentally shape the aggregate and distributional outcomes of reform. Studying these host-side dynamics is essential for designing refugee policies that are both economically efficient and politically sustainable in an era when mobility and displacement will define global development.

rooted in Arab societies and govern many social and professional transactions (Baranik et al., 2023). In Jordan, these personal connections play a substantial role in both job search and matching. In survey data I collected in the northern governorate of Mafraq, which was most affected by Syrian refugee arrival, I find that 48% of Jordanians found their last job through a family or friend, 96% of whom were also Jordanian.

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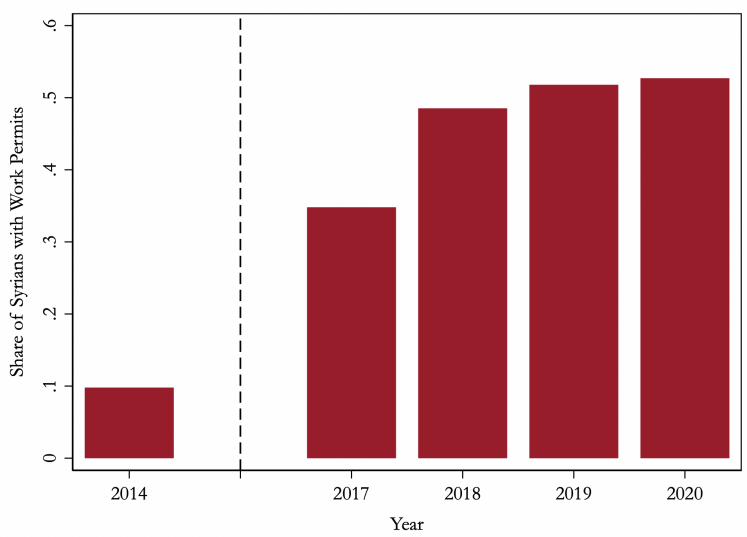
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Appendices

A Appendix: Additional Tables and Figures

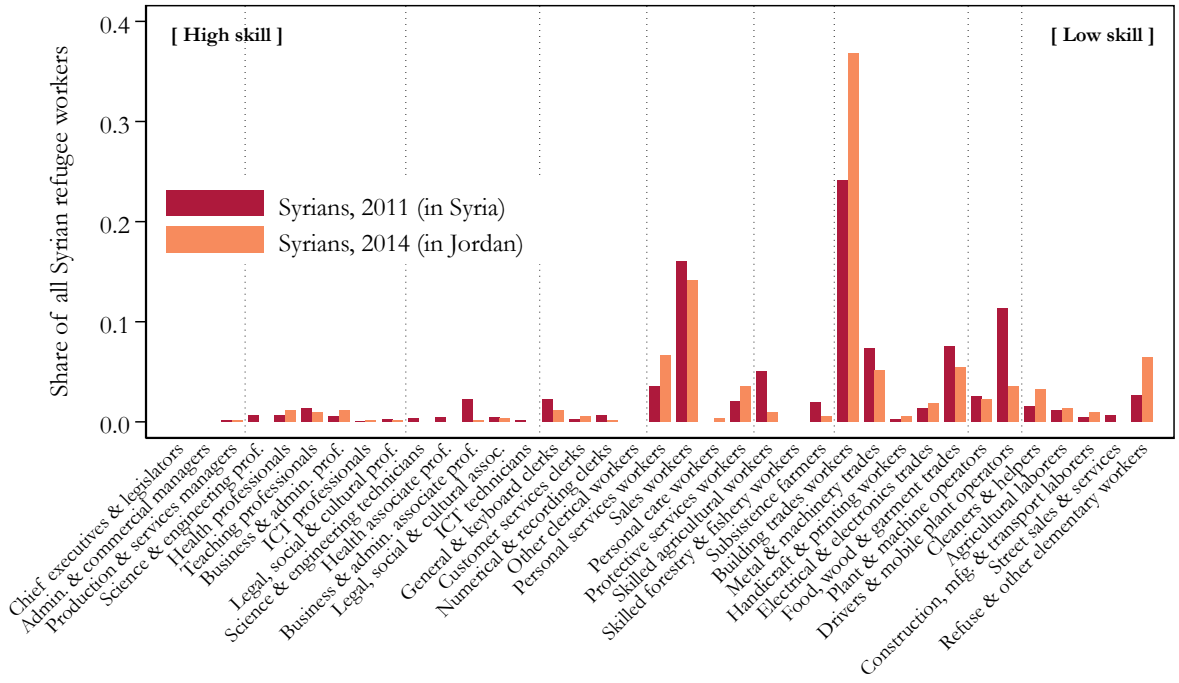
A.1 Empirical Results

Figure A1: Work Permit Coverage Among Employed Syrian Refugees



Note: sample restricted to employed working-aged male Syrian refugees. (Observations: 7,839)

Figure A2: Occupation Distribution of Syrian Refugees - Pre vs. Post Displacement



Note: the graph plots bar the share of Syrian refugees that had worked in an occupation group, based on 2-digit ISCO occupation code. The dark red bars represent the share of Syrian refugees in Syria working in an occupation prior to displacement. The orange bars represent the share of all Syrian refugees in Jordan working in an occupation, post displacement but before work permits. The x-axis corresponds to the 2-digit ISCO occupation code names, which are themselves ordered by skill level. (Observations: 1,886)

Table A1: Occupational Mobility of Jordanian Workers - Prior to Refugee Arrival (2008–2010)

	All	Tertiary-educated	Less than tertiary
Panel A: All employed men			
Switched jobs	0.17 (0.006)	0.19 (0.016)	0.17 (0.007)
Changed 2-digit occupation	0.10 (0.005)	0.09 (0.012)	0.10 (0.005)
Changed 1-digit occupation	0.08 (0.004)	0.06 (0.009)	0.09 (0.005)
Upgraded (1-digit)	0.04 (0.003)	0.03 (0.007)	0.05 (0.004)
Upgraded (wage ladder)	0.05 (0.004)	0.04 (0.008)	0.06 (0.004)
Downgraded (1-digit)	0.04 (0.003)	0.02 (0.006)	0.04 (0.003)
Downgraded (wage ladder)	0.04 (0.003)	0.05 (0.009)	0.04 (0.004)
Observations	3,815	609	3,206
Panel B: In a non-degree occupation (ISCO 3–9)			
Switched jobs	0.17 (0.007)	0.18 (0.030)	0.17 (0.007)
Changed 2-digit occupation	0.10 (0.005)	0.11 (0.024)	0.10 (0.005)
Changed 1-digit occupation	0.09 (0.005)	0.10 (0.023)	0.09 (0.005)
Upgraded (1-digit)	0.05 (0.004)	0.09 (0.022)	0.05 (0.004)
Upgraded (wage ladder)	0.06 (0.004)	0.09 (0.022)	0.06 (0.004)
Downgraded (1-digit)	0.04 (0.003)	0.01 (0.008)	0.04 (0.003)
Downgraded (wage ladder)	0.04 (0.004)	0.02 (0.010)	0.04 (0.004)
Observations	3,287	171	3,116

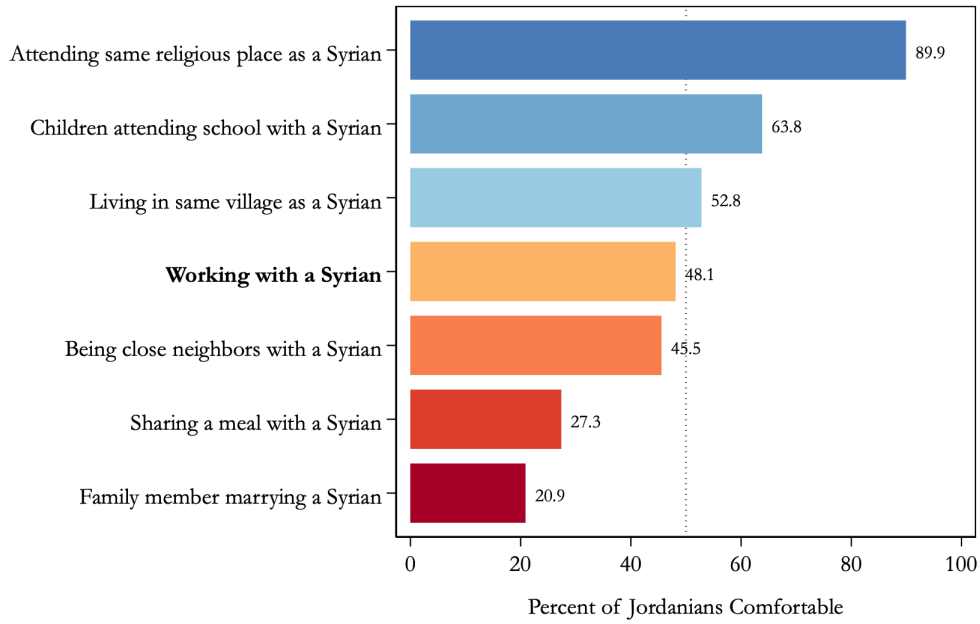
Note: Panel A's sample consists of working-aged Jordanian men who are employed in 2008, reconstructed from self-reported employment history in the JLMPS 2010 survey wave to measure job switching and occupational change over a two-year period. Results are consistent over longer-time windows (three- and four-year periods) but increasingly noisy. The sample is split in Columns 2 and 3 into workers who have at least a tertiary education and those that do not. Panel B further restricts the sample to Occupation Groups 3-9, which do not require a tertiary education based on ILO skill requirements to look at occupational mobility among those with college degrees that are not in positions that utilize them. Occupational upgrading (downgrading) is measured in two ways: 1) moving up (down) in the 1-digit ISCO codes, which corresponds to switching groups and is ordered to reflect skills and 2) moving up (down) the occupation wage ladder, which orders occupations based on the average wage earned in that occupation.

Table A2: Summary Statistics on Network-Based Job Search (2025 Mafraq Survey)

	Jordanian	Syrian refugees	Difference	p-value
Panel A: How respondents found their current or most recent job				
Through a friend	0.31 (0.029)	0.47 (0.026)	0.16 (0.039)	<0.001
Through a family member	0.19 (0.028)	0.13 (0.018)	-0.07 (0.033)	0.040
Observations	602	590		
Panel B: Nationality of the friend who helped (friend referrals)				
Friend is Jordanian	0.98 (0.011)	0.37 (0.036)	-0.61 (0.037)	<0.001
Friend is Syrian	0.01 (0.009)	0.61 (0.037)	0.60 (0.038)	<0.001
Observations	193	266		
Panel C: What the friend or family member did (friend or family referrals)				
Connected me to my employer	0.71 (0.038)	0.73 (0.030)	0.01 (0.049)	0.760
Shared information or advice	0.07 (0.017)	0.09 (0.021)	0.02 (0.027)	0.392
Gave me the job directly	0.17 (0.032)	0.15 (0.023)	-0.02 (0.040)	0.577
Observations	285	325		

Note: Panel A's sample consists of working-aged Jordanian and Syrian refugee men who were surveyed in spring 2025 in Mafraq governorate. Panel B is for the subset of respondents who received help from friends to find their current or most recent job. Panel C is for the subset of individuals who relied on either friends or family. The first and second columns report the share of Jordanian and Syrian refugee respondents respectively for each outcome of interest. The third column provides the differences and the fourth column reports the corresponding p-values from t-tests of the difference in weighted means, using robust standard errors. Standard errors in parentheses.

Figure A3: Jordanian Sentiments Toward Interacting With Syrian Refugees



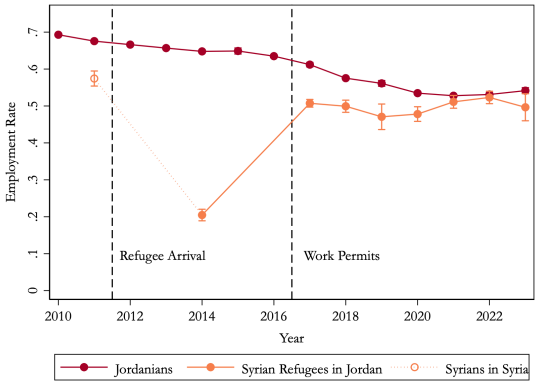
Note: the graph plots the share of Jordanian respondents who said they were comfortable in each of the following settings described. Data is from the 2014 ILO/Fafo Survey.

Table A3: Reason for Working Without a Work Permit

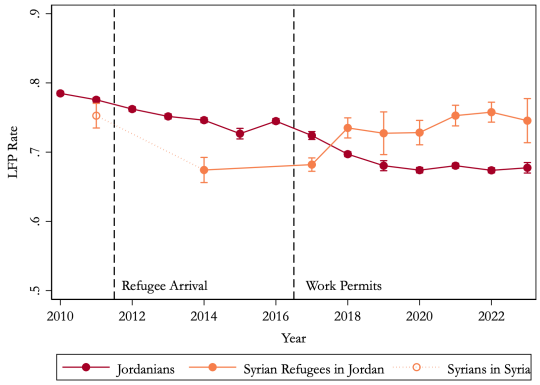
	Count	Percent
Not needed	1449	47.92
Applied but unsuccessful	650	21.49
Waiting to receive it	554	18.32
Employer won't support	180	5.95
Other	122	4.03
Too difficult a process	69	2.28
N	3024	

Note: sample restricted to working-aged male Syrian refugees employed in 2017 without work permits but after the work permit scheme has been introduced. Data comes from Fafo/DoS 2018 survey. (Observations: 3,024)

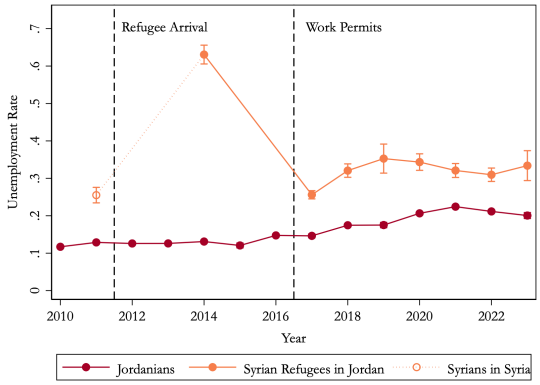
Figure A4: Labor Market Outcomes for Syrian Refugees and Jordanians



(a) Employment Rate



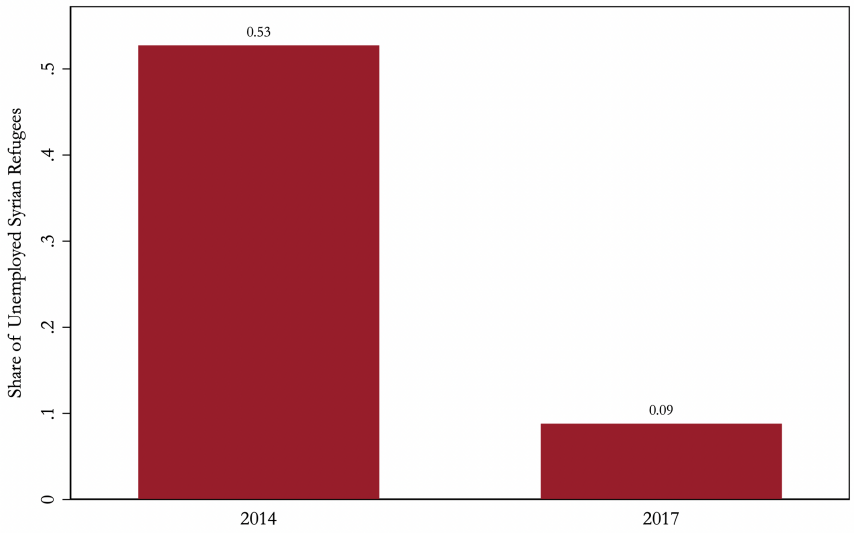
(b) Labor Force Participation Rate



(c) Unemployment Rate

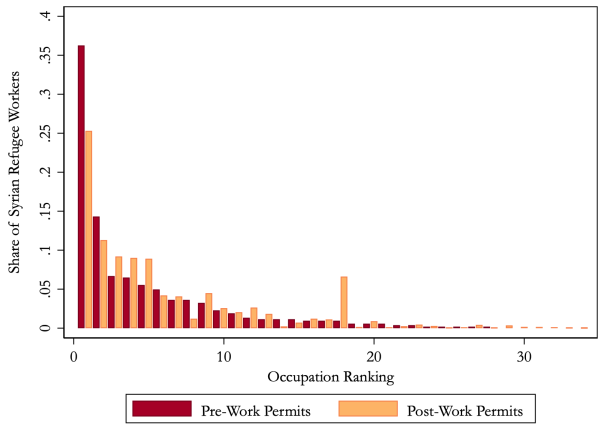
Notes: Sample includes working-aged male Syrian refugees and Jordanians. (A) Employment rate is the share of each group that is employed. (B) Labor force participation rate is the share of each group that is either employed or unemployed. (C) Unemployment rate is the conditional share of labor force participants in each group that is unemployed. (Observations: 448,325)

Figure A5: Labor Law Compliers as Share of Unemployed Syrians



Note: Sample restricted to unemployed working-aged male Syrian refugees. Data comes from ILO/Fafo 2014 and Fafo/DoS 2018 surveys. (Observations: 6,193)

Figure A6: Distribution of Syrian Refugee Occupation Shares (Ranked by 2014 Share Size)



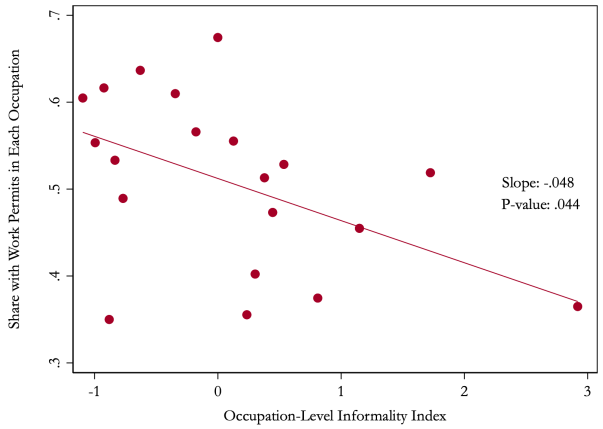
Note: the x-axis is the ranked order of 2-digit ISCO occupation codes based on the share of all employed working-aged male Syrian refugees in the occupation before work permits. The first bar refers to the share before work permits and the second bar refers to the average share in that occupation for the period after work permits are introduced. (Observations: 11,936)

Table A4: Characteristics of the Occupation Share Distributions

	Mean	Median	SD	Skewness
Pre-Work Permits	0.17	0.14	0.15	0.37
Post-Work Permits	0.12	0.09	0.09	0.63
Observations	11936			

Note: shares are defined as the share of all employed working-aged male Syrian refugees working in a 2-digit ISCO occupation codes in each period.

Figure A7: Work Permit Coverage by Occupation Informality Level



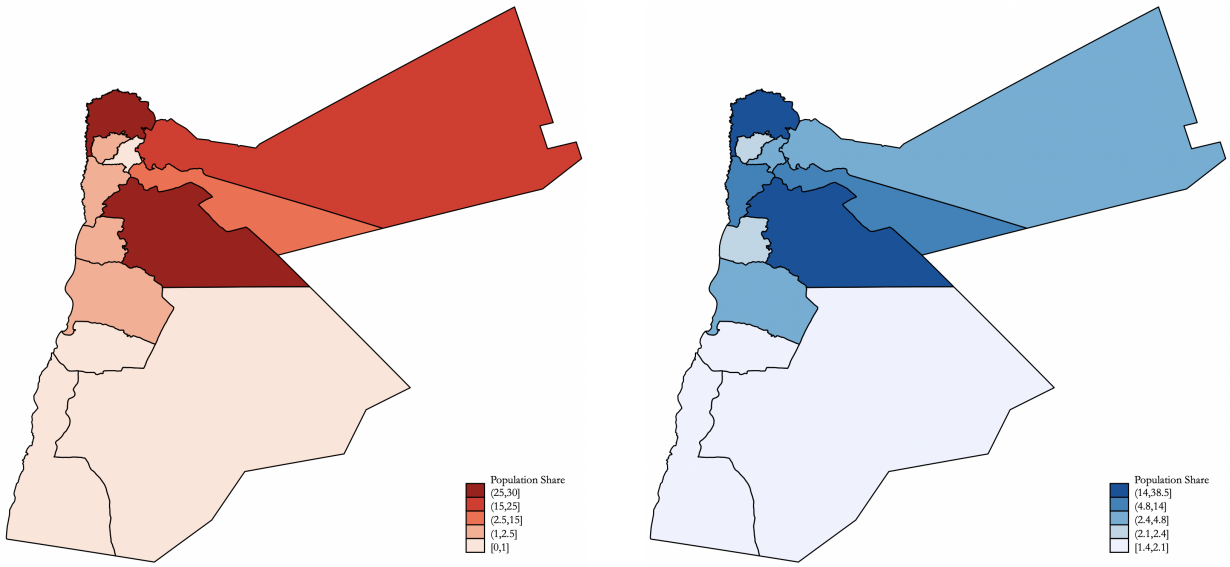
Note: this is a binned scatter plot from regressing the share of Syrian refugee workers that have work permits in a given 2-digit ISCO code occupation on the z-scored index of informality for the occupation. Regression includes year fixed effects. Sample conditioned to working-aged male Syrian refugees after the introduction of work permits. (Observations: 6,719)

Table A5: Wage Gap between Syrian Refugees and Jordanians

	Log Monthly Income	Monthly Income
	(1)	(2)
2014 x Refugee	-0.361	-69.734
	[0.032]***	[10.895]***
2017 x Refugee	-0.340	-60.272
	[0.042]***	[11.143]***
2018 x Refugee	-0.161	-29.405
	[0.036]***	[10.201]***
2019 x Refugee	-0.189	-40.314
	[0.035]***	[10.475]***
2020 x Refugee	-0.125	-23.867
	[0.039]***	[14.641]
2021 x Refugee	-0.138	-34.118
	[0.039]***	[11.481]***
2022 x Refugee	-0.106	-20.241
	[0.041]***	[14.223]
Year	Yes	Yes
Governorate	Yes	Yes
Occupation	Yes	Yes
Industry	Yes	Yes
Human Capital Controls	Yes	Yes
2014 Jordanian Mean	5.72	337.74
Obs	149044	149149

Note: Coefficients estimate the wage gap (measured in monthly income) between Syrian refugees and Jordanians in a given year. The independent variable is an indicator variable for whether the worker is a Syrian refugee and the year the worker is observed. Regressions include fixed effects for year, governorate, 2-digit iSCO occupation codes, 1-digit ISIC industry codes. Regression also include Human capital controls (age and education in years). Standard errors are clustered at the level of year x 2-digit occupation.

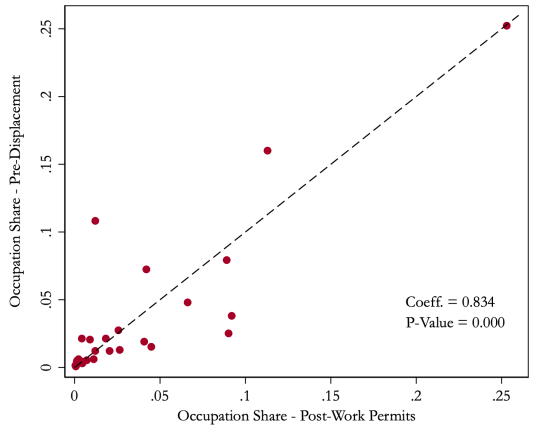
Figure A8: Spatial Distribution of Populations Across Jordan



(a) Syrian Refugees (b) Jordanians

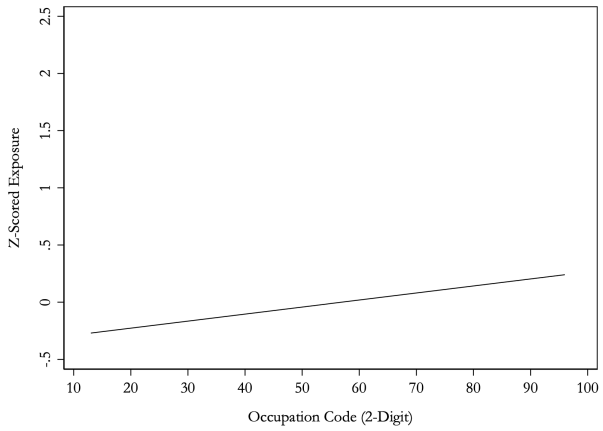
Note: Graphs plot the share of each population residing in each governorate in Jordan based on the Jordanian Department of Statistics' 2015 Census. Total population of Jordanians was 6,578,636 and total population of Syrian refugees was 953,289.

Figure A9: Syrian Refugees' Occupation Shares Pre-Displacement vs. After Work Permits



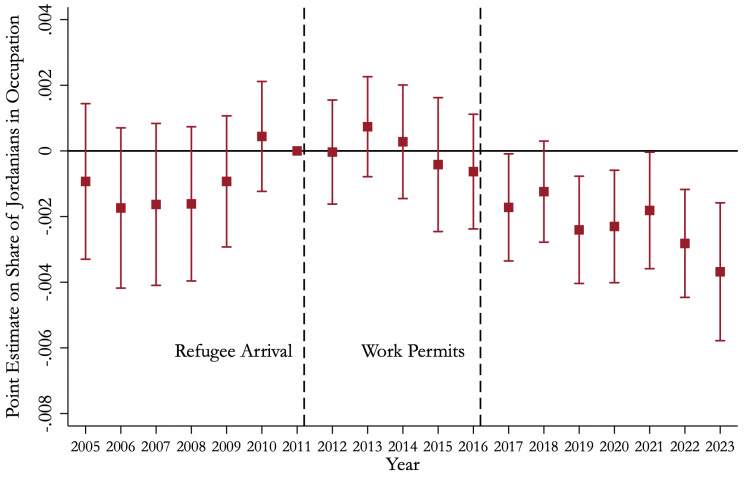
Note: the graph plots the z-score of the exposure measure averaged across locations for each 2-digit ISCO occupation code.

Figure A10: Average Z-Score Exposure by ISCO 2-Digit Occupation



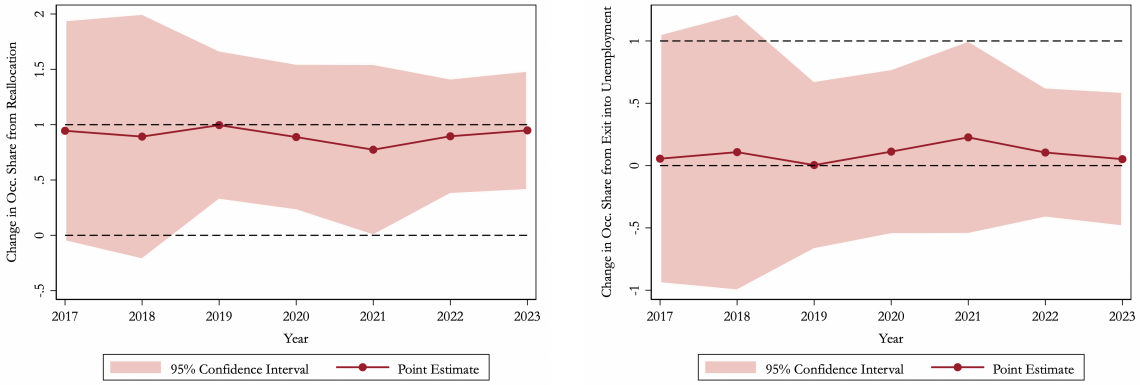
Note: The y-coordinate corresponds to the share of Syrian refugees that had worked in the occupation in Syria prior to displacement. The x-coordinate corresponds to the share of Syrian refugees that work in the occupation after work permits are introduced. (Observations: 12,620)

Figure A11: Effect of Exposure on Jordanian Occupational Shares (Excluding Unemployment)



Note: Point estimates from a regression of occupation shares, computed **excluding** unemployment from the denominator, for Jordanians on a z-scored shift-share measure of exposure to Syrian refugees, measured as share of refugees in that occupation x share of refugees in a given governorate. Regression includes year and occupation-by-governorate fixed effects. Standard errors are clustered at the unit of treatment: the occupation-governorate. (Observations: 6,696 occupation x governorates)

Figure A12: Decomposition of Effect on Occupation Shares - Reallocation vs. Unemployment



(a) Reallocation (≈ 85-90%)

(b) Unemployment (≈ 10-15%)

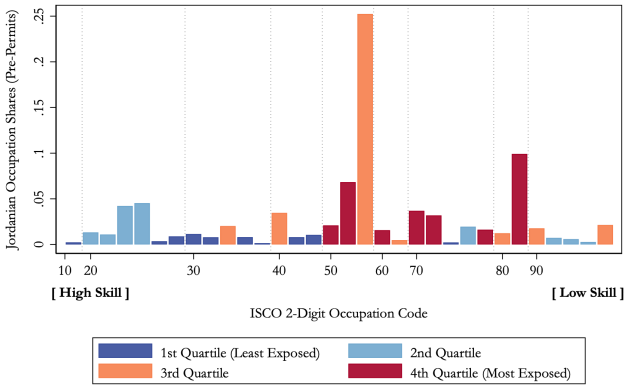
Note: Figures are based on the coefficient estimates of z-score exposure regressed on occupation shares where shares either **include** or **exclude** unemployment from the denominator. Recall inclusion of unemployment is the preferred specification include in the main text of the paper. The corresponding decomposition plotted above is then based on taking the absolute difference in these coefficients dived by the main specification coefficient to capture the component of the coefficient estimate attributable to unemployment. The amount attributable to reallocation is then 1-(unemployment amount), such that they together account for 100% of the variation. To construct standard errors, I use a delta method based on the standard errors for each specification's coefficients with the assumption of independence of sample variances.

Table A6: Full Decomposition of Estimated Exposure Coefficients for Occupation Shares

Year	$\beta_{t,u}$ (incl. unemployment)	$SE_{t,u}$	$\beta_{t,e}$ (excl. unemployment)	$SE_{t,e}$	Difference	Share Unemp.	Share Realloc.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
2005	-.00103	.00106	-.00093	.00121	.0001	.0970874	.9029126
2006	-.00182	.0011	-.00174	.00124	.00008	.043956	.956044
2007	-.00139	.00112	-.00163	.00126	-.00024	.1726619	.8273381
2008	-.00129	.00104	-.00161	.0012	-.00032	.248062	.751938
2009	-.000788	.000924	-.000929	.00102	-.000141	.178934	.821066
2010	.000308	.000862	.000441	.000852	.000133	.4318182	.5681818
2011	0	.	0	.	0	.	.
2012	-.0000211	.000823	-.0000349	.000807	-.0000138	.6540284	.3459716
2013	.000431	.000769	.000738	.000776	.000307	.712297	.287703
2014	-.0000599	.000852	.000278	.000881	.0003379	5.641068	-4.641068
2015	-.000482	.00101	-.000417	.00104	.000065	.1348548	.8651452
2016	-.000929	.000815	-.00063	.00089	.000299	.3218515	.6781485
2017	-.00163	.000778	-.00172	.000831	-.00009	.0552147	.9447853
2018	-.00139	.000757	-.00124	.000784	.00015	.1079137	.8920863
2019	-.00241	.000805	-.0024	.000832	1.00e-05	.0041494	.9958506
2020	-.00259	.000877	-.0023	.000872	.00029	.1119691	.8880309
2021	-.00234	.000942	-.00181	.000903	.00053	.2264957	.7735043
2022	-.00315	.000879	-.00282	.000837	.00033	.1047619	.8952381
2023	-.00388	.00105	-.00368	.00107	.0002	.0515464	.9484536

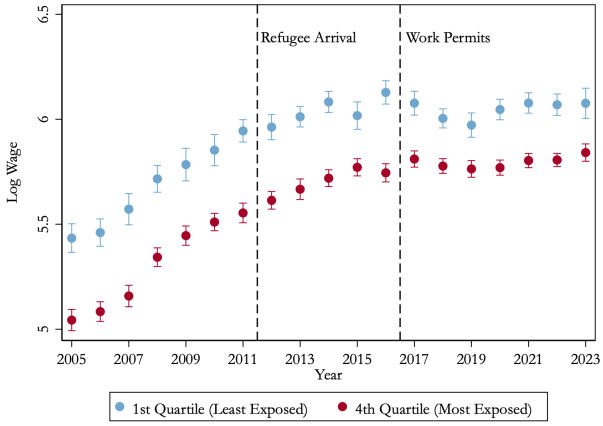
Note: Columns (1) and (3) correspond to the estimated coefficients on exposure in the regression on occupation shares, where (1) is the main specification estimate and includes unemployment in the denominator and (3) excludes it. Columns (2) and (4) correspond to the standard errors for each specification. Column (5) reports the difference in the point estimates between Columns (1) and (3). Column (6) and Column (7) correspond to decomposition of the main coefficient estimate into the share that can be attributed to unemployment and reallocation respectively. Values before 2016 are noisy because both specifications yield coefficients close to zero, mechanically inflating ratios. After 2016, the decomposition stabilizes. 2011 is the base year in the event study, so its coefficient is mechanically set to zero with no standard errors.

Figure A13: Jordanians' Occupation Distribution by Exposure to Syrian Refugees



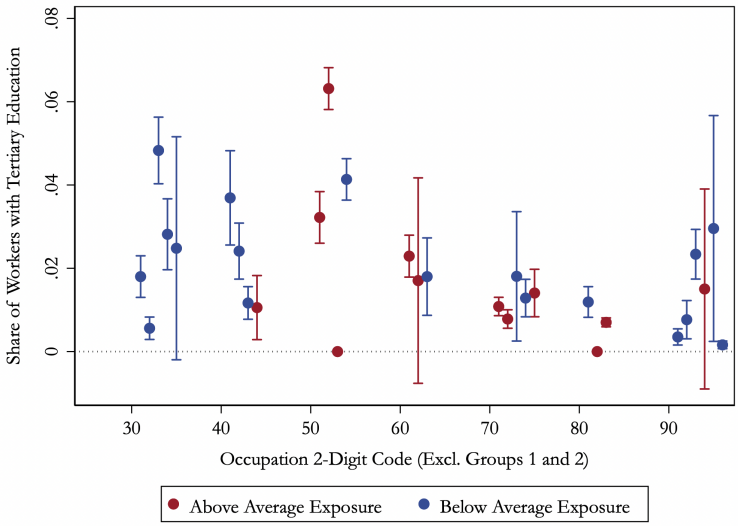
Note: the graph plots the share of Jordanians working in a given 2-digit ISCO occupation codes prior to the introduction of work permits. The color coding corresponds to the quartiles of the occupation-level exposure, defined by the pre-displacement occupation share of Syrian refugees. 36% of Jordanians work in an occupation in the highest exposure quartile prior to the work permit scheme.

Figure A14: Average Occupation-Level Wage by Exposure Quartile



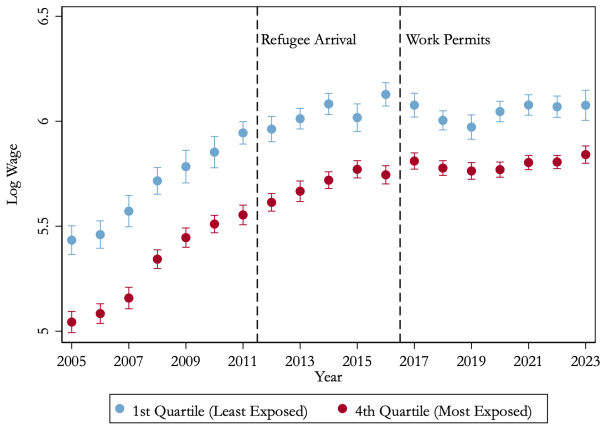
Note: the graph plots the evolution of the log average wage of Jordanians in occupations in the highest and lowest quartiles of exposure. Quartiles are based on the occupation-level exposure, defined by the pre-displacement occupation share of Syrian refugees. The highest level of exposure is the fourth quartile and the lowest is the first quartile.

Figure A16: Average Share of Workers with Tertiary Education by Exposure Level



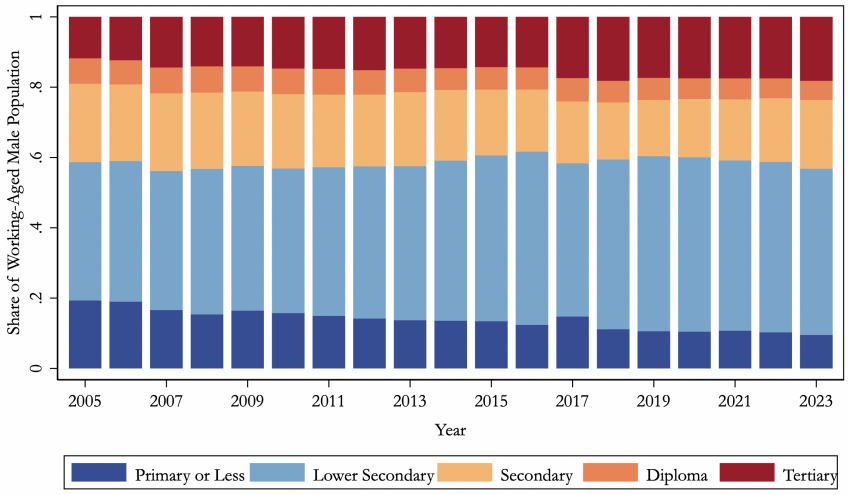
Note: the graph plots the average share of Jordanian workers who have tertiary education in each 2-digit ISCO occupation code (in each location), excluding Groups 1 and 2, which require tertiary education according to ILO skill requirements for each occupation. The remaining Groups 3-9, plotted here, do not require tertiary education. Data is is based on the years prior to the enactment of the work permits (pre-2016) before patterns of reallocation started. The colors of the bar correspond to whether the occupation was above or below the mean exposure level based on the shift-share measure of exposure. (Observations: 363,497)

Figure A15: Average Occupation-Level Wage by Exposure Quartile



Note: the graph plots the evolution of the log average wage of Jordanians in occupations in the highest and lowest quartiles of exposure. Quartiles are based on the occupation-level exposure, defined by the pre-displacement occupation share of Syrian refugees. The highest level of exposure is the fourth quartile and the lowest is the first quartile.

Figure A17: Highest Level of Education Completed by Year



Note: the graph plots the distribution of working-aged male Jordanians by different levels of educational attainment. Each year adds up to 1. "Primary or less" includes education up to and including grade 5. "Lower secondary" corresponds to education up to and including 10. "Secondary" corresponds to grades 11 and 12. "Diploma" corresponds to post-secondary education that does not result in a four-year undergraduate degree."Tertiary" includes all college-level education including undergraduate degrees, masters, and PhDs. (Observations: 601,558)

Table A7: DiD Effects of Exposure on Extensive Margin Outcomes by Education

	Employed (All)	Employed (All)	Unemployed (Only LFPs)	Unemployed (Only LFPs)	Exited Labor Force
	(1)	(2)	(3)	(4)	(5)
Exposure x Post	-0.006	-0.004	0.008	0.014	-0.005
	[0.011]	[0.019]	[0.011]	[0.021]	[0.011]
Post	-0.251	-0.355	0.078	0.105	0.235
	[0.035]***	[0.049]***	[0.016]***	[0.027]***	[0.028]***
Exposure x Post x Tertiary Educ		-0.070		0.016	
		[0.038]*		[0.053]	
Exposure x Post x Diploma		0.041		-0.024	
		[0.063]		[0.029]	
Exposure x Post x Secondary Educ		-0.012		-0.007	
		[0.023]		[0.026]	
Exposure x Post x Lower Secondary		0.017		-0.013	
		[0.021]		[0.019]	
Post x Tertiary Educ		0.146		-0.046	
		[0.049]***		[0.031]	
Post x Diploma		0.134		-0.049	
		[0.062]**		[0.034]	
Post x Secondary Education		0.120		-0.035	
		[0.053]**		[0.030]	
Post x Lower Secondary		0.118		-0.021	
		[0.043]***		[0.028]	
Individual FEs		Yes		Yes	
2010 Dep. Var Mean		.71		.11	
Obs	4834	4834	4339	4339	4834

Notes: Reported regressions implement a DiD on different dummy variables for labor force outcomes on a measure of exposure, constructed at the level of the ISCO 2-digit occupation code x governorate, that accounts for both the share of refugees in a given location and the share of refugees that have experience working in a given occupation. Regressions control for age and include individual fixed effects. Standard errors are clustered at the unit of exposure - the ISCO 2-digit occupation x governorate. Regressions control for age and are restricted to working-aged men. Columns (1) and (2) use the outcome variable employed, which is the share of all working-aged men working. Columns (3) and (4) use unemployed, which restricts the same to only active labor force participants.

A.2 Structural Results

Table A8: Occupation-Level Informality and Wedge Estimates

	Z-Scored Informality Index	Initial Tau	Change in Tau (2014 to 2022)
Managers and Professionals	-0.96	0.69	-0.08
Technicians	-0.95	0.78	-0.25
Clerical Workers	-1.08	0.71	-0.31
Service and Sales	-0.25	0.58	-0.19
Agriculture	1.75	0.56	-0.96
Craft Trades	0.81	0.35	-0.33
Machine Operators	0.28	0.65	-0.17
Elementary Jobs	0.40	0.41	-0.30

Notes: The rows correspond to each occupation in the model, excluding unemployment as it is assumed to be frictionless (no wedge). The first column corresponds to the Z-Scored Informality Index based on the informality of Jordanian workers in an occupation prior to the introduction of work permits. Values below zero correspond to below average levels of informality. The second column corresponds to the initial (2014) wedge estimate in the model, before refugees have work permits, starting in mid-2016. The last column reports the percentage point change in the estimated wedge between the initial and final period (2022).

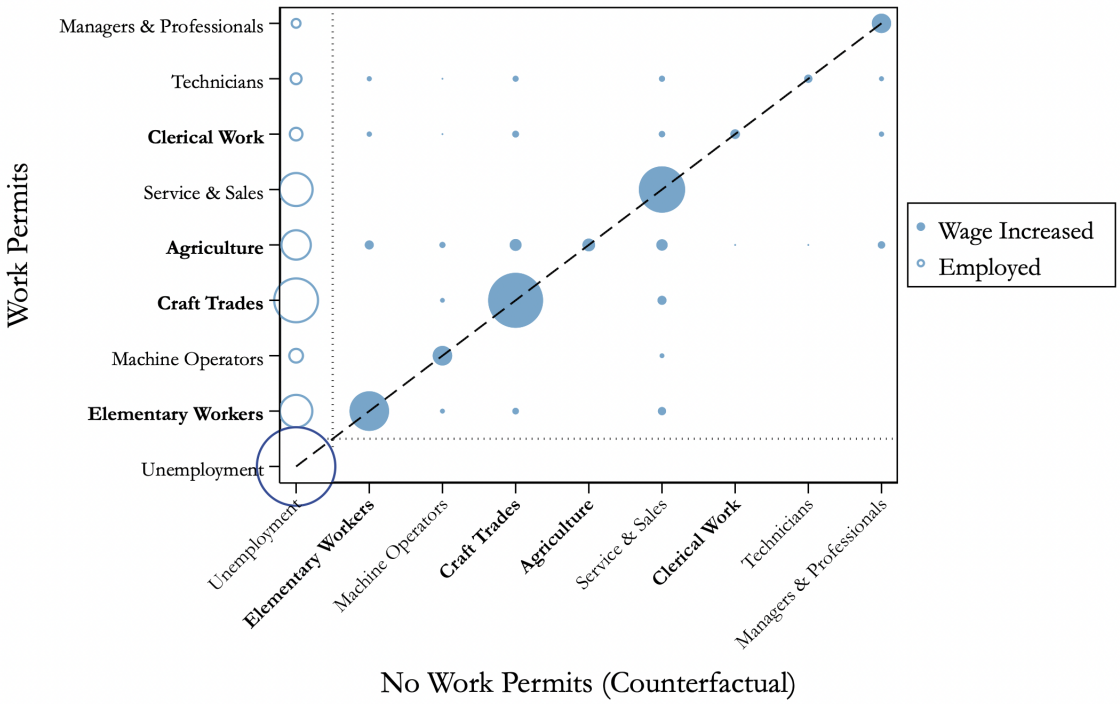
Table A9: Parameter Estimates for 2022 Data

Occupation	τ	w	α_{Syr}	α_{Jor}	H^S	H^S_{Syr}	H^S_{Jor}	A
Managers & Professionals	0.607	14.057	5.354	1.783	451.321	16.843	434.478	6.718
Technicians	0.529	10.506	5.910	2.229	156.258	8.600	147.658	2.554
Clerks	0.397	11.427	4.915	2.390	143.263	10.896	132.367	2.774
Services and Sales	0.395	19.238	5.989	2.647	514.675	54.432	460.242	11.486
Agriculture	-0.395	6.887	5.257	7.114	50.529	24.304	26.225	0.771
Craft Trades	0.022	14.577	5.013	3.946	237.659	69.313	168.346	5.148
Machine Operators	0.482	13.481	6.553	2.533	168.940	13.223	155.717	3.860
Elementary Jobs	0.111	11.914	6.026	4.162	181.430	42.706	138.723	3.323
Unemployment	0.000	12.942	5.632	4.180	523.763	101.781	421.982	N/A

$\lambda_{\text{Syr}} = 0.149$
$\lambda_{\text{Jor}} = 1.235$

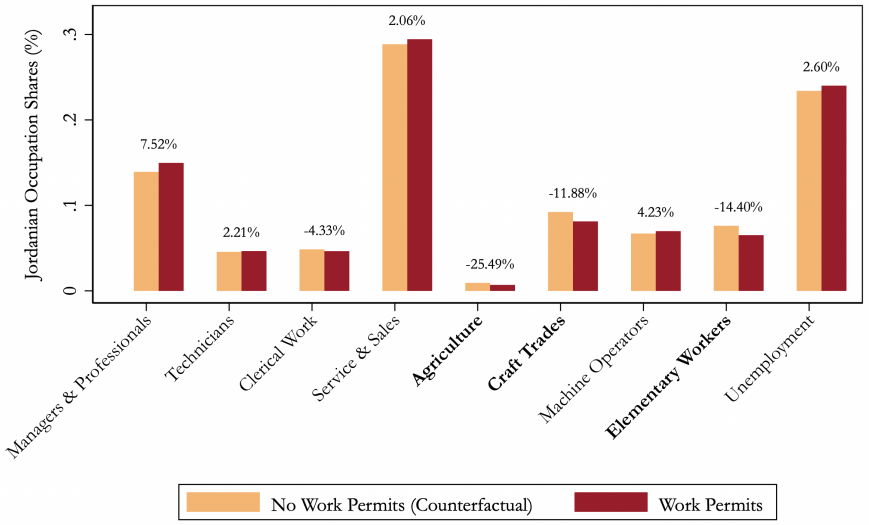
Notes: Columns correspond to parameter vectors for each occupation, denoted in rows. τ corresponds to the occupation-specific wedge faced by Syrian refugees. w is the wage per efficiency unit in each occupation. $\alpha_s yr$ and $\alpha_j o$ refer to the amenity-value each group has for each occupation. H^S refers to the total human capital units supplied to each occupation, disaggregated into the contribution by Syrians (H^S_{Syr}) and by Jordanians(H^S_{Jor}). A is the exogenous productivity of each occupation. Finally, λ correspond to the group-specific preferences for homophily, which are constant across occupations.

Figure A18: Syrian Occupation Transitions



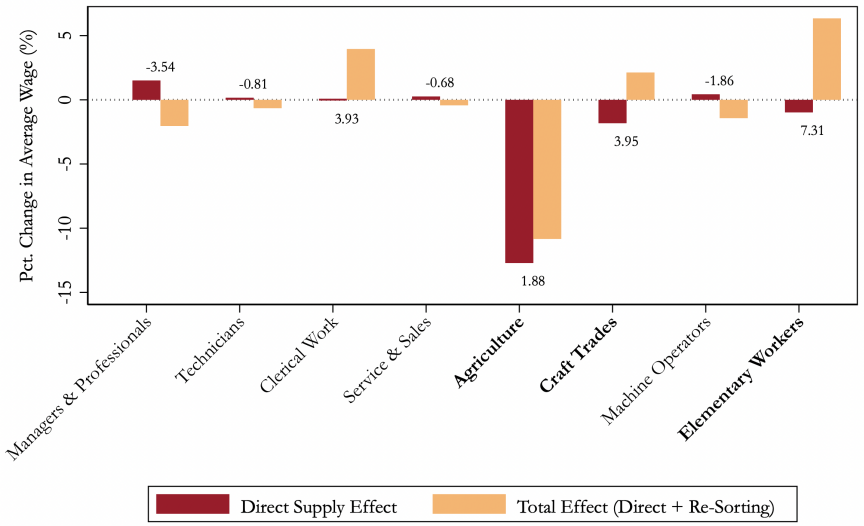
Note: The y-axis plots the occupational choice of simulated individuals after work permits and the x-axis plots the occupational choice in the counterfactual without work permits. The size of the circles correspond to the number of individuals in each transition cell. Those circles off the diagonal represent individuals who switched occupations in response to the work permit scheme. The open circles correspond to unemployment.

Figure A19: Estimated Effect of Work Permits on Jordanian Occupation Shares



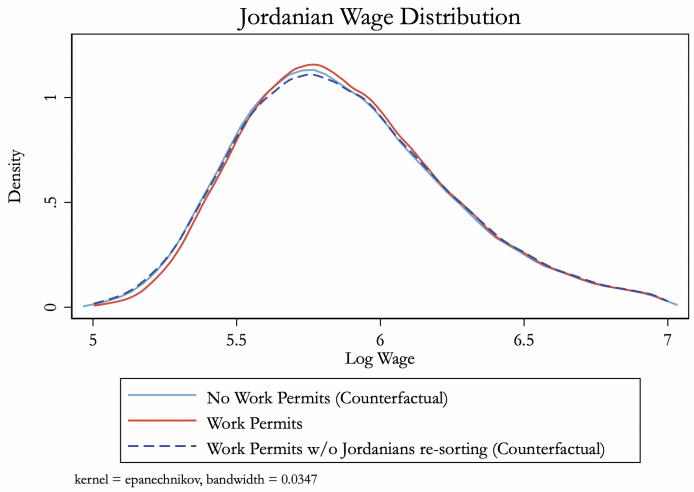
Note: The graph shows the share of Jordanians working in an occupation (or unemployment) under the counterfactual of the 2022 economy without work permits (maroon) and the observed 2022 economy with work permits. The difference between the bars corresponds to the percent change in that occupation's share without vs. with work permits. Bolded occupations correspond to those that received the largest increases of Syrian refugees from the introduction of work permits.

Figure A20: Estimated Effect of Work Permits on Jordanian Occupation-Average Wages



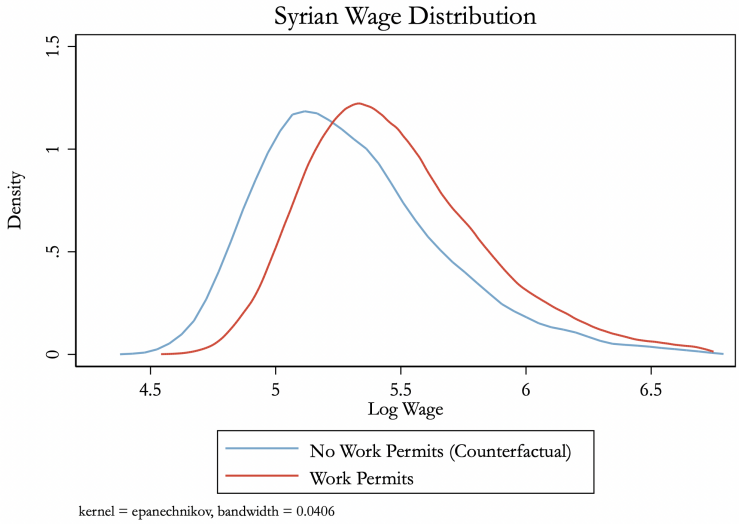
Note: The graph shows percent change in the average wage among Jordanians by occupation, accounting for only the direct supply effect of Syrian entry in response to work permits, as shown in maroon, and the total effect, which includes Jordanians' re-sorting, as shown in orange. The numbers represent the difference in the percent change between the two bars, corresponding to the change caused by the re-sorting effect alone. Bolded occupations correspond to those that received the largest increases of Syrian refugees from the introduction of work permits.

Figure A21: Simulated Wage Distributions for Jordanians



Note: The graph shows the kdensity plots of the simulated log wage distribution for Jordanians under different scenarios. The light blue corresponds to the counterfactual of the 2022 economy without work permits. The red line corresponds to the simulated version of the observed 2022 economy, where work permits exist. Finally, the dark blue dotted line represents the counterfactual of the 2022 economy with work permits, where Jordanians are not allowed to re-sort from their occupations determined in the without work permit counterfactual.

Figure A22: Simulated Wage Distributions for Syrians



Note: The graph shows the kdensity plots of the simulated log wage distribution for Jordanians under different scenarios. The light blue corresponds to the counterfactual of the 2022 economy without work permits. The red line corresponds to the simulated version of the observed 2022 economy, where work permits exist.

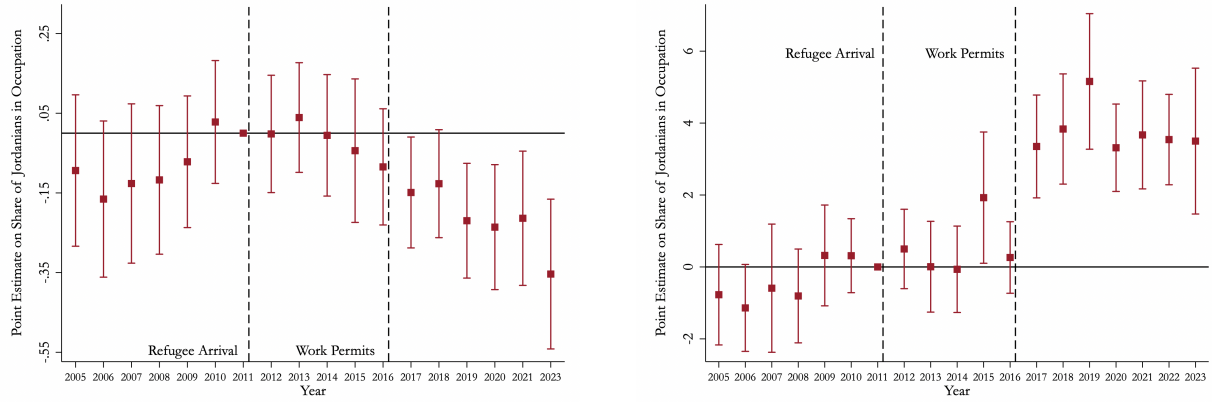
Figure A23: Simulated Effect of Re-Sorting on the Percent Change in Wages by Wage Percentile



Note: The y-axis measures the change in the wage caused by introducing work permits, measured in the local currency of Jordanian Dinars. The x-axis represents a simulated individuals' percentile in the wage distribution before work permits are introduced. The dots then correspond to the percent change in an individual Jordanian's wage, binned at the 0.1 percentile for visibility. The light blue corresponds to the change in wages under the counterfactual where Jordanians cannot re-sort after work permits are introduced while the orange dots correspond to when they can. The maroon and dark blue lines plot the best fit line resulting from a kernel-weighted local polynomial regression of the change in the wage relative on the percentile in the pre-work permit wage distribution.

B Alternative Shift-Share Specifications

Figure B1: Main Event Studies with Non-standardized Exposure Measure

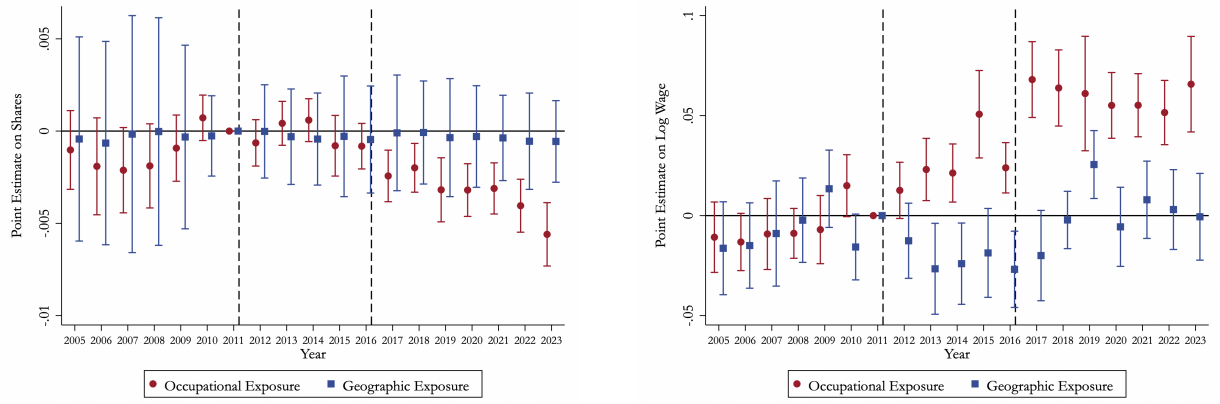


(a) Occupation Shares

(b) Occupation Log Average Wages

Note: Point estimates from regression of preferred specification for occupation shares (Panel A) and occupation log average wages (Panel B) for Jordanians on the main shift-share measure of exposure to Syrian refugees without standardization (not z-scored). Mean value of exposure is 0.39 percent – treatment effects should be scaled accordingly. Regression includes year and occupation-governorate fixed effects. Standard errors are clustered at the unit of treatment: occupation x governorate. (Observations: 6,696 occupation x governorates)

Figure B2: Decomposition of Effects into Geographic Exposure (Shift) and Occupation Exposure (Share)

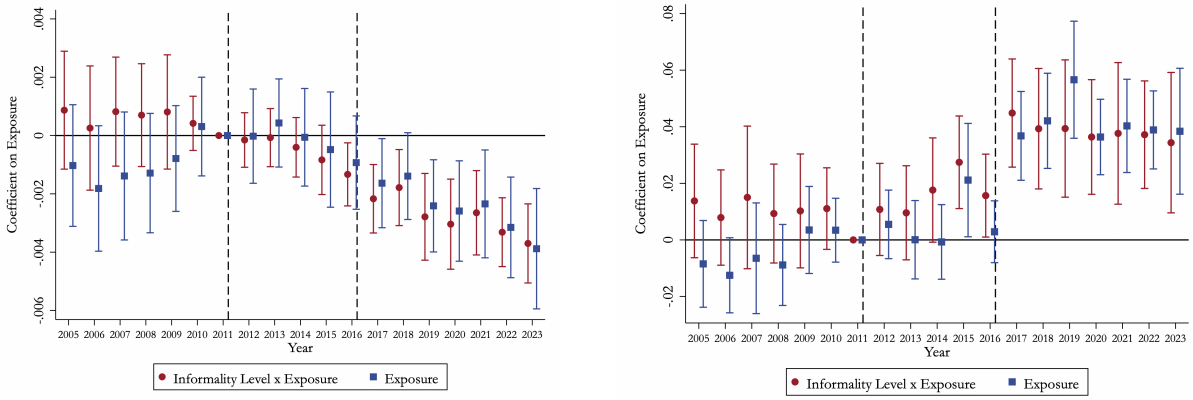


(a) Occupation Shares

(b) Occupation Log Average Wages

Note: Point estimates from regression of preferred specification for occupation shares (Panel A) and occupation log average wages (Panel B) for Jordanians on a the z-scored components of the shift-share measure of exposure to Syrian refugees. Occupation exposure is the z-score of the share of Syrians that worked in the occupation pre-displacement. Geographic Exposure is the z-score of the share of Syrians relative to the total population of the governorate. Regression includes year and occupation-governorate fixed effects. Standard errors are clustered at the unit of treatment: occupation x governorate. (Observations: 6,696 occupation x governorates)

Figure B3: Exposure Effects with Occupation-Level Informality



(a) Occupation Shares

(b) Occupation Log Average Wages

Note: Point estimates from regression of preferred specification for occupation shares (Panel A) and occupation log average wages (Panel B) for Jordanians on a the z-scored shift-share measure of exposure compare to the z-scored shift-share measure interacted with the informality index of the occupation. Regression includes year and occupation-governorate fixed effects. Standard errors are clustered at the unit of treatment: occupation x governorate (Observations: 6,696 occupation x governorates)

C Appendix: Additional Evidence of Selection in Sorting

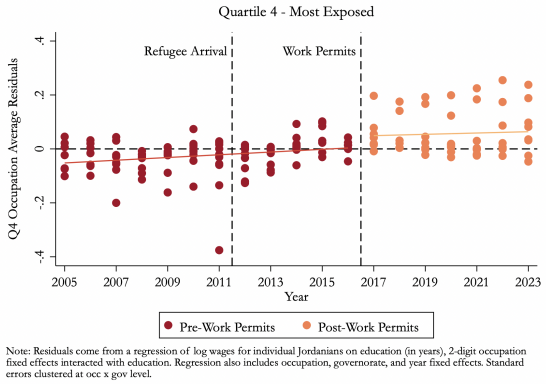
Without panel data for the majority of the study period, I cannot observe the evolution of which individuals exit from each occupation or where they reallocate. However, I can infer changes in the average ability or productivity of workers in a given occupation, using the residuals from the individual-level wage regression. Specifically, I implement the following regresion for Jordanian workers, controlling for occupation-level differences in productivity, returns to education overall and by occupation, governorate and year fixed effects,

$$Y_{i,t} = \beta_o \left[\mathbb{1}\left(\text{Occupation}_i = o\right) \times \text{Educ}_i\right] + \alpha \cdot \text{Educ}_i + \delta_o + \gamma_t + \epsilon_i, t \tag{19}$$

I then aggregate the resulting residuals from this regression to compute an occupation-level average, which proxies the average productivity level of the workers in that occupation in a given year. I then plot the evolution of these occupation average residuals over time by quartile. Figure C1a shows the evolution of these residuals for occupations in the highest quartile of exposure to Syrian refugee competition. I find that the occupation average residual increases following the introduction of work permits, consistent with a theory of Jordanians re-sorting in response to the supply shock. In particular, it should be the least productive Jordanians in these high exposure occupations that will exit, which would result in the average productivity of these occupations improving following work permits.

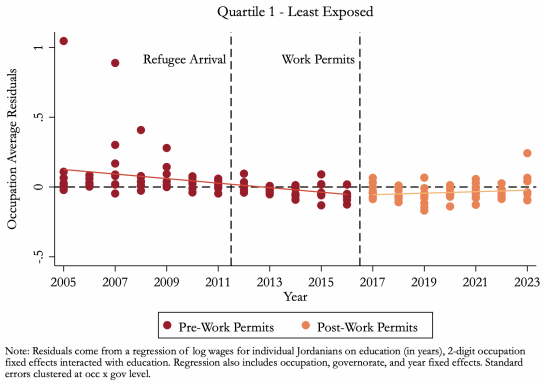
For comparison, Figure C1b plots the occupation-average residuals for the least exposed quaritle of occupations over time. If Jordanians are re-sorting to less exposed occupations, these entrants will be of lower productivity than the existing workers, thus bringing down the occupation average ability level. While the pattern is less pronounced, there is some evidence that the average residuals of the occupations in the least exposed quartile decline with the onset of work permits.

Figure C1: Occupation Average Wage Regression Residuals by Exposure Quartile



Note: Residuals come from a regression of log wages for individual Jordanians on education (in years), 2-digit occupation fixed effects interacted with education. Regression also includes occupation, governorate, and year fixed effects. Standard errors clustered at occ x gov level.

(a) Quartile 4 - Most Exposed



Note: Residuals come from a regression of log wages for individual Jordanians on education (in years), 2-digit occupation fixed effects interacted with education. Regression also includes occupation, governorate, and year fixed effects. Standard errors clustered at occ x gov level.

(b) Quartile 1 - Least Exposed

D Appendix: Detailed Description of Data

D.1 Description of Individual Data Sources

- **ERF EUS 2005-2016, excluding 2015:** This dataset comprises the old version of the Employment and Unemployment Survey that is run annually by DoS, made publicly available through a data sharing agreement with the Economic Research Forum in Cairo, Egypt. The survey is only representative for Jordanians and is implemented through a stratified sampling procedure using census data of households across the country. Because 2015 was the census year, that round was not shared with ERF. See OAMDI (2017) for description and access to data.

- **DoS EUS 2015:** A random 25% sample of the 2015 round of the Employment and Unemployment Survey was provided by DoS for use in this project. Given it was not shared with ERF, raising concerns of data quality, I take estimates from this year with a degree of caution. Similar to the other waves pre-2017, the data is only representative for Jordanians. See Department of Statistics (2024) for publicly available documentation and summary tables.

- **DoS EUS 2017-2023:** After Syrian refugees gained a legal pathway to work, the DoS Employment and Unemployment survey became representative for both Jordanians and Syrian refugees. The data is not public but DoS has given me access to a 25% random sample for the years of 2017 to 2023. Due to the COVID-19 pandemic, DoS conducted smaller-scale surveys in 2020 and 2021, leaving the resulting sub-sample provided under-powered for Syrian refugees. See Department of Statistics (2024) for publicly available documentation and summary tables.

- **ILO/Fafo 2014:** This dataset was originally collected by the International Labor Organization (ILO) in conjunction with the Fafo Institute in early 2014. Fafo has provided an anonymized version of the dataset for this project, although it is not publicly available. The dataset is representative for Syrian refugees and Jordanians living in Amman, Mafraq, and Irbid, which are the three governorates with the largest Syrian refugee populations. See Stave and Hillesund (2015) for the corresponding publicly available report.

- **JLMPS 2016:** The Jordan Labor Market Panel Survey 2016 round was collected by a team of researchers at ERF in conjunction with the Jordanian Department of Statistics (DoS) in early 2017. This data is publicly available on the ERF's data portal upon application. The dataset is representative for Syrian refugees and Jordanians across the entirety of Jordan. The sample for Syrian refugees is quite small once restricted to working-aged individuals (around 300 observations) so it is only used for the Jordanian dataset. See OAMDI (2018) for description and access to data.

- **Fafo/DoS 2018:** This dataset was originally collected by the Fafo Institute in conjunction with DoS in late 2017. Fafo has provided this dataset to me, although it is not publicly

available. The dataset only surveys Syrian refugees but is representative for all of Jordan. See Tiltnes et al. (2019) for the corresponding publicly available report.

D.2 Harmonization of Key Variables

To ensure measures are consistent across different datasets, I construct all variables using common survey questions found in all surveys.

Labor Force Status:

- **Working-aged:** An individual is working-aged if they are between the ages of 18-60 at the time of the survey, consistent with Jordan's minimum legal age for full time employment (18 years old) and the retirement age (60 years old).
- **Employed:** An individual is defined as employed if they answer that they have worked for at least one hour in the past 7 days.
- **Unemployed:** An individual is unemployed if they answer that they (i) want to be working and (ii) are actively looking for work, as defined as engaging in some job search activity in the past 30 days.
- **Labor Force Participant:** An individual who is either employed or unemployed and is of working-age.

Informality: Informality is defined along four dimensions in line with the ILO definition of informality – a worker is informal if they work for their own-account, work without a written contract, work on a temporary contract, or are not covered by the social security contribution scheme. Each of these components are elicited through survey questions in all surveys with the exception of DoS EUS pre-2016 and for some metrics after the 2020 adjustment for COVID questions. I take a more nuanced approach to measure inforamlity and construct an inverse-covariance weighted index of the four components of the ILO definition of informality to create a continuous measure of informality at the worker-level. Occupation-level informality is then the average index-value across Jordanian workers in a given occupation in a given year.

Wages: Wages are harmonized to be measured at the monthly-level consistent with Jordan's minimum wage, which is defined in terms of monthly wages. In the surveys, individuals are asked to report their last wage payment and the intervals (hourly, daily, weekly, monthly, etc.) that the payments are issued. If the wage is reported at the monthly level, I make no adjustments. If the wage is reported for a shorter interval of time, I scale the reported wage by the relevant hours worked (using reported hours worked each day or week) to construct a monthly measure. Final wages are winsorized at the 99th percentile. From 2005-2016, the Jordanian Department of Statistics' EUS survey collected monthly wage data at binned intervals. I assign individuals to the mean value of their corresponding wage bin for these years. I can benchmark this binning

against 2010 JLMPS and 2014 ILO/Fafo data, which have continuous measures of wages, and the distribution is consistent.

Occupation Average Wages: For harmonization in the event study, I assign individuals with continuous wages to corresponding wage bins to match the binned data. I then take the average value across individuals to construct the average wage measure at the unit of occupation x governorate x year. I find the results of the main event study are robust to using unbinned individual wage data where available, in the construction of this average.

Occupations: Occupations are defined using the ILO's ISCO 08 2-digit occupation codes. For the DoS EUS 2005-2009 data, occupations were categorized using the ISCO 88 codes. To harmonize these years' codes to the ISCO 08 codes, I use the crosswalk provided by ILO entitled "Correspondence ISCO-08 to ISCO-88" available here. However, the crosswalk is defined at the 4-digit level and my data is available only at the 3-digit level, requiring manual matching based on corresponding descriptions of occupations and work functions that fall under the given occupation code.

D.3 Summary Statistics for Final Harmonized Dataset

Table D1: Number of Observations by Year

	Data Sources		Observations	
Year	**Jordanians**	**Syrians**	**Jordanians**	**Syrians**
2005	ERF EUS		51,349	
2006	ERF EUS		50,593	
2007	ERF EUS		63,288	
2008	ERF EUS		62,271	
2009	ERF EUS		60,513	
2010	ERF EUS; JLMPS		65,637	
2011	ERF EUS	ILO/Fafo *Recall*	60,898	2,560
2012	ERF EUS		58,292	
2013	ERF EUS		54,551	
2014	ERF EUS; ILO/Fafo	ILO/Fafo	55,122	2,560
2015	DoS EUS		13,037	
2016	ERF EUS		53,338	
2017	DoS EUS; JLMPS	DoS EUS; Fafo/DoS	8,909	
2018	DoS EUS	DoS EUS	66,220	3,551
2019	DoS EUS	DoS EUS	15,883	803
2020	DoS EUS	DoS EUS	54,460	2,447
2021	DoS EUS	DoS EUS	64,151	3,220
2022	DoS EUS	DoS EUS	62,111	3,397
2023	DoS EUS		14,609	

Notes: The table above documents the number of working-aged males in each year by population group - Jordanians and Syrian refugees. The sources correspond to the data sources described in Appendix Subsection D.1. (Jordanian Obs: 950,111. Syrian Obs: 27,447)

Table D2: Jordanian Labor Market Summary Statistics (2005-2014)

	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Total
Observations	51349	50593	63288	62271	60513	65637	60898	58292	54551	55122	58659.9
	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(4819.7)
Age	32.54	32.57	32.86	32.96	33.03	33.22	33.38	33.60	33.85	34.00	33.20
	(11.46)	(11.51)	(11.36)	(11.35)	(11.32)	(11.37)	(11.37)	(11.50)	(11.59)	(11.65)	(11.45)
Urban Area	0.674	0.665	0.724	0.730	0.725	0.722	0.720	0.722	0.719	0.714	0.713
	(0.469)	(0.472)	(0.447)	(0.444)	(0.447)	(0.448)	(0.449)	(0.448)	(0.449)	(0.452)	(0.452)
LFP Rate	0.785	0.781	0.785	0.784	0.800	0.785	0.776	0.762	0.752	0.748	0.776
	(0.411)	(0.413)	(0.411)	(0.411)	(0.400)	(0.411)	(0.417)	(0.426)	(0.432)	(0.434)	(0.417)
Unemployment Rate	0.159	0.151	0.123	0.118	0.120	0.117	0.129	0.126	0.126	0.133	0.129
	(0.366)	(0.358)	(0.328)	(0.323)	(0.325)	(0.322)	(0.335)	(0.332)	(0.332)	(0.340)	(0.335)
Employment Rate	0.660	0.663	0.689	0.692	0.704	0.693	0.676	0.666	0.657	0.648	0.676
	(0.474)	(0.473)	(0.463)	(0.462)	(0.457)	(0.461)	(0.468)	(0.472)	(0.475)	(0.478)	(0.468)
Education (in Years)	10.81	10.87	11.16	11.20	11.18	11.27	11.28	11.31	11.30	11.28	11.17
	(3.709)	(3.720)	(3.647)	(3.568)	(3.566)	(3.559)	(3.519)	(3.479)	(3.423)	(3.372)	(3.560)
Current Student	0.104	0.105	0.107	0.109	0.105	0.107	0.106	0.107	0.110	0.111	0.107
	(0.305)	(0.306)	(0.310)	(0.312)	(0.306)	(0.310)	(0.307)	(0.309)	(0.313)	(0.314)	(0.309)
Monthly Wage	198.1	204.3	227.1	267.4	293.6	306.8	323.7	343.5	350.9	362.5	288.5
	(117.4)	(120.1)	(134.6)	(141.9)	(146.8)	(145.3)	(141.9)	(142.4)	(138.9)	(139.2)	(148.7)
Hours Per Week	45.76	44.64	47.11	43.66	44.43	40.36	41.99	41.67	42.24	42.61	43.42
	(14.59)	(14.40)	(13.23)	(12.47)	(13.68)	(16.10)	(13.54)	(11.89)	(11.63)	(11.18)	(13.54)

Notes: Columns correspond to years in the data with total being the average for the entire period. Rows correspond to variables. Means of each variable for each year are reported in each cell with standard deviations in parentheses below. Observations corresponds to the number of working-aged men in the sample for a given year, which does not have a corresponding standard deviation. Urban area corresponds to individual living in an urban area within Jordan. LFP rate measures the labor force participation rate of working-aged men. Unemployment rate is computed by the share of labor force participants not employed. Employment rate is the *effective* employment rate of all working-aged men, including non-labor force participants. Education is measured in years. Current student is the share of working-aged men that are enrolled as full-time students. Monthly wage is measured in Jordanian dinar. Hours per week is the average number of hours that employed individuals work in a given 7 day period. (Observations: 582,514)

Table D3: Jordanian Labor Market Summary Statistics (2015-2023)

	2015	2016	2017	2018	2019	2020	2021	2022	2023	Total
Observations	13037	53338	23788	66220	15883	54460	64151	62111	14609	52695.1
	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(17245.6)
Age	34.28	34.25	34.16	34.54	34.72	34.99	34.96	35.01	35.12	34.71
	(11.84)	(12.05)	(11.87)	(12.19)	(12.21)	(12.29)	(12.36)	(12.37)	(12.36)	(12.22)
Urban Area	0.716	0.711	0.769	0.765	0.761	0.763	0.760	0.753	0.746	0.752
	(0.451)	(0.453)	(0.421)	(0.424)	(0.426)	(0.426)	(0.427)	(0.431)	(0.435)	(0.432)
LFP Rate	0.727	0.745	0.724	0.697	0.680	0.674	0.680	0.674	0.677	0.695
	(0.446)	(0.436)	(0.447)	(0.460)	(0.466)	(0.469)	(0.466)	(0.469)	(0.467)	(0.460)
Unemployment Rate	0.121	0.147	0.146	0.175	0.175	0.207	0.224	0.212	0.201	0.187
	(0.326)	(0.355)	(0.353)	(0.380)	(0.380)	(0.405)	(0.417)	(0.408)	(0.400)	(0.390)
Employment Rate	0.638	0.635	0.612	0.575	0.561	0.535	0.528	0.531	0.542	0.565
	(0.481)	(0.481)	(0.487)	(0.494)	(0.496)	(0.499)	(0.499)	(0.499)	(0.498)	(0.496)
Education (in Years)	11.47	11.29	11.54	11.80	11.78	11.79	11.79	11.80	11.90	11.70
	(2.997)	(3.323)	(3.335)	(3.006)	(2.932)	(2.958)	(2.985)	(2.940)	(2.939)	(3.055)
Current Student	0.104	0.100	0.0855	0.0945	0.0864	0.0945	0.101	0.105	0.108	0.0981
	(0.305)	(0.300)	(0.280)	(0.292)	(0.281)	(0.293)	(0.301)	(0.306)	(0.311)	(0.297)
Monthly Wage	368.7	377.5	387.1	374.5	371.6	376.1	379.6	383.8	395.8	378.8
	(142.3)	(136.4)	(152.6)	(132.9)	(126.2)	(129.0)	(132.0)	(131.3)	(138.9)	(134.5)
Hours Per Week	43.67	42.25	35.27	44.81	44.17	42.97	44.29	44.69	44.71	43.31
	(8.833)	(10.02)	(19.14)	(9.481)	(9.291)	(9.768)	(8.490)	(8.633)	(7.859)	(10.47)

Notes: Columns correspond to years in the data with total being the average for the entire period. Rows correspond to variables. Means of each variable for each year are reported in each cell with standard deviations in parentheses below. Observations corresponds to the number of working-aged men in the sample for a given year, which does not have a corresponding standard deviation. Urban area corresponds to individual living in an urban area within Jordan. LFP rate measures the labor force participation rate of working-aged men. Unemployment rate is computed by the share of labor force participants not employed. Employment rate is the *effective* employment rate of all working-aged men, including non-labor force participants. Education is measured in years. Current student is the share of working-aged men that are enrolled as full-time students. Monthly wage is measured in Jordanian dinar. Hours per week is the average number of hours that employed individuals work in a given 7 day period. (Observations: 27,447)

Table D4: Syrian Refugee Labor Market Summary Statistics (2011, 2014, 2017-2022)

	2011	2014	2017	2018	2019	2020	2021	2022	Total
Observations	2560	2560	8909	3551	803	2447	3220	3397	4868.6
	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(2848.0)
Age	32.75	32.75	33.06	34.11	33.74	34.16	34.18	34.21	33.53
	(10.71)	(10.71)	(11.16)	(11.05)	(11.30)	(11.48)	(11.64)	(11.76)	(11.25)
Urban Area	.	0.804	0.906	0.904	0.924	0.925	0.932	0.918	0.905
	(.)	(0.397)	(0.293)	(0.295)	(0.265)	(0.264)	(0.251)	(0.275)	(0.293)
LFP Rate	0.745	0.671	0.682	0.735	0.727	0.728	0.753	0.758	0.717
	(0.436)	(0.470)	(0.466)	(0.441)	(0.446)	(0.445)	(0.431)	(0.429)	(0.451)
Unemployment Rate	0.274	0.631	0.256	0.321	0.353	0.343	0.321	0.310	0.320
	(0.446)	(0.483)	(0.437)	(0.467)	(0.478)	(0.475)	(0.467)	(0.462)	(0.467)
Employment Rate	0.518	0.207	0.507	0.499	0.471	0.478	0.511	0.523	0.478
	(0.500)	(0.405)	(0.500)	(0.500)	(0.499)	(0.500)	(0.500)	(0.500)	(0.500)
Education (in Years)	.	8.361	8.041	8.495	8.372	8.500	8.640	8.581	8.351
	(.)	(3.223)	(3.065)	(2.907)	(2.854)	(2.944)	(2.974)	(3.026)	(3.032)
Current Student	.	0.0141	0.0274	0.0225	0.0374	0.0343	0.0438	0.0445	0.0310
	(.)	(0.118)	(0.163)	(0.148)	(0.190)	(0.182)	(0.205)	(0.206)	(0.173)
Monthly Wage	.	196.2	224.7	254.9	237.9	262.5	258.3	276.1	243.3
	(.)	(133.0)	(231.9)	(164.2)	(103.7)	(206.5)	(108.2)	(291.2)	(212.5)
Hours Per Week	.	49.53	38.92	44.03	44.12	43.96	44.83	45.41	42.61
	(.)	(23.08)	(25.03)	(12.98)	(11.84)	(12.96)	(11.52)	(11.74)	(18.99)

Notes: Columns correspond to years in the data with total being the average for the entire period. Rows correspond to variables. Means of each variable for each year are reported in each cell with standard deviations in parentheses below. Observations corresponds to the number of working-aged men in the sample for a given year, which does not have a corresponding standard deviation. Urban area corresponds to individual living in an urban area within Jordan. LFP rate measures the labor force participation rate of working-aged men. Unemployment rate is computed by the share of labor force participants not employed. Employment rate is the *effective* employment rate of all working-aged men, including non-labor force participants. Education is measured in years. Current student is the share of working-aged men that are enrolled as full-time students. Monthly wage is measured in Jordanian dinar. Hours per week is the average number of hours that employed individuals work in a given 7 day period. 2011 recall data has limited variables, explaining missing values. (Observations: 367,597)

E Appendix: Model Derivations and Proofs

E.1 Probability of Choosing an Occupation

Since all individuals within an occupation draw from the same distribution of ϵ's, I drop the individual subscript i for clarity in the remainder of the derivation of this model. The probability that an individual in group g chooses occupation o can be defined as $p_{o,g}$ such that:

$$\begin{aligned} p_{o,g} &= \mathbb{P}\left(\tilde{w}_{o,g}\epsilon_o > \tilde{w}_{k,g}\epsilon_k\right) \quad \forall k \neq o \\ &= \mathbb{P}\left(\epsilon_k < \frac{\tilde{w}_{o,g}\epsilon_o}{\tilde{w}_{k,g}}\right) \quad \forall k \neq o \\ &= \int f_{o,g}(\alpha_1\epsilon, ..., \alpha_m\epsilon)d\epsilon \end{aligned} \tag{20}$$

where $\alpha_m = \frac{\tilde{w}_{o,g}}{\tilde{w}_{m,g}}$. From the properties of the Frechet distribution, $f_{o,g}$ can be derived as the derivative of the joint CDF of F_g with respect to its k-th argument:

$$F_g(\epsilon_1, ..., \epsilon_m) = \exp\left(\sum_{j=1}^{m} \epsilon_j^{-\theta}\right) \tag{21}$$

$$\Rightarrow \frac{\partial F_g}{\partial \epsilon_o} = f_{o,g}(\alpha_1\epsilon, ..., \alpha_m\epsilon) = \theta\epsilon^{-\theta-1}\exp\left(\sum_{k=1}^{m} \alpha_k^{-\theta}\epsilon^{-\theta}\right) \tag{22}$$

Evaluating the integral in Equation (10)

$$\begin{aligned} p_{o,g} &= \int \theta\epsilon^{-\theta-1}\exp\left(\sum_{k=1}^{m} \alpha_k^{-\theta}\epsilon^{-\theta}\right) \\ &= \frac{1}{\sum_k \alpha_k^{-\theta}} \int \sum_k \alpha_k^{-\theta}\theta\epsilon^{-\theta-1}\exp\left(\sum_{k=1}^{m} \alpha_k^{-\theta}\epsilon^{-\theta}\right) \\ &= \frac{1}{\sum_k \alpha_k^{-\theta}} \int dF_g(\epsilon) \end{aligned}$$

Recalling that $\alpha_m = \frac{\tilde{w}_{o,g}}{\tilde{w}_{m,g}}$, I get a final expression for the probability of an individual in group g choosing occupation o.

$$\therefore p_{o,g} = \frac{\tilde{w}_{o,g}^{\theta}}{\sum_{k=1}^{m} \tilde{w}_{k,g}^{\theta}} \quad \text{where} \quad \tilde{w}_{o,g} = z_o(1-\tau_{o,g})w_o\phi_o(1-s)s^{\phi_o-1}$$

E.2 Occupation-Specific Average Quality

This appendix presents the full derivation to arrive at the expression for average worker quality by occupation in the model. Beginning from the human capital accumulation equation, the

efficiency units of labor for a given individual in group g in occupation o can be defined as:

$$h_{o,g}\epsilon_o = s^{\phi_o}\epsilon_o$$

The average quality of workers (including human capital and talent) when workers sort based on their ability draw ϵ_o is then:

$$e^{\mathbb{E}[ln(h_{o,g}\epsilon_o)|\text{choose } o]} = s^{\phi_o} e^{\mathbb{E}[ln(\epsilon_o)|\text{choose } o]}$$

Let ϵ^* denote the ability of an individual in their chosen occupation. Post-distortion income from labor can then be defined as:

$$W^* = \max_o W_o = \max_o \tilde{w}_o\epsilon_o = \tilde{w}_o^*\epsilon_o^*$$

As individuals choose their occupation to maximize W_o and thus it inherits the extreme value distribution of ϵ_o

$$\begin{aligned}
\mathbb{P}(Y^* < x) &= \mathbb{P}(W_o < x) \quad \forall o \\
&= \mathbb{P}(\epsilon_o < \frac{x}{\tilde{w}_o}) \quad \forall o \\
&= F(\frac{x}{\tilde{w}_1}, ..., \frac{z}{\tilde{w}_m}) \\
&= \exp\left(-\sum_s \tilde{w}_s^\theta x^{-\theta} \right) \\
\Rightarrow \mathbb{P}(\epsilon^* < v) &= \mathbb{P}(\frac{Y^*}{\tilde{w}^*} < v) \\
&= \mathbb{P}(Y^* < v\tilde{w}^*) \\
&= \exp\left(-\sum_s \tilde{w}_s^\theta (v\tilde{w}^*)^{-\theta} \right) \\
&= \exp\left(-\sum_s \left(\frac{\tilde{w}_s}{\tilde{w}^*}\right)^\theta v^{-\theta} \right)
\end{aligned}$$

Then the CDF of ϵ^*, the ability of workers in their observed (chosen) occupation, is:

$$\begin{aligned}
G(v) = \mathbb{P}(\epsilon^* < v) &= \exp\left(-\sum_s \left(\frac{\tilde{w}_s}{\tilde{w}^*}\right)^\theta v^{-\theta} \right) \\
&= \exp\left(\frac{1}{p^*} v^{-\theta}\right) \quad \text{where} \quad p^* = \sum_s \left(\frac{\tilde{w}^*}{\tilde{w}_s}\right)^\theta
\end{aligned}$$

Integrating over the PDF of ϵ^* to get the expected value of ϵ^*

$$\begin{aligned}\mathbb{E}[\epsilon^*] &= \int \epsilon^* dG(\epsilon^*) = \int \epsilon^* g(\epsilon^*) d\epsilon^* \\ &= \int \theta\left(\frac{1}{p^*}\right)\epsilon^{*\,-\theta} \cdot \exp\left[\left(\frac{-1}{p^*}\right)\epsilon^{*\,-\theta}\right]\end{aligned}$$

Applying a Change of Variables where $x = \left(\frac{-1}{p^*}\right)\epsilon^{*\,-\theta}$, we can rewrite the above integral as:

$$\mathbb{E}[\epsilon^*] = \left(\frac{-1}{p^*}\right)^{\frac{1}{\theta}} \int x^{\frac{-1}{\theta}} e^{-x} dx$$

Recalling the Gamma Function: $\Gamma(\alpha) = \int x^{\alpha-1} e^{-x} dx$

$$\begin{aligned}\mathbb{E}[\epsilon^*] &= \left(\frac{1}{p^*}\right)^{\frac{1}{\theta}} \Gamma(1 - \frac{1}{\theta}) \\ \therefore \mathbb{E}[\epsilon_i | \text{choose } i] &= \left(\frac{1}{p_{i,g}}\right)^{\frac{1}{\theta}} \Gamma(1 - \frac{1}{\theta})\end{aligned}$$

Using properties of the Frechet distribution, if $x \sim Frechet(\theta)$, then it follows that $ln(x) \sim Gumbel(\frac{1}{\theta})$ and $\mathbb{E}[ln(x)] = \frac{\gamma_{em}}{\theta} \equiv \tilde{\Gamma}$ where γ_{em} is the Euler-Mascheroni constant. Applying this to Equation (15), I get the geometric mean ability for individuals given that occupational sorting is done based on one's ability draw.

$$e^{\mathbb{E}[ln(h_{o,g}\epsilon_o)|\text{choose } o]} = s^{\phi_o}\left[\left(\frac{1}{p_{o,g}}\right)^{\frac{1}{\theta}} \tilde{\Gamma}\right]$$

This concludes the derivation of the average worker quality in a given occupation.

E.3 Homophily as an amplifier of re-sorting

Proposition 1 (Homophily amplifies re-sorting)**.** *Suppose the aggregate production function is* $Y = [\sum_o (A_o H_o)^\rho]^{1/\rho}$ *with* $0 < \rho < 1$*, and that* $\frac{\partial H_o}{\partial p_{o,S}} > 0$*. Then, for any occupation* o *that is not the only one contributing to aggregate output,*

$$\frac{\partial w_o}{\partial p_{o,S}} < 0 \quad \textit{and} \quad \frac{\partial z_{o,J}}{\partial p_{o,S}} < 0.$$

Consequently, the utility of Jordanians in occupation o *satisfies*

$$\frac{\partial U_{o,i,J}}{\partial p_{o,S}} = \left(\frac{\partial z_{o,J}}{\partial p_{o,S}} w_o + z_{o,J} \frac{\partial w_o}{\partial p_{o,S}}\right) s_{i,o}^{\phi_o} \epsilon_{i,o,J} < 0,$$

and the magnitude of this decline is strictly greater than in the case without homophily ($\lambda_J = 0$*). Homophily thus amplifies re-sorting by deepening the utility loss for Jordanians in high-exposure*

occupations.

Here I provide a proof that homophily acts as an amplifier of re-sorting by exacerbating the decline in utility beyond the case of wage-based sorting with exogenous preferences. I show that an increase in the share of Syrians in an occupation ($p_{o,S}$) pushes down both (i) the wage per efficiency unit in that occupation and (ii) the non-wage value of working in that occupation for Jordanians.

I begin with the utility function for an individual i from group g in occupation o:

$$U_{o,i,g} = \underbrace{\alpha_{o,g}\left(\frac{N_J p_{o,J}}{\sum_g N_g p_{o,g}}\right)^{\lambda_g}}_{\equiv z_{o,g}} \underbrace{w_o s_i^{\phi_o} \epsilon_{i,o,g}}_{\equiv \text{wage}_{i,o,g}}, \qquad g \in \{S, J\}.$$

Now imagine that Syrian refugees enter occupation o, such that the occupation share of Syrian refugees $p_{o,S}$ increases. The derivative of Jordanians' utility with respect to this change is:

$$\begin{aligned}\frac{\partial U_{o,i,J}}{\partial p_{o,S}} &= \left(\frac{\partial z_{o,J}}{\partial p_{o,S}}\right) w_o s_i^{\phi_o} \epsilon_{i,o,J} + z_{o,J}\left(\frac{\partial w_o}{\partial p_{o,S}}\right) s_i^{\phi_o} \epsilon_{i,o,J} \\ &= \left[\left(\frac{\partial z_{o,J}}{\partial p_{o,S}}\right) w_o + \left(\frac{\partial w_o}{\partial p_{o,S}}\right) z_{o,J}\right]\left(s_i^{\phi_o} \epsilon_{i,o,J}\right).\end{aligned}$$

Consistent with wages falling as total human capital in an occupation increases, the wage per efficiency unit should be decreasing with respect to the share of Syrians in that occupation. Recall that the wage per efficiency unit follows from the firm's first order condition under the aggregate production function

$$Y = \left[\sum_o (A_o H_o)^\rho\right]^{\frac{1}{\rho}}.$$

Taking the derivative of the first-order condition for occupation o,

$$w_o = A_o^\rho H_o^{\rho-1} Y^{1-\rho},$$

with respect to $p_{o,S}$ gives:

$$\frac{\partial w_o}{\partial p_{o,S}} = w_o\left[(\rho-1)\frac{1}{H_o}\frac{\partial H_o}{\partial p_{o,S}} + (1-\rho)\frac{1}{Y}\frac{\partial Y}{\partial p_{o,S}}\right].$$

Because $\frac{\partial Y}{\partial p_{o,S}} = \frac{\partial Y}{\partial H_o}\frac{\partial H_o}{\partial p_{o,S}} = w_o \frac{\partial H_o}{\partial p_{o,S}}$, we can simplify to:

$$\frac{\partial w_o}{\partial p_{o,S}} = \frac{\partial H_o}{\partial p_{o,S}} w_o(1-\rho)\left(\frac{w_o}{Y} - \frac{1}{H_o}\right).$$

The partial derivative of human capital in occupation o with respect to the share of Syrians in occupation o is positive, since adding more members of either group raises total efficiency units of

labor:

$$\frac{\partial H_o}{\partial p_{o,S}} > 0.$$

Hence, the sign of $\frac{\partial w_o}{\partial p_{o,S}}$ depends on the term in parentheses:

$$\text{sign}\left(\frac{\partial w_o}{\partial p_{o,S}}\right) = \text{sign}\left(\frac{w_o}{Y} - \frac{1}{H_o}\right).$$

Using $w_o = A_o^\rho H_o^{\rho-1} Y^{1-\rho}$, we have

$$\frac{w_o}{Y} - \frac{1}{H_o} = \frac{A_o^\rho H_o^{\rho-1}}{Y^\rho} - \frac{1}{H_o} = \frac{1}{H_o}\left[\frac{(A_o H_o)^\rho}{Y^\rho} - 1\right].$$

Since $Y^\rho = \sum_k (A_k H_k)^\rho$, we know that $(A_o H_o)^\rho < Y^\rho$ whenever occupation o is not the only one in production. Thus,

$$\frac{\partial w_o}{\partial p_{o,S}} = \frac{\partial H_o}{\partial p_{o,S}}\, w_o(1-\rho)\left(\frac{w_o}{Y} - \frac{1}{H_o}\right) < 0.$$

Next, to determine whether this effect is amplified or attenuated by homophily, consider the derivative of the preference term with respect to the share of Syrians in the occupation:

$$\begin{aligned}\frac{\partial z_{o,J}}{\partial p_{o,S}} &= \alpha_{o,J}\lambda_J \left(\frac{N_J p_{o,J}}{\sum_g N_g p_{o,g}}\right)^{\lambda_J - 1} \frac{\partial}{\partial p_{o,S}}\left(\frac{N_J p_{o,J}}{\sum_g N_g p_{o,g}}\right) \\ &= -\, z_{o,J}\, \lambda_J\, \frac{N_S p_{o,S}}{(\sum_g N_g p_{o,g}) p_{o,J}} < 0.\end{aligned}$$

Thus, an increase in the share of Syrians in occupation o reduces both the Jordanian preference value for that occupation and the wage rate in that occupation, negatively affecting Jordanian utility:

$$\frac{\partial U_{o,i,J}}{\partial p_{o,S}} = \left(\frac{\partial z_{o,J}}{\partial p_{o,S}} w_o + z_{o,J}\frac{\partial w_o}{\partial p_{o,S}}\right) s_{i,o}^{\phi_o} \epsilon_{i,o,J} \quad \Rightarrow \quad \frac{\partial U_{o,i,J}}{\partial p_{o,S}} < 0.$$

Because homophily introduces an additional negative channel via $\frac{\partial z_{o,J}}{\partial p_{o,S}} < 0$, the overall decline in utility is strictly larger than in a model without homophily. Homophily therefore acts as an amplifier of re-sorting.

E.4 Demand-Side Wage Effects from Labor Supply Shocks

This appendix derives the equilibrium effects on the wage per efficiency unit across occupations in response to a labor supply shock.

E.4.1 Wage Effect from Direct Supply Shock

I prove here that the labor demand is downward sloping – an increase in the labor supply to an occupation will result in a decrease in its wage per efficiency unit. Here, I derive the derivative of the wage per efficiency unit w_o with respect to total human capital H_o in occupation o, under the CES production function. The firm's first-order condition implies that the wage per efficiency unit in occupation o is

$$w_o = A_o^\rho H_o^{\rho-1} Y^{1-\rho}.$$

Taking the derivative of w_o with respect to H_o and using the fact that Y itself depends on H_o gives

$$\frac{\partial w_o}{\partial H_o} = A_o^\rho(\rho-1)H_o^{\rho-2}Y^{1-\rho} + A_o^\rho H_o^{\rho-1}(1-\rho)Y^{-\rho}\frac{dY}{dH_o}.$$

From the production function, $\frac{dY}{dH_o} = w_o$. Substituting this into the expression above yields:

$$\frac{\partial w_o}{\partial H_o} = A_o^\rho H_o^{\rho-2}Y^{1-\rho}(1-\rho)\left(\frac{(A_oH_o)^\rho}{Y^\rho} - 1\right).$$

Equivalently, using $w_o = A_o^\rho H_o^{\rho-1}Y^{1-\rho}$, this can be written as:

$$\frac{\partial w_o}{\partial H_o} = w_o(1-\rho)\left(\frac{w_o}{Y} - \frac{1}{H_o}\right) = w_o(1-\rho)\frac{1}{H_o}\left[\frac{(A_oH_o)^\rho}{Y^\rho} - 1\right].$$

Sign of the derivative. All multiplicative terms preceding the bracket are positive:

$$w_o > 0, \qquad 1-\rho > 0, \qquad H_o > 0.$$

The term in brackets is strictly negative whenever occupation o is not the sole contributor to aggregate output:

$$\frac{(A_oH_o)^\rho}{Y^\rho} - 1 = \frac{(A_oH_o)^\rho}{\sum_k (A_kH_k)^\rho} - 1 < 0.$$

Therefore,

$$\frac{\partial w_o}{\partial H_o} < 0 \quad \text{for all } o \text{ such that } (A_oH_o)^\rho < Y^\rho.$$

As an elasticity. Multiplying by H_j/w_o gives

$$\varepsilon_{w_o,H_o} = (1-\rho)\left(\frac{w_oH_o}{Y} - 1\right).$$

Interpretation. An increase in total efficiency units of labor H_o in a given occupation o, where the total number of occupations used in the production function is greater than one, reduces the marginal product of labor per efficiency unit (and thus the wage per efficiency unit). This property ensures that the model exhibits diminishing marginal returns to occupation-specific human capital,

consistent with the standard CES production function structures.

The corresponding elasticity is negative, unless occupation o accounts for the entire labor input. Notably, it also varies with respect to the substitution parameter ρ. As $\rho \to 1$, occupations become near-perfect substitutes so the firm does not care about the relative supply of human capital across occupations, resulting in the wage per efficiency unit becoming less elastic to changes in the human capital supply. Lower values of ρ will result in the wage per efficiency unit being more responsive to changes in the relative supply of labor.

E.4.2 Wage Effect from an Indirect Supply Shock

Next, I see how a labor supply shock to one occupation impacts the remaining occupations through general equilibrium adjustments in the wage per efficiency unit. To do this, I derive the cross derivative of w_o with respect to H_j $(j \neq o)$.

Since H_o does not depend on H_j for $j \neq o$, the dependence runs only through Y:

$$\frac{\partial w_o}{\partial H_j} = A_o^{\rho} H_o^{\rho-1} (1-\rho)\, Y^{-\rho} \frac{\partial Y}{\partial H_j}.$$

Using $\dfrac{\partial Y}{\partial H_j} = w_j = A_j^{\rho} H_j^{\rho-1} Y^{\,1-\rho}$, the result cross-derivative is positive:

$$\frac{\partial w_o}{\partial H_j} = (1-\rho)\, \frac{w_o\, w_j}{Y} \;>\; 0 \qquad (j \neq o).$$

As an elasticity. Multiplying by H_j / w_o gives

$$\varepsilon_{w_o, H_j} = (1-\rho)\, \frac{w_j H_j}{Y}$$

Interpretation. This derivation shows that the wage per efficiency unit of occupations that do not receive the labor supply shock will also be affected. In a CES production function when types of labor are substitutable ($\rho > 0$) with constant returns to scale, an increase in the labor supply to one occupation will strictly raise the equilibrium wages in all other occupations. This result comes from the fact that output is strictly increasing with respect to human capital units supplied and the wage per efficiency unit scales with output. Moreover, since the CES is homogeneous of degree one, $\sum_k w_k H_k = Y$, the term $\frac{w_j H_j}{Y}$ is occupation j's income share. Hence, the cross elasticity is positive and proportional to j's share.

Together, these two results have important implications for the model's results with respect to an increase in human capital supplied by refugees. First, occupations that receive relatively higher human capital supply shocks will experience a decrease in the wage per efficiency unit, which will dominate. Second, an increase in human capital units overall will increase output and this creates a positive indirect effect on the wage per efficiency unit of all occupations. Thus, all wages per efficiency unit will adjust in response to a supply shock.

F Appendix: Timeline of Jordan Compact

Table F1: Detailed Timeline of Work Permit Policy Changes (2016-2021)

	Policy Change
February 2016	Jordan Compact signed
April 2016	Work permit fees waived for Syrian refugees
October 2016	Beginning of flexible work permits (not tied to an employer) in the agriculture sector
June 2017	Beginning of flexible work permits in the construction sector
August 2017	Syrian refugees in Zaatari refugee camp are able to obtain work permits
October 2017	The Ministry of Labor waives the requirement of the Recognition of Prior Learning (RPL) certificate for construction work permits
February 2018	Syrian refugees in Azraq refugee camp are able to obtain work permit
October 2018	Syrian refugees with work permits are allowed to move freely between any industrial sector
November 2018	Syrian refugees are allowed to obtain work permits for home-based and independent businesses in three sectors - food processing, handicrafts, and tailoring
January 2019	Work permits for Syrian refugees become automatically renewed
September 2019	Syrian refugee workers and their employers are exempted from fines associated with working with an expired work permit
December 2020	The Ministry of Labor publishes updated list of open occupations for non-Jordanian workers.
June 2021	The Ministry of Labor introduces flexible work permits for all occupations that are open to Syrian refugees.

Note: This table was adapted from a table published in Kattaa et al. (2021) with additional verification from ILO (2022) and Tobin and Alahmed (2019).

G Appendix: Occupations Closed to Non-Jordanian Workers

According to a formal document issued in October 2016 by the then Minister of Labor Dr. Nidal Murdi Al Katamine, the following 19 occupations were closed to foreign workers (Jordanian Ministry of Labor, 2016). This in an affirmation of restrictions first set out in the 1996 Labor Law and 2007 Regulations for Foreign Workers (Jordan et al., 2023).

1. Administrative and accounting professions
2. Clerical work including typing and secretarial work
3. Switchboards, telephones, and connections work
4. Warehouse work
5. Sales work, including all groups
6. Decoration work *
7. Fuel selling in main cities *
8. Electricity professions *
9. Mechanical and car repair professions *
10. Drivers (except drivers who work in companies where the Jordanian government is a partner)
11. Guards and servants
12. Medical professions **
13. Engineering professions **
14. Haircutting (case-by-case exceptions considered via consultation with Ministry of Labor)
15. Teaching professions (case-by-case exceptions considered via Civil Service Bureau)
16. Loading and unloading workers in fruit and vegetable markets (except in central market)
17. Loading and unloading workers in malls and supermarkets
18. Cleaning workers in private schools and hotels
19. Regional offices for foreign companies (unless they are the regional or deputy regional coordinator)

Occupations marked with * have the following exceptions for work renewals and companies that provide training programs for Jordanian workers. Occupations with marked with ** are open to foreign workers only if no Jordanian staff are available to fill the position and after consultation with the related ministries and departments overseeing this occupation.

Below is a proposed matching between these occupations and the 9 major occupation groups in the ISCO8 Codes, excluding "workers in regional offices for foreign companies" as that could span multiple groups(ILO, 2008).

Table G1: Potential Matching Between ISCO8 Occupation Groups and Closed Occupations

Occupation Group	Corresponding Closed Occupations	Count
Manager	*None*	0
Professional	Medical professionals; engineering professionals; teaching professionals	3
Technicians	Administrative and accounting professions; switchboard, telephones, and connections work; mechanical and car repair professions; electricity professions	4
Clerical Support	Clerical work	1
Service and Sales	Sales work; decoration work; fuel selling in main cities; haircutting; guards and servants	5
Skilled Agriculture	*None*	0
Craft and Trade	*None*	0
Machine Operator	Drivers; warehouse work	2
Elementary Work	Loading and unloading workers in fruit/vegetable markets; loading and unloading workers in malls/supermarkets; cleaning workers in private schools and hotels	3

H Appendix: Review of 2-Digit ISCO Occupation Codes

Table H1: ISCO-08 Two-Digit Occupations with Specific Examples

ISCO Code	Occupation Title	Common Examples in Jordan
1. Managers		
11	Chief Executives, Senior Officials and Legislators	Company directors, municipal officials, NGO executives
12	Administrative and Commercial Managers	Business managers, marketing managers, shop managers
13	Production and Specialized Services Managers	Construction site managers, factory supervisors, hotel managers
14	Hospitality, Retail and Other Services Managers	Restaurant managers, tourism managers, retail branch managers
2. Professionals		
21	Science and Engineering Professionals	Civil engineers, architects, environmental specialists
22	Health Professionals	Doctors, pharmacists, nurses
23	Teaching Professionals	School teachers, university lecturers
24	Business and Administration Professionals	Accountants, auditors, financial analysts
25	Information and Communications Technology Professionals	Software developers, systems analysts, network engineers
26	Legal, Social and Cultural Professionals	Lawyers, social workers, journalists, artists
3. Technicians and Associate Professionals		
31	Science and Engineering Associate Professionals	Engineering technicians, surveyors, lab technicians
32	Health Associate Professionals	Medical lab assistants, dental hygienists, nursing aides
33	Business and Administration Associate Professionals	Office supervisors, procurement clerks, customs officers
34	Legal, Social, Cultural and Related Associate Professionals	Paralegals, teaching assistants, community workers
35	Information and Communications Technicians	IT support staff, telecommunications technicians
4. Clerical Support Workers		
41	General and Keyboard Clerks	Secretaries, receptionists, data-entry clerks
42	Customer Services Clerks	Bank tellers, call center staff, hotel front-desk agents
43	Numerical and Material Recording Clerks	Cashiers, stock clerks, shipping clerks
5. Service and Sales Workers		
51	Personal Service Workers	Hairdressers, waiters, childcare workers
52	Sales Workers	Shop assistants, street vendors, cashiers
53	Personal Care Workers	Housekeepers, elderly care aides
54	Protective Services Workers	Police officers, security guards, firefighters
6. Skilled Agricultural, Forestry and Fishery Workers		
61	Market-Oriented Skilled Agricultural Workers	Crop farmers, livestock herders, greenhouse growers
62	Market-Oriented Skilled Forestry, Fishery and Hunting Workers	Fishermen, forestry technicians (limited in Jordan)
63	Subsistence Farmers, Fishers, Hunters and Gatherers	Smallholder farmers, traditional herders
7. Craft and Related Trades Workers		
71	Building and Related Trades Workers (excluding Electricians)	Bricklayers, masons, plumbers, painters
72	Metal, Machinery and Related Trades Workers	Welders, mechanics, machine fitters
73	Handicraft and Printing Workers	Tailors, jewelers, artisans
74	Electrical and Electronic Trades Workers	Electricians, appliance repairers, telecom installers
75	Food Processing, Woodworking, Garment and Other Craft Workers	Bakers, carpenters, textile workers, upholsterers
8. Plant and Machine Operators, and Assemblers		
81	Stationary Plant and Machine Operators	Factory machine operators, textile operators
82	Assemblers	Electronics assemblers, furniture assemblers
83	Drivers and Mobile Plant Operators	Truck drivers, taxi drivers, forklift operators
9. Elementary Occupations		
91	Cleaners and Helpers	Janitors, domestic workers, hotel cleaners
92	Agricultural, Forestry and Fishery Laborers	Farm laborers, fruit pickers, packers
93	Laborers in Mining, Construction, Manufacturing and Transport	Construction helpers, loaders, porters
94	Food Preparation Assistants	Kitchen helpers, fast-food workers
95	Street and Related Sales and Service Workers	Street vendors, hawkers
96	Refuse Workers and Other Elementary Workers	Street sweepers, garbage collectors, maintenance workers
0. Armed Forces Occupations		
01	Commissioned Armed Forces Officers	Military officers
02	Non-Commissioned Armed Forces Officers	Sergeants, training officers
03	Armed Forces Occupations, Other Ranks	Soldiers, military support staff

Notes: Occupation titles follow ISCO-08 two-digit classification. Examples reflect typical jobs found in Jordan's labor market.

I Appendix: GMM Estimation Results

I.1 Full Parameter Estimates

Table I1: Structural Estimates for Occupation-Specific Wedges (τ)

Parameter	2014	2017	2018	2019	2020	2021	2022
τ - Managers & Professionals	0.689	0.673	0.630	0.642	0.656	0.600	0.607
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Technicians	0.783	0.611	0.610	0.542	0.542	0.579	0.529
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Clerks	0.711	0.610	0.557	0.525	0.485	0.467	0.397
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Services and Sales	0.583	0.472	0.470	0.458	0.424	0.408	0.395
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Agriculture	0.560	0.101	-0.320	-0.497	-0.345	-0.214	-0.395
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Craft Trades	0.354	0.105	-0.004	0.049	-0.005	-0.109	0.022
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Machine Operators	0.651	0.485	0.522	0.540	0.528	0.523	0.482
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)
τ - Elementary Jobs	0.413	0.122	0.159	0.251	0.198	0.109	0.111
	(0.022)	(0.011)	(0.019)	(0.041)	(0.024)	(0.018)	(0.022)

Table I2: Structural Estimates for Wage Per Efficiency Unit (w)

Parameter	2014	2017	2018	2019	2020	2021	2022
w - Managers & Professionals	13.566	15.221	13.397	12.806	13.468	13.806	14.057
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Technicians	9.096	10.590	9.869	10.034	9.875	10.038	10.506
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Clerks	12.567	12.952	11.318	10.519	10.831	10.944	11.427
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Services and Sales	16.986	19.266	18.799	19.054	18.848	18.717	19.238
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Agriculture	7.561	6.535	6.249	5.868	5.813	6.522	6.887
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Craft Trades	15.682	14.919	13.711	12.724	13.006	13.269	14.577
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Machine Operators	15.717	14.396	13.538	13.050	12.633	12.803	13.481
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Elementary Jobs	8.445	10.599	11.107	11.024	10.583	11.453	11.914
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)
w - Unemployment	15.901	10.932	12.250	12.159	12.018	12.670	12.942
	(0.616)	(0.310)	(0.090)	(0.172)	(0.112)	(0.114)	(0.115)

Table I3: Structural Estimates for Homophilic Preference Parameters (λ)

Parameter	2014	2017	2018	2019	2020	2021	2022
λ - Syrians	0.079	0.153	0.128	0.130	0.130	0.143	0.149
	(0.006)	(0.007)	(0.005)	(0.013)	(0.005)	(0.008)	(0.008)
λ - Jordanians	2.253	2.117	1.709	1.094	1.239	1.803	1.235
	(0.006)	(0.007)	(0.005)	(0.013)	(0.005)	(0.008)	(0.008)

Table I4: Structural Estimates for Exogenous Preferences - Syrians (α_{Syr})

Parameter	2014	2017	2018	2019	2020	2021	2022
α_{Syr} - Managers & Professionals	3.945	4.875	5.028	4.991	5.342	4.878	5.354
	(0.202)	(0.116)	(0.120)	(0.326)	(0.155)	(0.124)	(0.137)
α_{Syr} - Technicians	6.259	5.890	5.918	4.874	5.632	5.908	5.910
	(0.359)	(0.131)	(0.164)	(0.313)	(0.162)	(0.187)	(0.189)
α_{Syr} - Clerks	5.687	6.016	5.916	5.578	5.381	5.356	4.915
	(0.378)	(0.177)	(0.143)	(0.331)	(0.164)	(0.128)	(0.135)
α_{Syr} - Services and Sales	6.006	5.776	6.030	5.951	5.654	5.690	5.989
	(0.247)	(0.081)	(0.061)	(0.149)	(0.065)	(0.080)	(0.090)
α_{Syr} - Agriculture	5.643	6.365	5.120	5.447	5.617	5.288	5.257
	(0.431)	(0.224)	(0.195)	(0.256)	(0.133)	(0.129)	(0.083)
α_{Syr} - Craft Trades	5.370	4.656	4.786	5.343	4.934	4.593	5.013
	(0.172)	(0.097)	(0.063)	(0.130)	(0.083)	(0.066)	(0.144)
α_{Syr} - Machine Operators	5.693	5.925	6.531	7.016	6.715	6.822	6.553
	(0.551)	(0.122)	(0.152)	(0.287)	(0.151)	(0.141)	(0.206)
α_{Syr} - Elementary Jobs	7.907	5.851	5.928	6.454	6.220	5.820	6.026
	(0.249)	(0.097)	(0.090)	(0.146)	(0.095)	(0.094)	(0.105)
α_{Syr} - Unemployment	4.217	5.155	5.284	5.509	5.667	5.415	5.632
	(0.125)	(0.054)	(0.045)	(0.079)	(0.059)	(0.045)	(0.048)

Table I5: Structural Estimates for Exogenous Preferences - Jordanians (α_{Jor})

Parameter	2014	2017	2018	2019	2020	2021	2022
α_{Jor} - Managers & Professionals	1.261 (0.000)	1.230 (0.000)	1.624 (0.000)	1.677 (0.000)	1.627 (0.000)	1.658 (0.000)	1.783 (0.000)
α_{Jor} - Technicians	1.695 (0.000)	1.715 (0.000)	2.001 (0.000)	1.990 (0.000)	2.042 (0.000)	2.066 (0.000)	2.229 (0.000)
α_{Jor} - Clerks	1.809 (0.000)	1.699 (0.000)	2.119 (0.000)	2.144 (0.000)	2.146 (0.000)	2.399 (0.000)	2.390 (0.000)
α_{Jor} - Services and Sales	2.192 (0.000)	2.160 (0.000)	2.469 (0.000)	2.338 (0.000)	2.427 (0.000)	2.687 (0.000)	2.647 (0.000)
α_{Jor} - Agriculture	2.320 (0.000)	5.933 (0.000)	7.014 (0.000)	7.081 (0.000)	7.163 (0.000)	7.048 (0.000)	7.114 (0.000)
α_{Jor} - Craft Trades	2.949 (0.000)	4.516 (0.000)	4.311 (0.000)	3.658 (0.000)	3.815 (0.000)	4.540 (0.000)	3.946 (0.000)
α_{Jor} - Machine Operators	1.886 (0.000)	2.154 (0.000)	2.396 (0.000)	2.373 (0.000)	2.405 (0.000)	2.542 (0.000)	2.533 (0.000)
α_{Jor} - Elementary Jobs	4.008 (0.000)	6.630 (0.000)	4.188 (0.000)	3.398 (0.000)	3.624 (0.000)	4.477 (0.000)	4.162 (0.000)
α_{Jor} - Unemployment	6.641 (0.000)	4.247 (0.000)	4.472 (0.000)	3.927 (0.000)	4.196 (0.000)	4.495 (0.000)	4.180 (0.000)

Table I6: Structural Estimates for Occupation-Specific Productivity (A)

Parameter	2014	2017	2018	2019	2020	2021	2022
A - Managers & Professionals	6.025	7.237	6.282	5.800	6.414	6.650	6.719
A - Technicians	2.054	2.570	2.311	2.513	2.346	2.456	2.555
A - Clerks	3.953	3.289	2.797	2.457	2.509	2.631	2.775
A - Services and Sales	9.338	11.462	11.701	12.036	11.867	11.519	11.488
A - Agriculture	0.859	0.642	0.636	0.578	0.635	0.693	0.771
A - Craft Trades	6.357	5.374	4.553	4.190	4.397	4.286	5.149
A - Machine Operators	5.853	4.429	4.119	3.940	3.567	3.689	3.861
A - Elementary Jobs	1.569	2.653	2.805	2.908	2.616	3.065	3.324

I.2 GMM Model Fit

2014 GMM Estimation - Model Fit

Table I7: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.490	-4.550	0.0599
Log Shares - Syrian Technicians	-6.741	-6.669	-0.0722
Log Shares - Syrian Clerks	-5.132	-5.231	0.0989
Log Shares - Syrian Services and Sales	-2.614	-2.621	0.0067
Log Shares - Syrian Agriculture	-5.488	-5.719	0.2307
Log Shares - Syrian Craft Trades	-1.893	-1.857	-0.0363
Log Shares - Syrian Machine Operators	-4.102	-4.104	0.0018
Log Shares - Syrian Elementary Jobs	-3.323	-3.353	0.0300
Log Shares - Syrian Unemployment	-0.355	-0.358	0.0029
Log Shares - Jordanian Managers & Professionals	-2.158	-2.146	-0.0122
Log Shares - Jordanian Technicians	-3.148	-3.155	0.0068
Log Shares - Jordanian Clerks	-2.470	-2.455	-0.0152
Log Shares - Jordanian Services and Sales	-1.270	-1.268	-0.0019
Log Shares - Jordanian Agriculture	-4.550	-4.418	-0.1315
Log Shares - Jordanian Craft Trades	-1.961	-2.006	0.0446
Log Shares - Jordanian Machine Operators	-2.124	-2.124	-0.0004
Log Shares - Jordanian Elementary Jobs	-3.179	-3.146	-0.0329
Log Shares - Jordanian Unemployment	-1.809	-1.808	-0.0015

Table I8: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.561	5.788	-0.2273
Log Wage - Syrian Technicians	5.521	5.463	0.0588
Log Wage - Syrian Clerks	5.342	5.436	-0.0941
Log Wage - Syrian Services and Sales	5.291	5.308	-0.0161
Log Wage - Syrian Agriculture	5.091	5.275	-0.1841
Log Wage - Syrian Craft Trades	5.367	5.331	0.0352
Log Wage - Syrian Machine Operators	5.412	5.411	0.0012
Log Wage - Syrian Elementary Jobs	4.905	4.940	-0.0344
Log Wage - Jordanian Managers & Professionals	6.299	6.282	0.0163
Log Wage - Jordanian Technicians	5.946	5.952	-0.0055
Log Wage - Jordanian Clerks	5.929	5.911	0.0183
Log Wage - Jordanian Services and Sales	5.797	5.793	0.0036
Log Wage - Jordanian Agriculture	5.889	5.826	0.0629
Log Wage - Jordanian Craft Trades	5.698	5.788	-0.0900
Log Wage - Jordanian Machine Operators	5.892	5.893	-0.0014
Log Wage - Jordanian Elementary Jobs	5.448	5.430	0.0182

Table I9: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.738	-0.491	-0.2475
Log Wage Ratio - Technicians	-0.425	-0.452	0.0276
Log Wage Ratio - Clerks	-0.587	-0.523	-0.0640
Log Wage Ratio - Services and Sales	-0.506	-0.481	-0.0244
Log Wage Ratio - Agriculture	-0.798	-0.518	-0.2793
Log Wage Ratio - Craft Trades	-0.331	-0.485	0.1538
Log Wage Ratio - Machine Operators	-0.480	-0.482	0.0019
Log Wage Ratio - Elementary Jobs	-0.543	-0.492	-0.0511

2017 GMM Estimation - Model Fit

Table I10: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.257	-4.267	0.0095
Log Shares - Syrian Technicians	-4.699	-4.767	0.0674
Log Shares - Syrian Clerks	-4.611	-4.597	-0.0138
Log Shares - Syrian Services and Sales	-1.811	-1.810	-0.0004
Log Shares - Syrian Agriculture	-3.512	-3.484	-0.0279
Log Shares - Syrian Craft Trades	-1.226	-1.218	-0.0075
Log Shares - Syrian Machine Operators	-3.191	-3.188	-0.0029
Log Shares - Syrian Elementary Jobs	-1.737	-1.751	0.0138
Log Shares - Syrian Unemployment	-1.336	-1.337	0.0009
Log Shares - Jordanian Managers & Professionals	-1.919	-1.917	-0.0017
Log Shares - Jordanian Technicians	-3.014	-2.981	-0.0335
Log Shares - Jordanian Clerks	-2.898	-2.901	0.0026
Log Shares - Jordanian Services and Sales	-1.136	-1.136	-0.0000
Log Shares - Jordanian Agriculture	-4.562	-4.701	0.1389
Log Shares - Jordanian Craft Trades	-2.315	-2.397	0.0816
Log Shares - Jordanian Machine Operators	-2.436	-2.438	0.0017
Log Shares - Jordanian Elementary Jobs	-3.002	-2.869	-0.1337
Log Shares - Jordanian Unemployment	-1.707	-1.705	-0.0019

Table I11: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.827	5.879	-0.0522
Log Wage - Syrian Technicians	5.383	5.576	-0.1932
Log Wage - Syrian Clerks	5.806	5.583	0.2230
Log Wage - Syrian Services and Sales	5.483	5.470	0.0133
Log Wage - Syrian Agriculture	5.270	5.169	0.1007
Log Wage - Syrian Craft Trades	5.507	5.472	0.0348
Log Wage - Syrian Machine Operators	5.475	5.456	0.0184
Log Wage - Syrian Elementary Jobs	5.195	5.221	-0.0253
Log Wage - Jordanian Managers & Professionals	6.325	6.322	0.0023
Log Wage - Jordanian Technicians	6.073	6.033	0.0398
Log Wage - Jordanian Clerks	6.026	6.030	-0.0039
Log Wage - Jordanian Services and Sales	5.913	5.913	-0.0003
Log Wage - Jordanian Agriculture	5.554	5.578	-0.0243
Log Wage - Jordanian Craft Trades	5.865	5.897	-0.0317
Log Wage - Jordanian Machine Operators	5.896	5.901	-0.0051
Log Wage - Jordanian Elementary Jobs	5.733	5.716	0.0172

Table I12: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.497	-0.449	-0.0489
Log Wage Ratio - Technicians	-0.690	-0.461	-0.2292
Log Wage Ratio - Clerks	-0.221	-0.436	0.2152
Log Wage Ratio - Services and Sales	-0.430	-0.445	0.0149
Log Wage Ratio - Agriculture	-0.284	-0.431	0.1465
Log Wage Ratio - Craft Trades	-0.359	-0.437	0.0787
Log Wage Ratio - Machine Operators	-0.422	-0.442	0.0204
Log Wage Ratio - Elementary Jobs	-0.538	-0.435	-0.1026

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Table I13: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.020	-4.106	0.0857
Log Shares - Syrian Technicians	-4.891	-5.063	0.1722
Log Shares - Syrian Clerks	-4.348	-4.429	0.0812
Log Shares - Syrian Services and Sales	-1.876	-1.915	0.0387
Log Shares - Syrian Agriculture	-2.969	-2.928	-0.0409
Log Shares - Syrian Craft Trades	-1.339	-1.316	-0.0231
Log Shares - Syrian Machine Operators	-3.623	-3.609	-0.0135
Log Shares - Syrian Elementary Jobs	-2.165	-2.151	-0.0137
Log Shares - Syrian Unemployment	-1.040	-1.042	0.0020
Log Shares - Jordanian Managers & Professionals	-1.917	-1.900	-0.0171
Log Shares - Jordanian Technicians	-3.156	-3.104	-0.0515
Log Shares - Jordanian Clerks	-3.039	-2.988	-0.0506
Log Shares - Jordanian Services and Sales	-1.133	-1.083	-0.0501
Log Shares - Jordanian Agriculture	-4.550	-5.180	0.6299
Log Shares - Jordanian Craft Trades	-2.409	-2.686	0.2773
Log Shares - Jordanian Machine Operators	-2.487	-2.480	-0.0069
Log Shares - Jordanian Elementary Jobs	-2.860	-2.944	0.0842
Log Shares - Jordanian Unemployment	-1.612	-1.577	-0.0343

Table I14: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.588	5.809	-0.2208
Log Wage - Syrian Technicians	5.290	5.575	-0.2855
Log Wage - Syrian Clerks	5.307	5.549	-0.2420
Log Wage - Syrian Services and Sales	5.409	5.475	-0.0660
Log Wage - Syrian Agriculture	5.575	5.437	0.1380
Log Wage - Syrian Craft Trades	5.677	5.537	0.1397
Log Wage - Syrian Machine Operators	5.503	5.457	0.0466
Log Wage - Syrian Elementary Jobs	5.387	5.353	0.0342
Log Wage - Jordanian Managers & Professionals	6.198	6.197	0.0014
Log Wage - Jordanian Technicians	6.021	6.015	0.0059
Log Wage - Jordanian Clerks	5.989	5.987	0.0029
Log Wage - Jordanian Services and Sales	5.898	5.895	0.0034
Log Wage - Jordanian Agriculture	5.597	5.615	-0.0184
Log Wage - Jordanian Craft Trades	5.815	5.842	-0.0278
Log Wage - Jordanian Machine Operators	5.863	5.866	-0.0029
Log Wage - Jordanian Elementary Jobs	5.723	5.727	-0.0039

Table I15: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.610	-0.443	-0.1680
Log Wage Ratio - Technicians	-0.731	-0.426	-0.3058
Log Wage Ratio - Clerks	-0.682	-0.402	-0.2801
Log Wage Ratio - Services and Sales	-0.489	-0.407	-0.0827
Log Wage Ratio - Agriculture	-0.022	-0.429	0.4073
Log Wage Ratio - Craft Trades	-0.138	-0.409	0.2713
Log Wage Ratio - Machine Operators	-0.360	-0.390	0.0297
Log Wage Ratio - Elementary Jobs	-0.336	-0.399	0.0636

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Table I16: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.212	-4.528	0.3155
Log Shares - Syrian Technicians	-4.500	-4.965	0.4647
Log Shares - Syrian Clerks	-4.346	-4.592	0.2466
Log Shares - Syrian Services and Sales	-1.873	-1.897	0.0247
Log Shares - Syrian Agriculture	-2.708	-2.628	-0.0798
Log Shares - Syrian Craft Trades	-1.424	-1.388	-0.0361
Log Shares - Syrian Machine Operators	-3.807	-3.704	-0.1028
Log Shares - Syrian Elementary Jobs	-2.400	-2.410	0.0101
Log Shares - Syrian Unemployment	-0.949	-0.952	0.0030
Log Shares - Jordanian Managers & Professionals	-1.966	-1.951	-0.0149
Log Shares - Jordanian Technicians	-3.056	-3.007	-0.0488
Log Shares - Jordanian Clerks	-3.121	-3.100	-0.0207
Log Shares - Jordanian Services and Sales	-1.105	-1.095	-0.0106
Log Shares - Jordanian Agriculture	-4.850	-5.415	0.5651
Log Shares - Jordanian Craft Trades	-2.487	-2.600	0.1127
Log Shares - Jordanian Machine Operators	-2.493	-2.494	0.0009
Log Shares - Jordanian Elementary Jobs	-2.777	-2.755	-0.0226
Log Shares - Jordanian Unemployment	-1.600	-1.590	-0.0098

Table I17: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.359	5.784	-0.4254
Log Wage - Syrian Technicians	5.347	5.664	-0.3169
Log Wage - Syrian Clerks	5.316	5.565	-0.2489
Log Wage - Syrian Services and Sales	5.456	5.509	-0.0533
Log Wage - Syrian Agriculture	5.421	5.363	0.0574
Log Wage - Syrian Craft Trades	5.559	5.442	0.1169
Log Wage - Syrian Machine Operators	5.548	5.447	0.1008
Log Wage - Syrian Elementary Jobs	5.297	5.304	-0.0068
Log Wage - Jordanian Managers & Professionals	6.166	6.163	0.0024
Log Wage - Jordanian Technicians	6.040	6.028	0.0121
Log Wage - Jordanian Clerks	5.960	5.956	0.0035
Log Wage - Jordanian Services and Sales	5.900	5.899	0.0010
Log Wage - Jordanian Agriculture	5.548	5.578	-0.0299
Log Wage - Jordanian Craft Trades	5.780	5.799	-0.0196
Log Wage - Jordanian Machine Operators	5.842	5.844	-0.0016
Log Wage - Jordanian Elementary Jobs	5.703	5.702	0.0008

Table I18: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.807	-0.499	-0.3071
Log Wage Ratio - Technicians	-0.693	-0.568	-0.1251
Log Wage Ratio - Clerks	-0.644	-0.469	-0.1749
Log Wage Ratio - Services and Sales	-0.445	-0.403	-0.0421
Log Wage Ratio - Agriculture	-0.127	-0.382	0.2550
Log Wage Ratio - Craft Trades	-0.221	-0.383	0.1621
Log Wage Ratio - Machine Operators	-0.294	-0.356	0.0624
Log Wage Ratio - Elementary Jobs	-0.406	-0.394	-0.0115

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Table I19: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.310	-4.322	0.0120
Log Shares - Syrian Technicians	-4.568	-4.613	0.0449
Log Shares - Syrian Clerks	-4.266	-4.343	0.0768
Log Shares - Syrian Services and Sales	-1.807	-1.851	0.0445
Log Shares - Syrian Agriculture	-2.738	-2.708	-0.0293
Log Shares - Syrian Craft Trades	-1.400	-1.376	-0.0233
Log Shares - Syrian Machine Operators	-4.000	-3.937	-0.0626
Log Shares - Syrian Elementary Jobs	-2.418	-2.403	-0.0143
Log Shares - Syrian Unemployment	-0.970	-0.974	0.0037
Log Shares - Jordanian Managers & Professionals	-1.921	-1.916	-0.0044
Log Shares - Jordanian Technicians	-3.159	-3.145	-0.0136
Log Shares - Jordanian Clerks	-3.217	-3.185	-0.0325
Log Shares - Jordanian Services and Sales	-1.160	-1.126	-0.0340
Log Shares - Jordanian Agriculture	-4.603	-4.968	0.3651
Log Shares - Jordanian Craft Trades	-2.451	-2.596	0.1453
Log Shares - Jordanian Machine Operators	-2.638	-2.647	0.0086
Log Shares - Jordanian Elementary Jobs	-2.911	-2.949	0.0374
Log Shares - Jordanian Unemployment	-1.447	-1.431	-0.0166

Table I20: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.790	5.824	-0.0342
Log Wage - Syrian Technicians	5.523	5.638	-0.1146
Log Wage - Syrian Clerks	5.432	5.630	-0.1974
Log Wage - Syrian Services and Sales	5.431	5.541	-0.1103
Log Wage - Syrian Agriculture	5.377	5.333	0.0442
Log Wage - Syrian Craft Trades	5.739	5.519	0.2201
Log Wage - Syrian Machine Operators	5.650	5.487	0.1625
Log Wage - Syrian Elementary Jobs	5.375	5.337	0.0386
Log Wage - Jordanian Managers & Professionals	6.202	6.202	0.0001
Log Wage - Jordanian Technicians	6.014	6.012	0.0021
Log Wage - Jordanian Clerks	5.996	5.991	0.0057
Log Wage - Jordanian Services and Sales	5.909	5.905	0.0035
Log Wage - Jordanian Agriculture	5.553	5.575	-0.0219
Log Wage - Jordanian Craft Trades	5.806	5.817	-0.0115
Log Wage - Jordanian Machine Operators	5.841	5.844	-0.0035
Log Wage - Jordanian Elementary Jobs	5.696	5.700	-0.0036

Table I21: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.411	-0.379	-0.0321
Log Wage Ratio - Technicians	-0.490	-0.395	-0.0955
Log Wage Ratio - Clerks	-0.564	-0.422	-0.1420
Log Wage Ratio - Services and Sales	-0.478	-0.410	-0.0679
Log Wage Ratio - Agriculture	-0.176	-0.396	0.2197
Log Wage Ratio - Craft Trades	-0.066	-0.405	0.3386
Log Wage Ratio - Machine Operators	-0.191	-0.374	0.1833
Log Wage Ratio - Elementary Jobs	-0.321	-0.374	0.0531

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Table I22: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-3.931	-4.038	0.1071
Log Shares - Syrian Technicians	-4.882	-4.981	0.0983
Log Shares - Syrian Clerks	-3.978	-4.205	0.2267
Log Shares - Syrian Services and Sales	-1.794	-1.884	0.0898
Log Shares - Syrian Agriculture	-3.172	-3.133	-0.0395
Log Shares - Syrian Craft Trades	-1.398	-1.343	-0.0556
Log Shares - Syrian Machine Operators	-3.954	-3.968	0.0135
Log Shares - Syrian Elementary Jobs	-2.067	-2.033	-0.0337
Log Shares - Syrian Unemployment	-1.041	-1.038	-0.0025
Log Shares - Jordanian Managers & Professionals	-1.945	-1.933	-0.0112
Log Shares - Jordanian Technicians	-3.101	-3.084	-0.0169
Log Shares - Jordanian Clerks	-3.238	-3.104	-0.1342
Log Shares - Jordanian Services and Sales	-1.259	-1.168	-0.0902
Log Shares - Jordanian Agriculture	-4.593	-4.989	0.3964
Log Shares - Jordanian Craft Trades	-2.418	-2.823	0.4051
Log Shares - Jordanian Machine Operators	-2.615	-2.602	-0.0136
Log Shares - Jordanian Elementary Jobs	-2.770	-2.907	0.1372
Log Shares - Jordanian Unemployment	-1.372	-1.352	-0.0203

Table I23: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.613	5.892	-0.2792
Log Wage - Syrian Technicians	5.478	5.674	-0.1963
Log Wage - Syrian Clerks	5.375	5.594	-0.2193
Log Wage - Syrian Services and Sales	5.451	5.546	-0.0953
Log Wage - Syrian Agriculture	5.471	5.444	0.0271
Log Wage - Syrian Craft Trades	5.690	5.625	0.0658
Log Wage - Syrian Machine Operators	5.472	5.494	-0.0218
Log Wage - Syrian Elementary Jobs	5.453	5.413	0.0401
Log Wage - Jordanian Managers & Professionals	6.237	6.235	0.0023
Log Wage - Jordanian Technicians	6.027	6.026	0.0011
Log Wage - Jordanian Clerks	6.018	6.009	0.0089
Log Wage - Jordanian Services and Sales	5.921	5.916	0.0048
Log Wage - Jordanian Agriculture	5.642	5.660	-0.0177
Log Wage - Jordanian Craft Trades	5.799	5.816	-0.0167
Log Wage - Jordanian Machine Operators	5.849	5.848	0.0009
Log Wage - Jordanian Elementary Jobs	5.726	5.732	-0.0054

Table I24: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.624	-0.402	-0.2221
Log Wage Ratio - Technicians	-0.549	-0.392	-0.1574
Log Wage Ratio - Clerks	-0.643	-0.399	-0.2441
Log Wage Ratio - Services and Sales	-0.470	-0.392	-0.0783
Log Wage Ratio - Agriculture	-0.171	-0.368	0.1972
Log Wage Ratio - Craft Trades	-0.109	-0.386	0.2772
Log Wage Ratio - Machine Operators	-0.377	-0.364	-0.0136
Log Wage Ratio - Elementary Jobs	-0.273	-0.342	0.0684

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Table I25: Targeted Moments - Occupation Shares

Target	Data	Model Estimate	Difference
Log Shares - Syrian Managers & Professionals	-4.050	-4.115	0.0649
Log Shares - Syrian Technicians	-4.628	-4.711	0.0830
Log Shares - Syrian Clerks	-3.980	-4.228	0.2487
Log Shares - Syrian Services and Sales	-1.864	-1.871	0.0066
Log Shares - Syrian Agriculture	-2.787	-2.732	-0.0551
Log Shares - Syrian Craft Trades	-1.424	-1.405	-0.0186
Log Shares - Syrian Machine Operators	-3.794	-3.776	-0.0180
Log Shares - Syrian Elementary Jobs	-2.031	-2.035	0.0043
Log Shares - Syrian Unemployment	-1.071	-1.073	0.0025
Log Shares - Jordanian Managers & Professionals	-1.903	-1.899	-0.0036
Log Shares - Jordanian Technicians	-3.078	-3.068	-0.0105
Log Shares - Jordanian Clerks	-3.139	-3.071	-0.0677
Log Shares - Jordanian Services and Sales	-1.228	-1.223	-0.0050
Log Shares - Jordanian Agriculture	-4.636	-4.978	0.3428
Log Shares - Jordanian Craft Trades	-2.452	-2.511	0.0585
Log Shares - Jordanian Machine Operators	-2.662	-2.662	0.0001
Log Shares - Jordanian Elementary Jobs	-2.743	-2.732	-0.0112
Log Shares - Jordanian Unemployment	-1.433	-1.427	-0.0062

Table I26: Targeted Moments - Occupation Average Wages

Target	Data	Model Estimate	Difference
Log Wage - Syrian Managers & Professionals	5.712	5.921	-0.2088
Log Wage - Syrian Technicians	5.571	5.731	-0.1600
Log Wage - Syrian Clerks	5.351	5.768	-0.4170
Log Wage - Syrian Services and Sales	5.549	5.612	-0.0634
Log Wage - Syrian Agriculture	5.535	5.493	0.0414
Log Wage - Syrian Craft Trades	5.774	5.598	0.1761
Log Wage - Syrian Machine Operators	5.658	5.597	0.0604
Log Wage - Syrian Elementary Jobs	5.440	5.440	-0.0005
Log Wage - Jordanian Managers & Professionals	6.236	6.236	0.0009
Log Wage - Jordanian Technicians	6.038	6.036	0.0020
Log Wage - Jordanian Clerks	6.036	6.030	0.0055
Log Wage - Jordanian Services and Sales	5.931	5.931	0.0002
Log Wage - Jordanian Agriculture	5.670	5.694	-0.0239
Log Wage - Jordanian Craft Trades	5.860	5.917	-0.0577
Log Wage - Jordanian Machine Operators	5.920	5.929	-0.0090
Log Wage - Jordanian Elementary Jobs	5.763	5.763	0.0002

Table I27: Targeted Moments - Average Wage Ratios

Target	Data	Model Estimate	Difference
Log Wage Ratio - Managers & Professionals	-0.524	-0.347	-0.1771
Log Wage Ratio - Technicians	-0.467	-0.364	-0.1028
Log Wage Ratio - Clerks	-0.685	-0.471	-0.2145
Log Wage Ratio - Services and Sales	-0.382	-0.326	-0.0559
Log Wage Ratio - Agriculture	-0.135	-0.326	0.1913
Log Wage Ratio - Craft Trades	-0.085	-0.300	0.2152
Log Wage Ratio - Machine Operators	-0.262	-0.305	0.0421
Log Wage Ratio - Elementary Jobs	-0.324	-0.323	-0.0009